CompTIA Cybersecurity Analyst (CySA+)

System & Network Security Implementation Concepts

Introduction

CS0-003
Splunk Enterprise
Multifactor Authentication
Encryption
BitLocker

Welcome to the **System & Network Security Implementation Concepts** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Log Collection with Splunk
- Exercise 2 - Encrypting Sensitive Data
- Exercise 3 - Enable Multifactor Authentication

After completing these modules, you should be able to:

- Install and Configure the Splunk Enterprise Application
- Configure a Splunk Client
- Collect Logs using Splunk Enterprise
- Install the BitLocker Drive Encryption Feature
- Encrypt a Local Drive on a Server
- Enable Multifactor Authentication
- Verify Multifactor Authentication

Exam Objectives

The following exam objectives are covered in this module:

1.1 Explain the importance of system and network architecture concepts in security operations.

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **ACIDC01 -** Windows Server 2022 - Domain Controller
- **ACIDM01 -** Windows Server 2022 - Domain Member Server

- **ACIWIN11 -** Windows 11 - Domain Member Workstation
- **ACIALMA -** ALMA Linux Standalone Workstation
- **ACIKALI -** Kali Linux Standalone Workstation

Click **Next** to proceed to the first exercise.

Exercise 1 - Log Collection with Splunk

The Splunk Enterprise application can be used to collect logs of devices on the network. Using the application, a centralized point of all the logs of the devices can be monitored.

In this exercise, the Splunk Enterprise application will be installed and configured to collect the logs of devices on the network.

Learning Outcomes

After completing this exercise, you should be able to:

- Install and Configure the Splunk Enterprise Application
- Configure a Splunk Client
- Collect Logs using Splunk Enterprise

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01 -** Windows Server 2022 - Domain Controller
- **ACIDM01 -** Windows Server 2022 - Domain Member Server
- **ACIWIN11 -** Windows 11 - Domain Member Workstation

Task 1 - Install and Configure the Splunk Enterprise Application

The Splunk Enterprise application collects logs from devices on the network. In this task, the Splunk Enterprise application will be installed and configured.

Step 1

Connect to **ACIWIN11**.

Open **File Explorer** from the **Taskbar**.

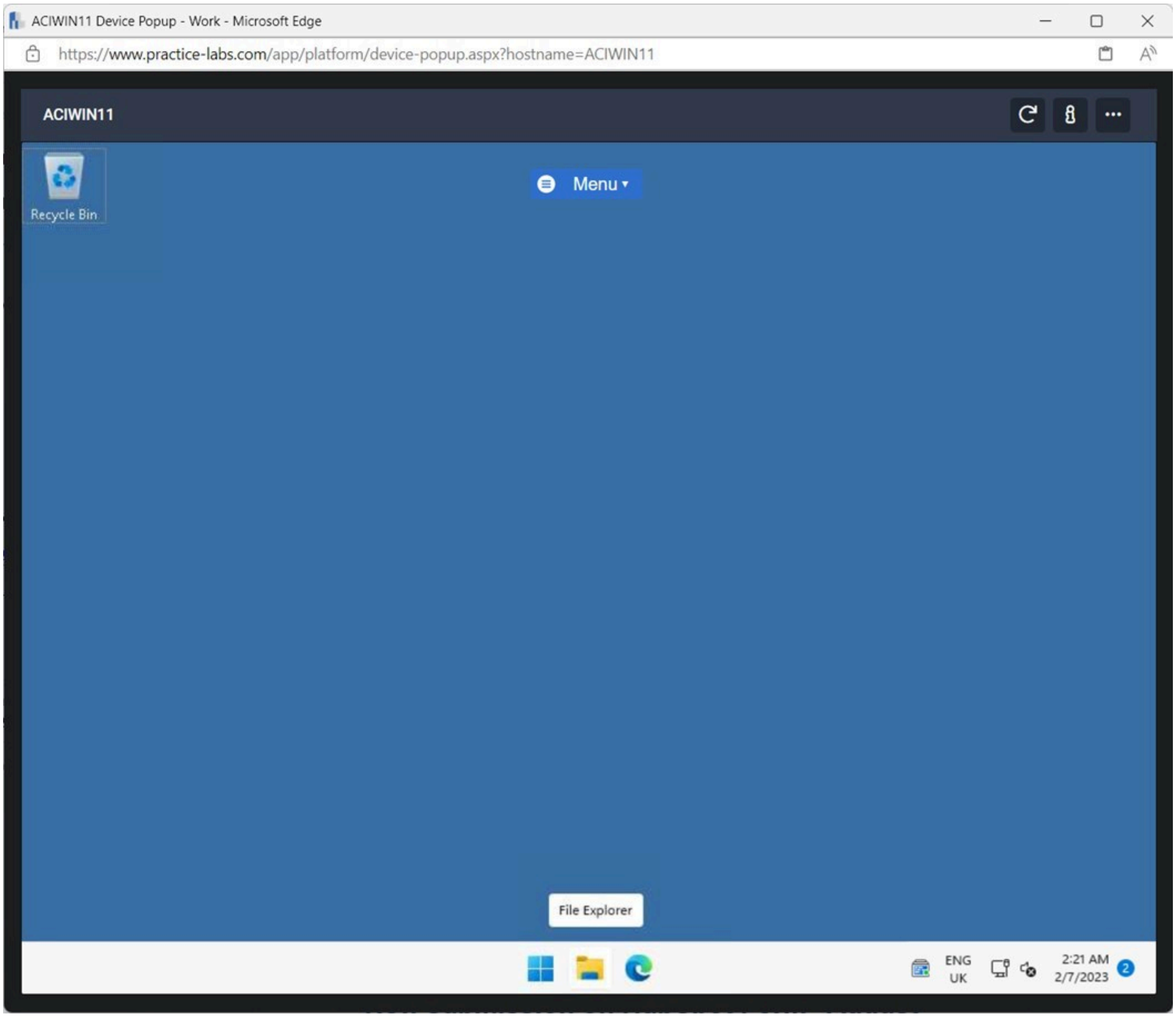


Figure 1.1 Screenshot of ACIWIN11: Displaying opening File Explorer from the Taskbar.

Step 2

In **File Explorer**, click the **Downloads** folder.

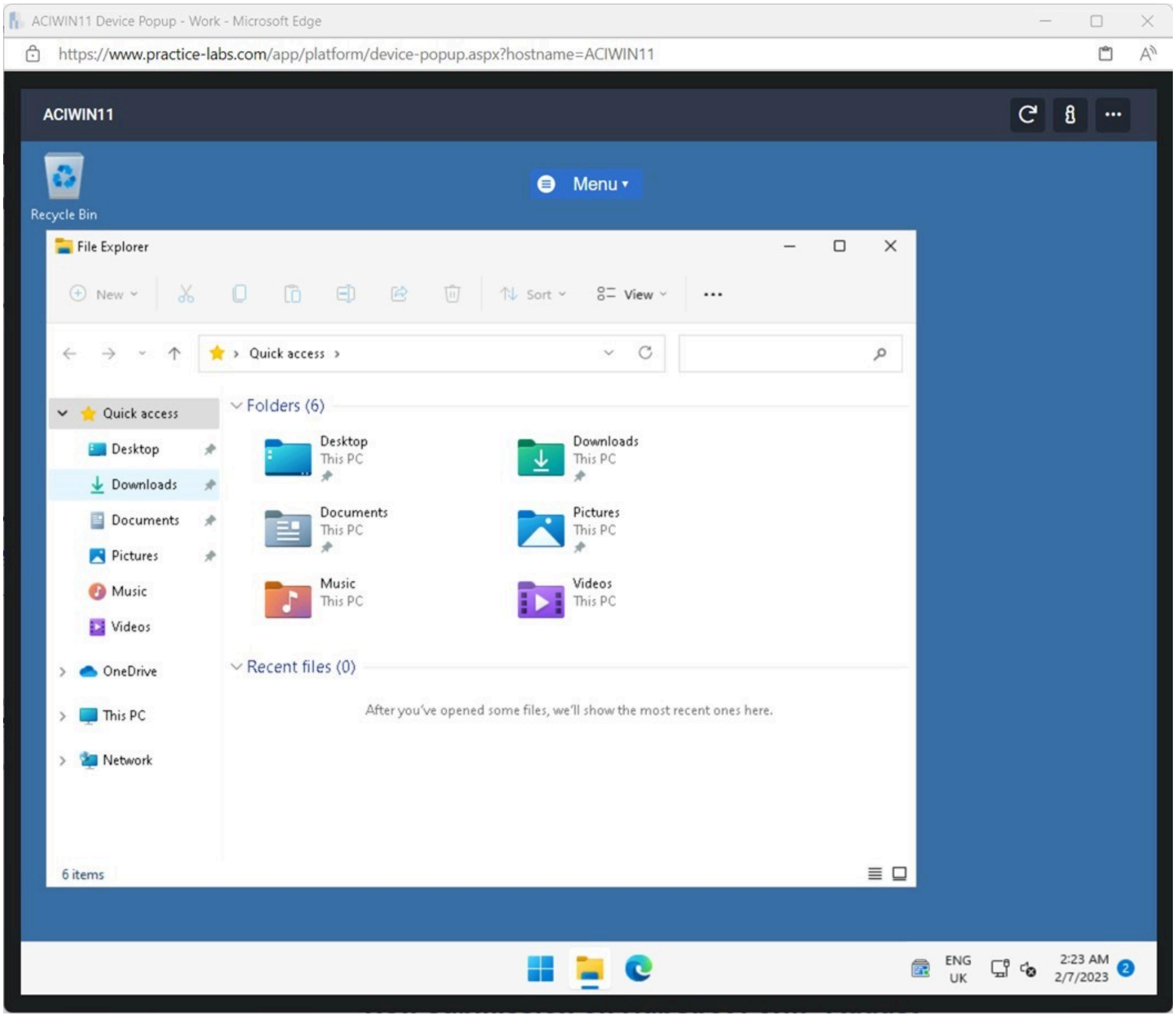


Figure 1.2 Screenshot of ACIWIN11: Displaying clicking the Downloads folder in File Explorer.

Step 3

In **File Explorer** in the **Downloads** folder, double click the **splunk-9.0.3-dd** installation file.

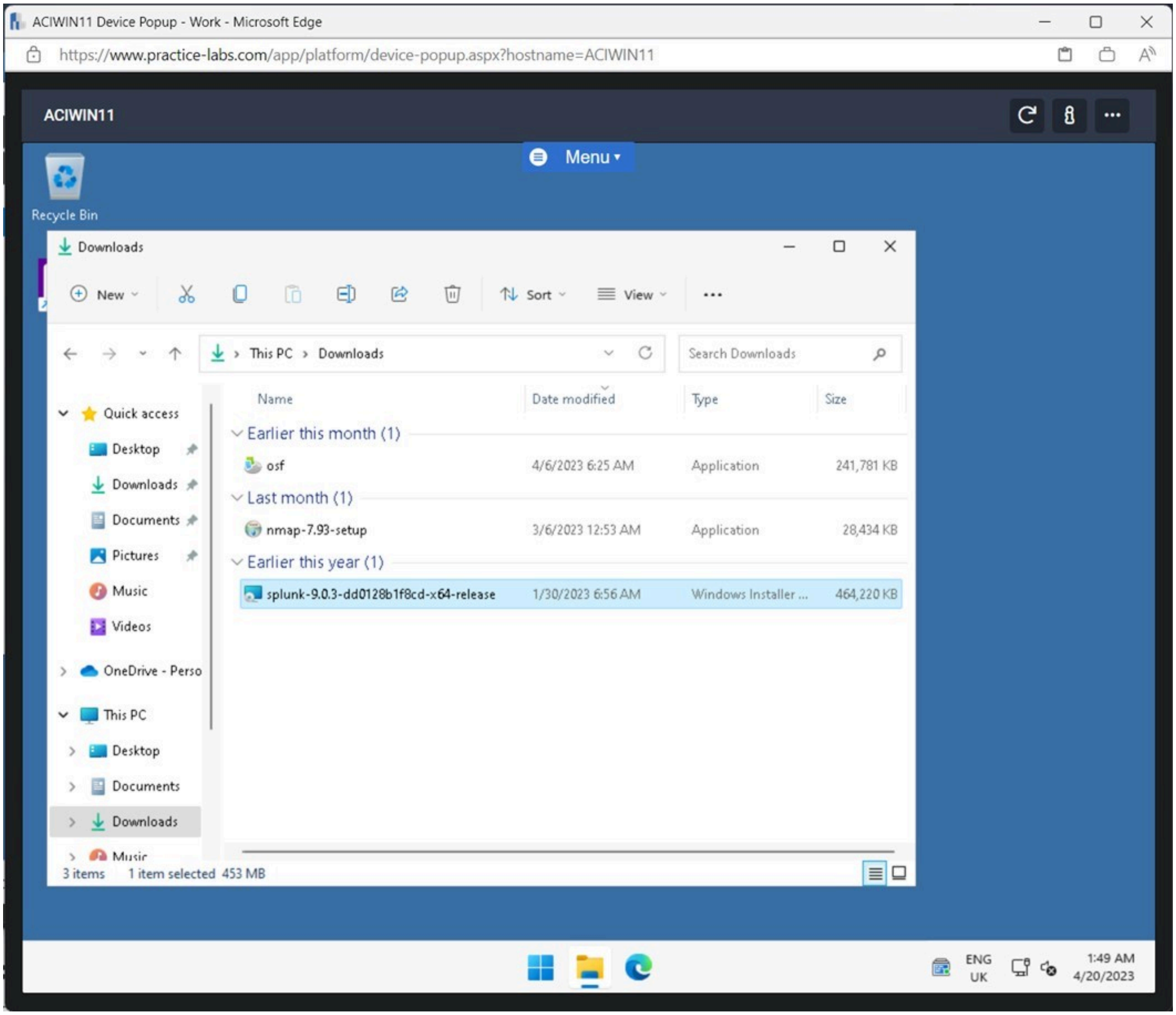


Figure 1.3 Screenshot of ACIWIN11: Displaying double-clicking the splunk-9.0.3-dd installation file.

Step 4

On the **Splunk Enterprise Installer** pop-up window, tick the **Check this box to accept the License Agreement** tick box

Click **Customize Options**.

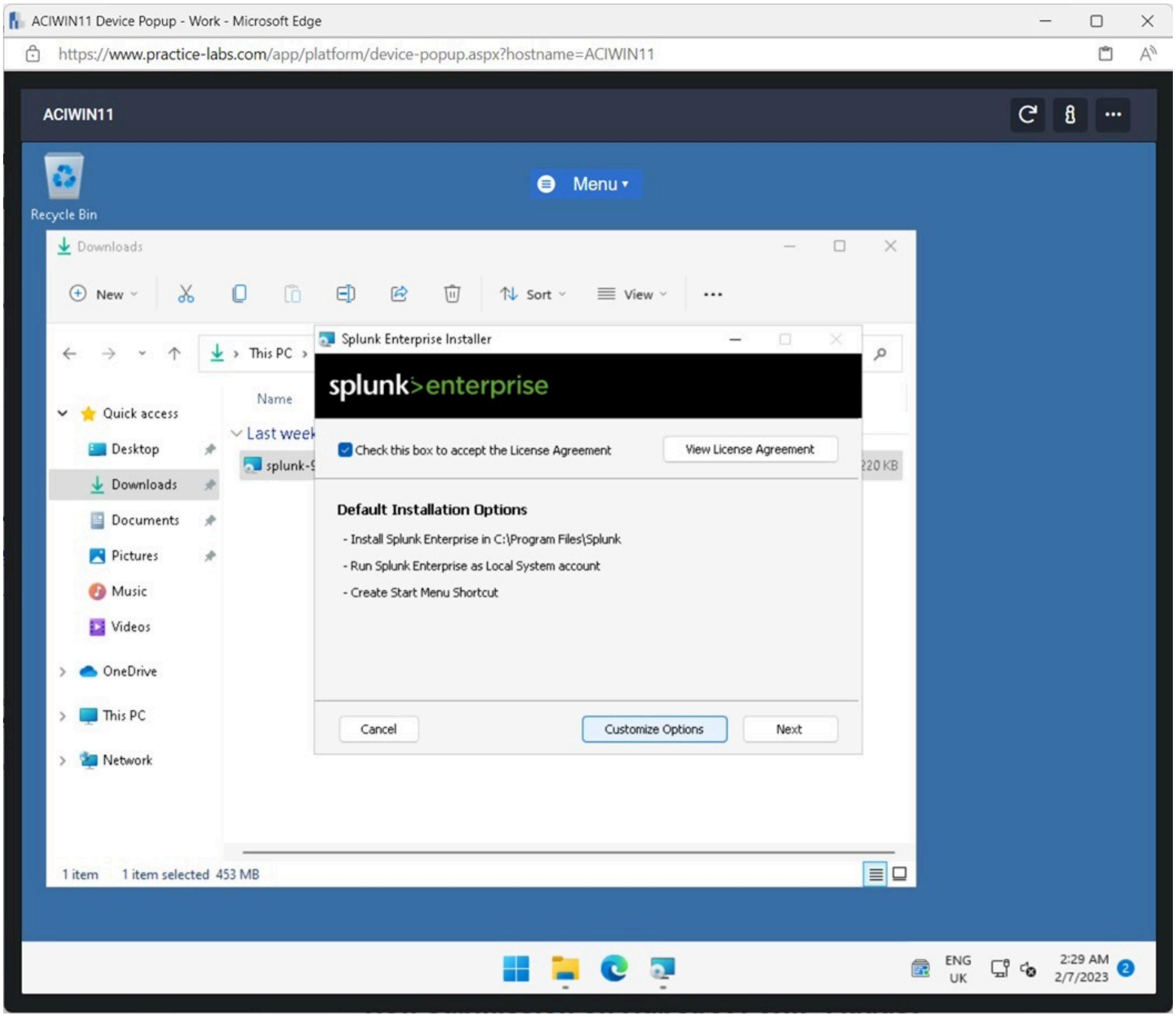


Figure 1.4 Screenshot of ACIWIN11: Displaying to checking the checkbox and selecting Customize Options.

Step 5

Click **Next** on the **Install Splunk Enterprise** screen.

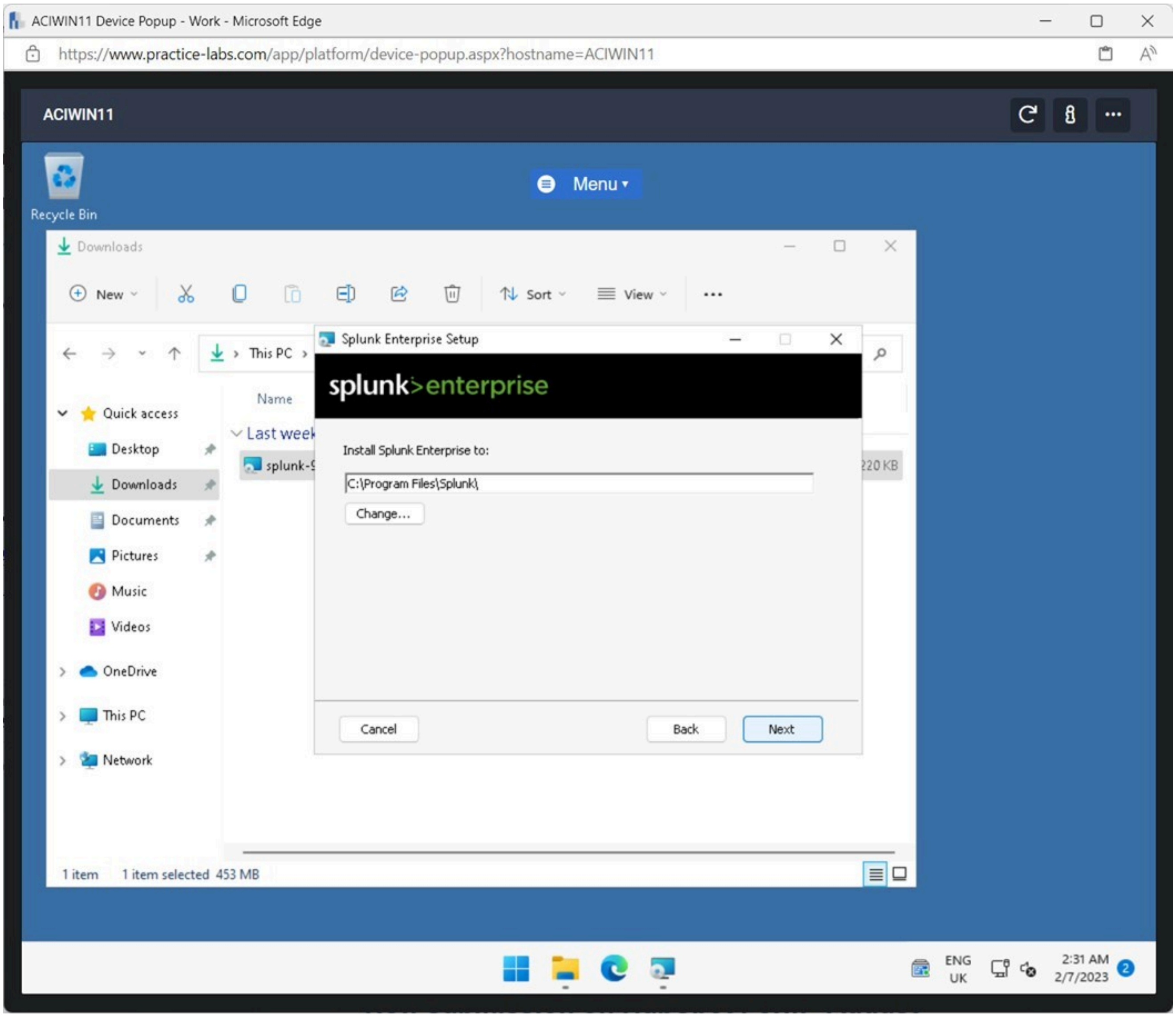


Figure 1.5 Screenshot of ACIWIN11: Displaying clicking Next on the Install Splunk Enterprise to screen.

Step 6

Select **Domain Account** on the **Install Splunk Enterprise as** screen.

Click **Next**.

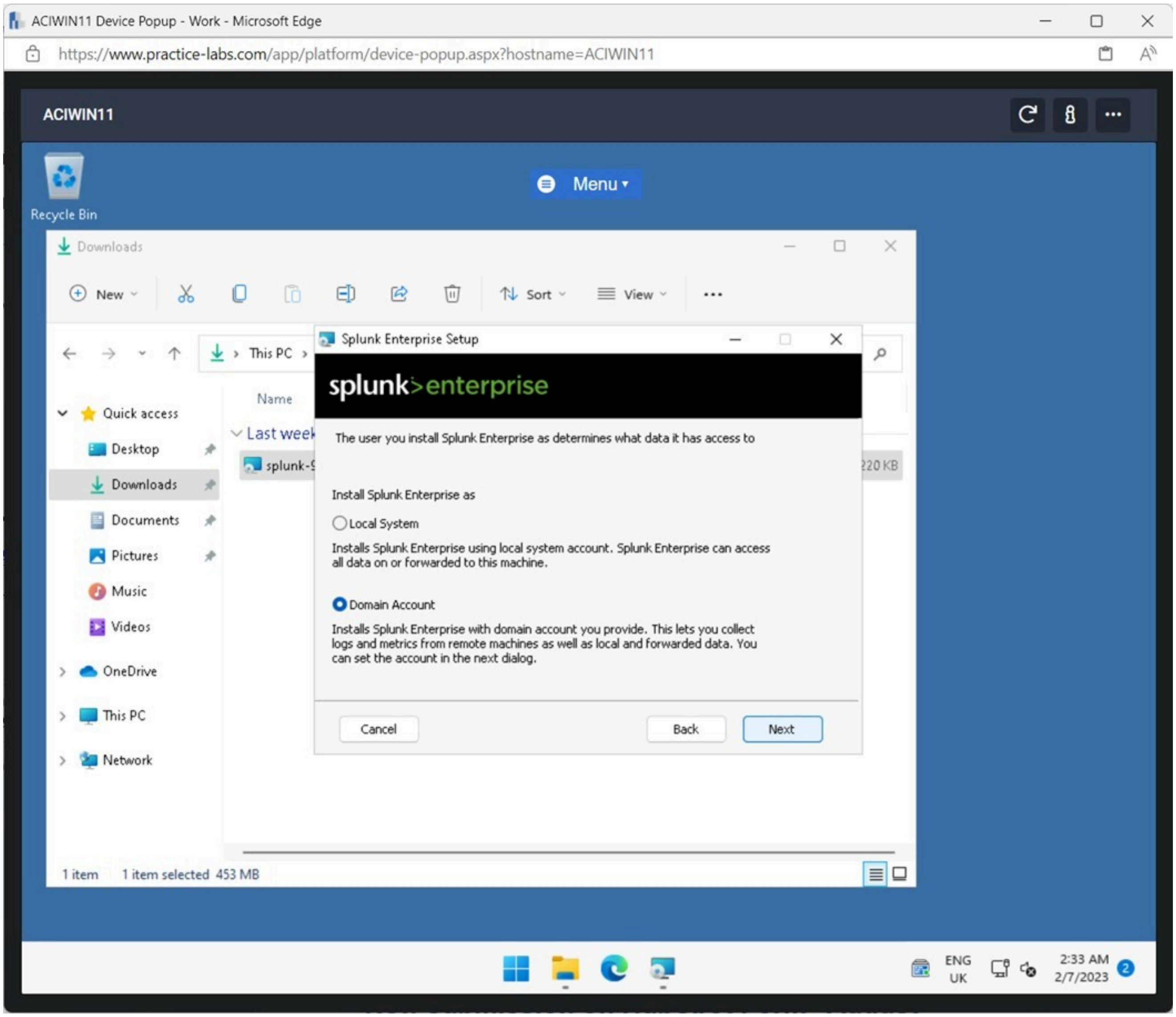


Figure 1.6 Screenshot of ACIWIN11: Displaying selecting the Domain Account radio button and clicking Next.

Step 7

Enter the following details on the **Splunk Enterprise Setup** screen:

```
Username:       aciplab\administrator
Password:       Passw0rd
Confirm password:   Passw0rd
```

Click **Next**.

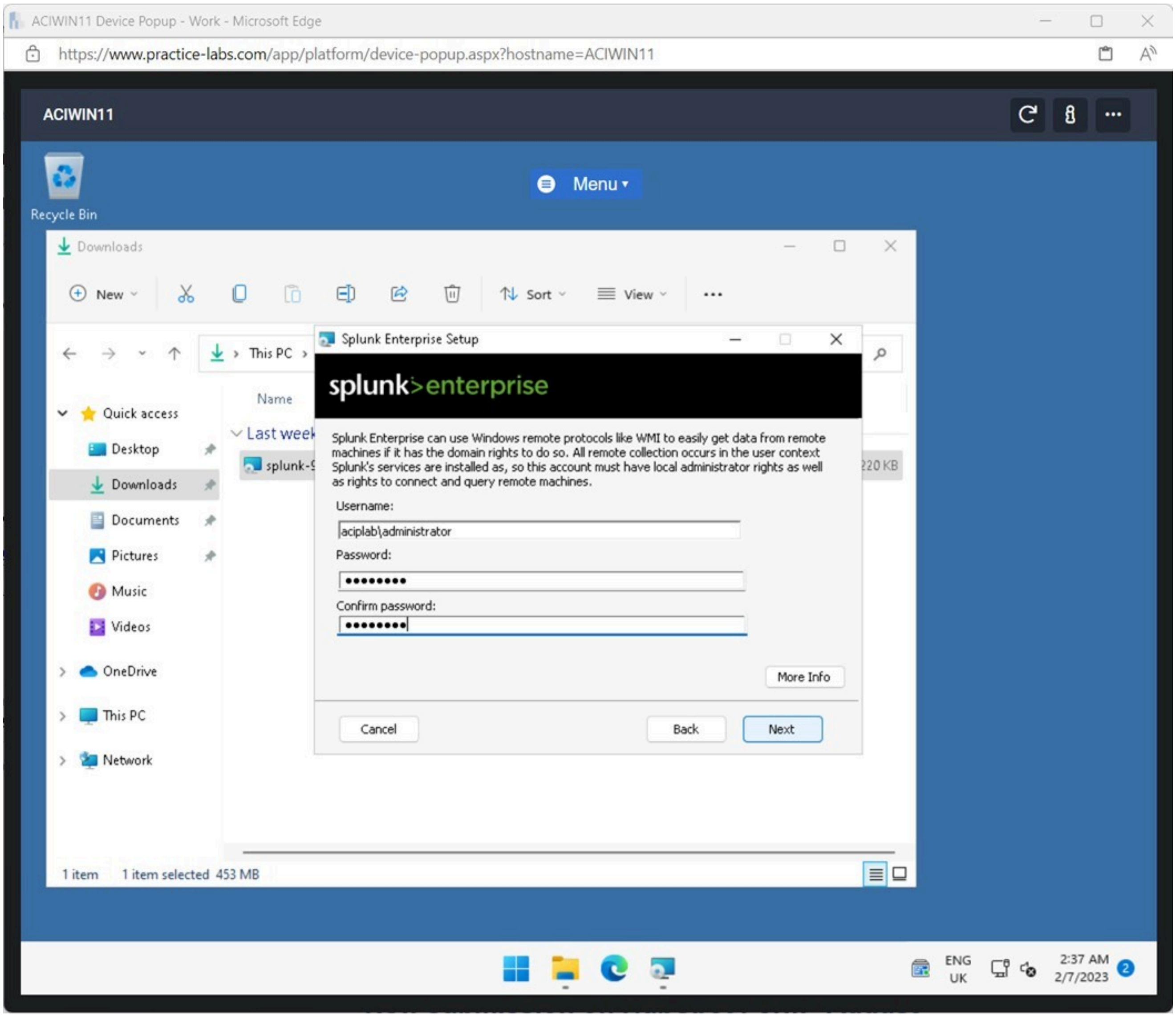


Figure 1.7 Screenshot of ACIWIN11: Displaying entering the domain credentials and password and clicking Next.

> ***Note:*** *Using the Domain Administrator account for installation of Splunk Enterprise will ensure the application will be able to collect logs from devices on the network*

Step 8

Enter the following details on the **Splunk Enterprise Setup** screen:

```
Username:      splunkadmin
Password:      Passw0rd
```

```
Confirm password:  Passw0rd
```

Click **Next.**

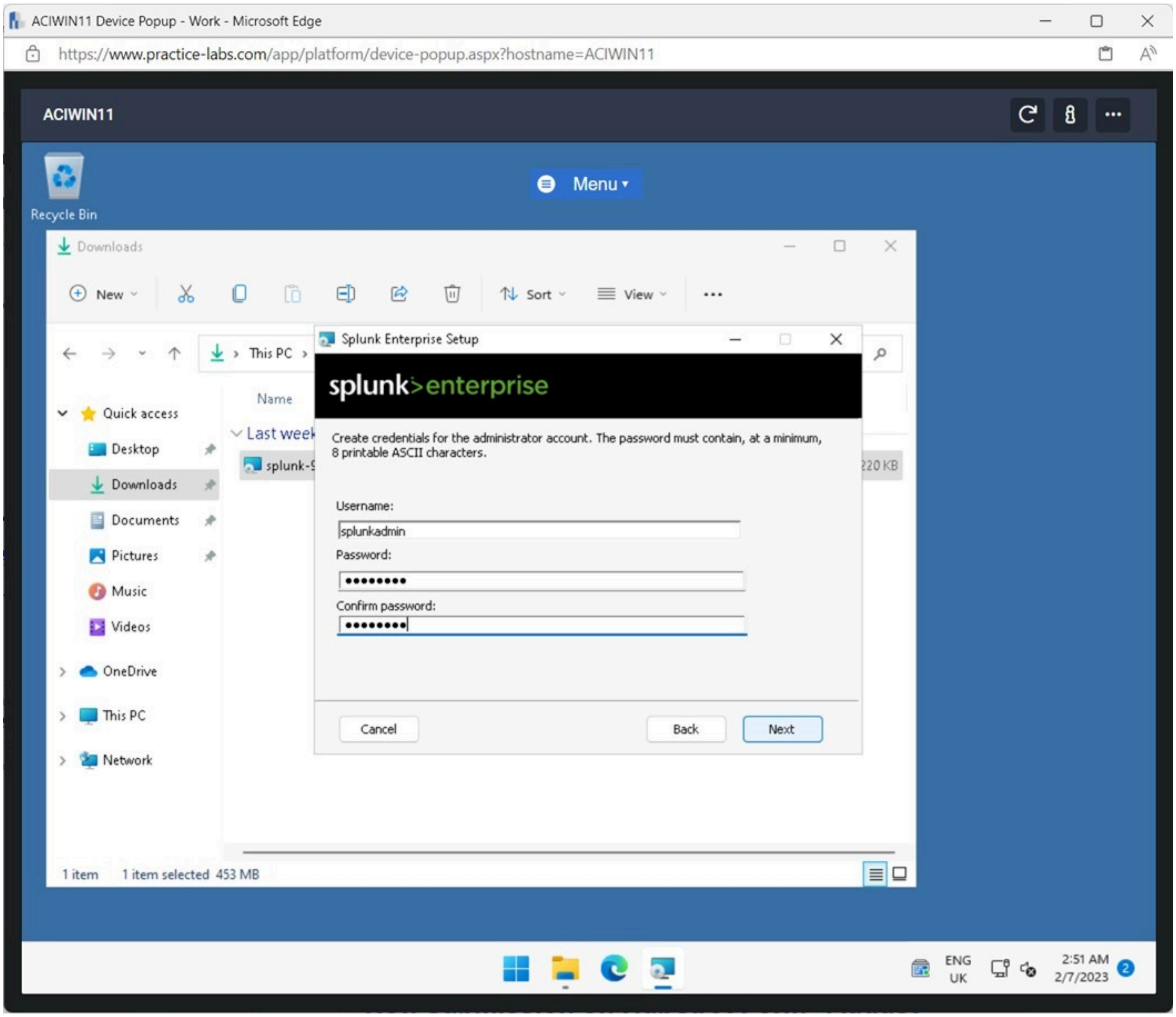


Figure 1.8 Screenshot of ACIWIN11: Displaying entering the credentials and password on the installation screen.

Note: *A local account with a password needs to be created that will be used to log into the Splunk Enterprise web application.*

Step 9

Click **Install** on the **Splunk Enterprise Setup** screen.

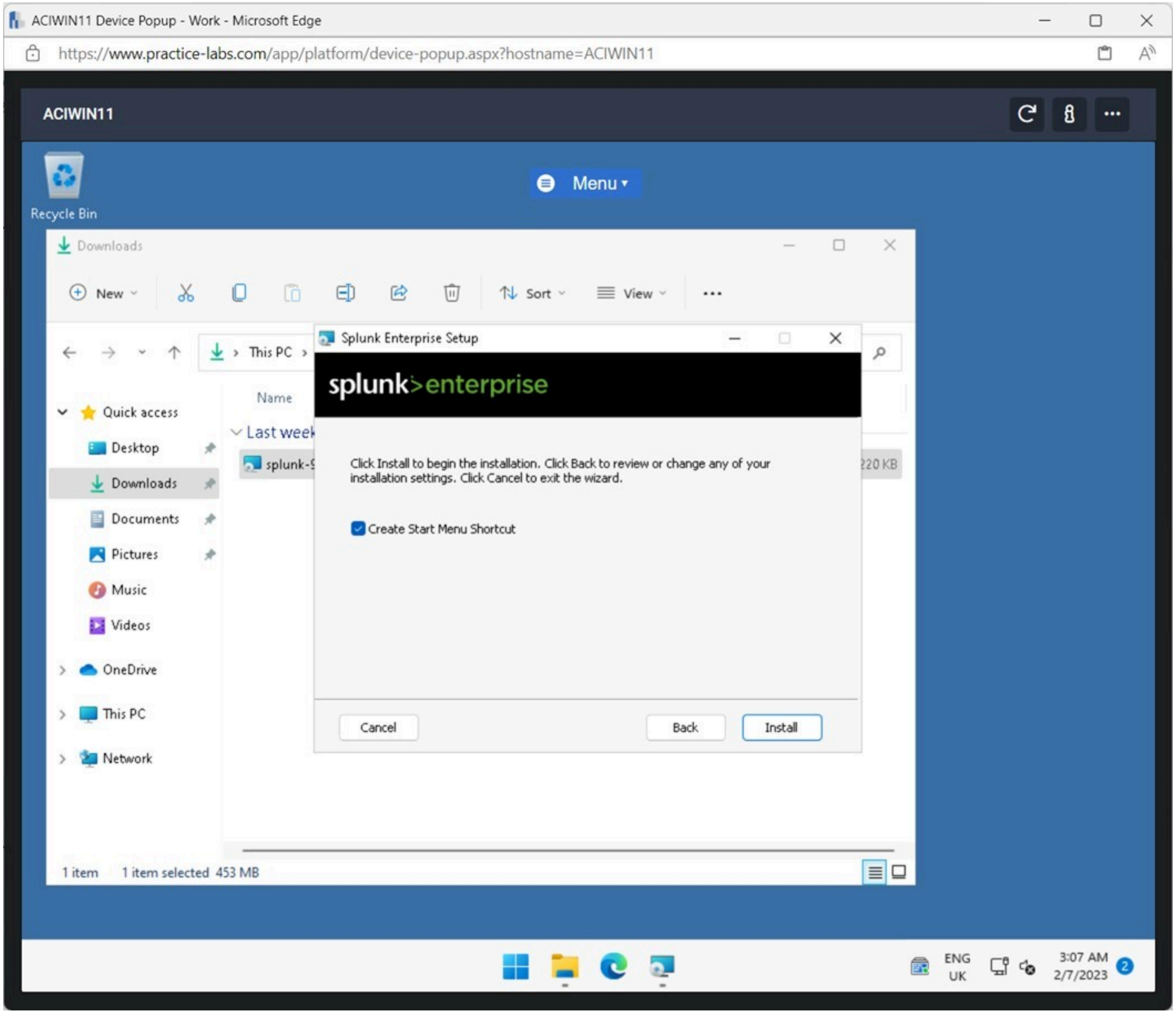


Figure 1.9 Screenshot of ACIWIN11: Displaying clicking Install on the Splunk Enterprise Setup screen.

Note: *The installation will take a couple of minutes to complete.*

Step 10

Click **Finish** on the **Splunk Enterprise Setup** screen.

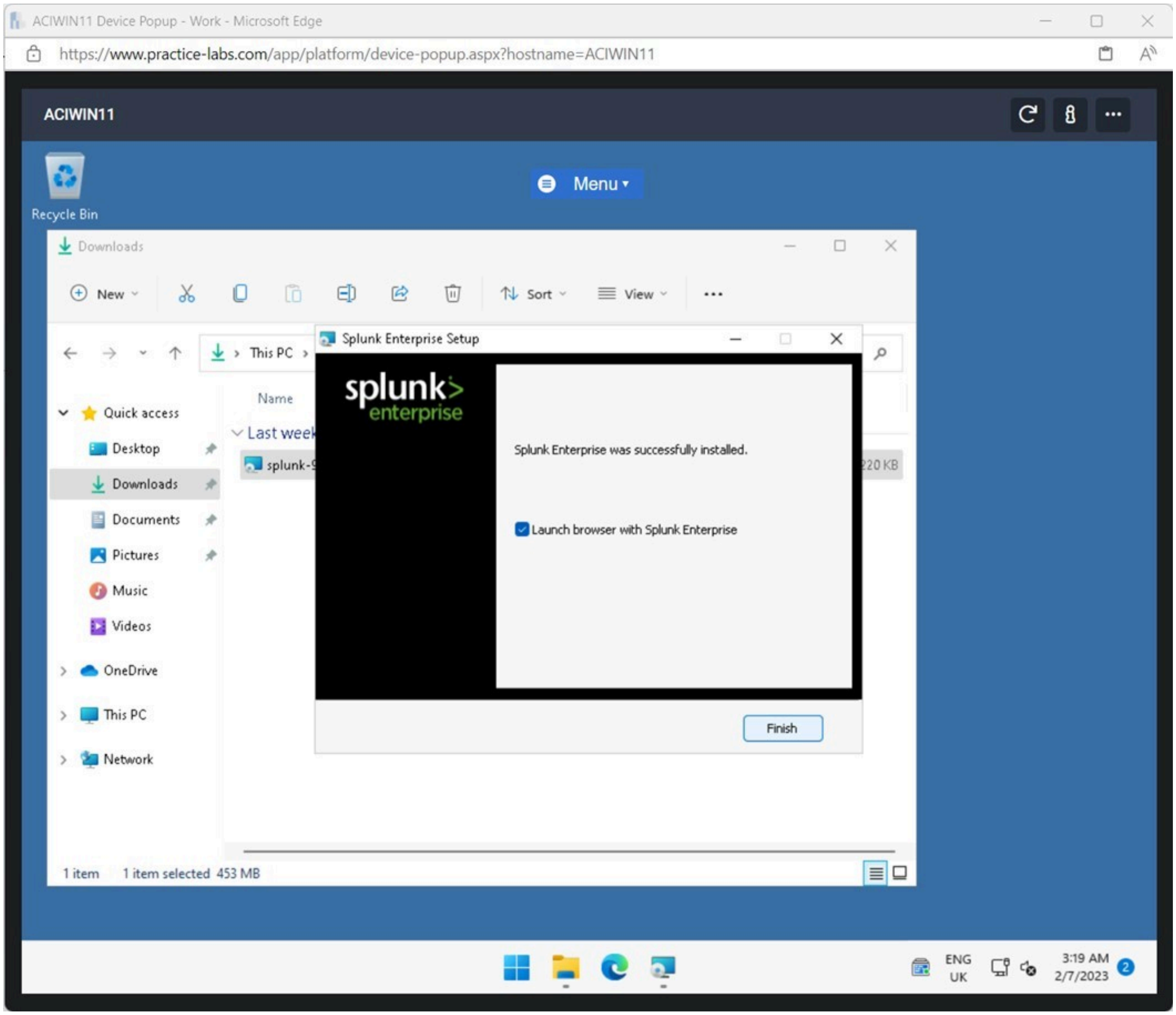


Figure 1.10 Screenshot of ACIWIN11: Displaying clicking Finish on the Splunk Enterprise Setup screen.

***Note:** The **Splunk Enterprise** web application will open automatically in the Microsoft Edge browser.*

Step 11

If the **Microsoft Edge** browser window does not prompt up, click on Microsoft Edge from the taskbar and enter the following credentials:

```
Username: splunkadmin
```

```
Password: Passw0rd
```

Click **Sign In**.

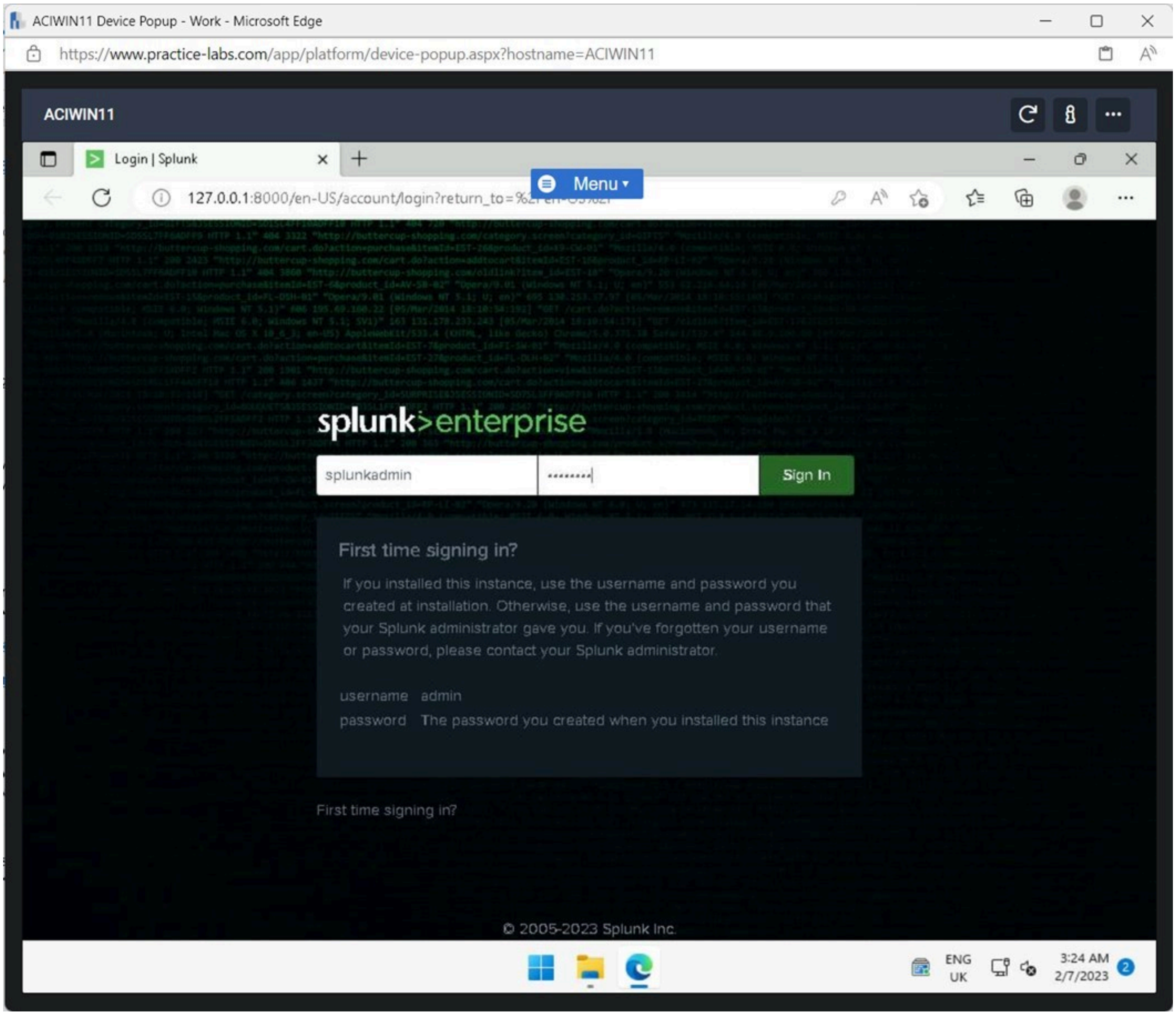


Figure 1.11 Screenshot of ACIWIN11: Displaying signing into the Splunk Enterprise web application.

Step 12

Click **Never** on the **Save password** pop-up window.

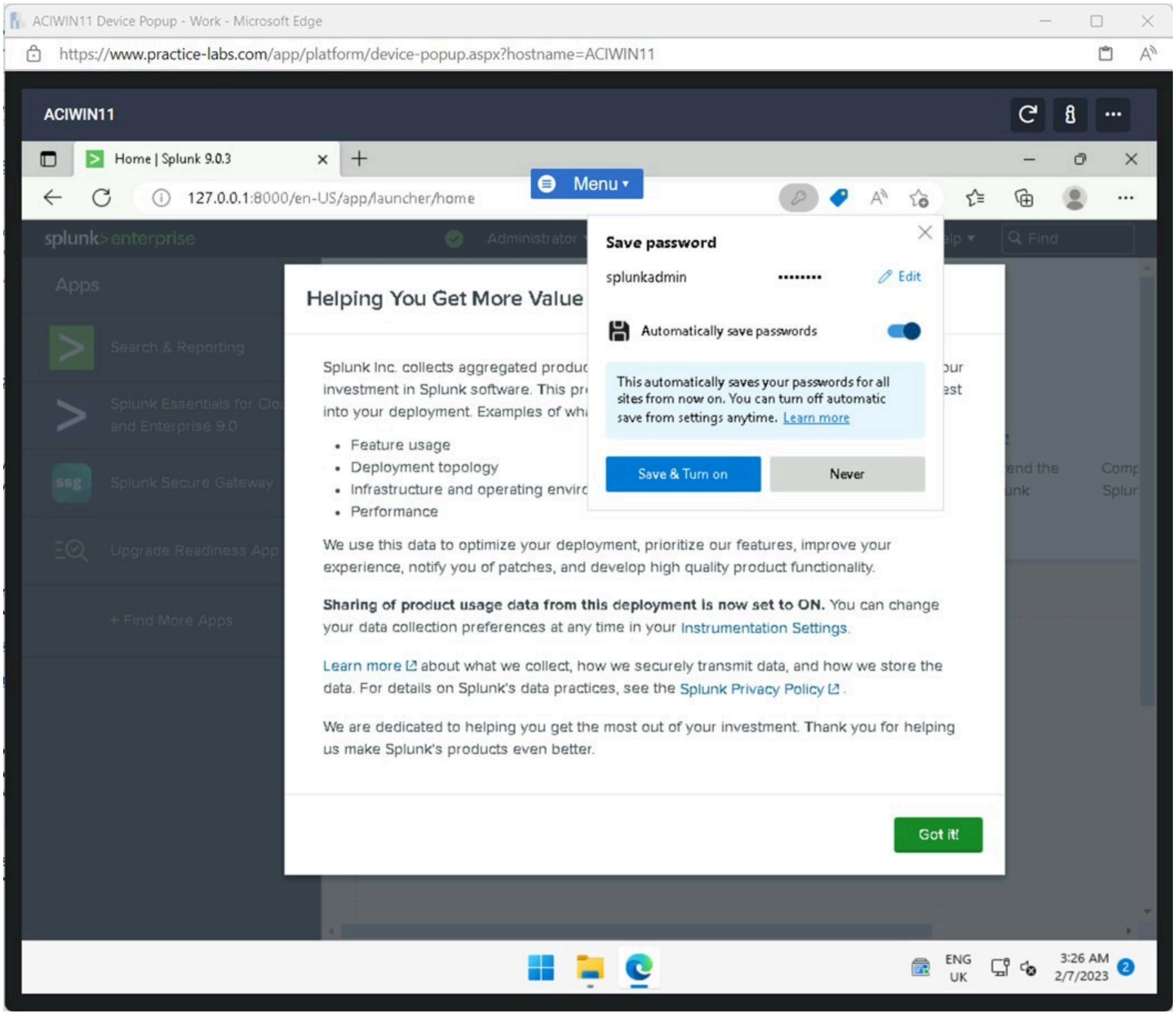


Figure 1.12 Screenshot of ACIWIN11: Displaying clicking Never on the pop-up window.

Step 13

Click **Got it** on the **Helping You Get More Value from Splunk Software** pop-up window.

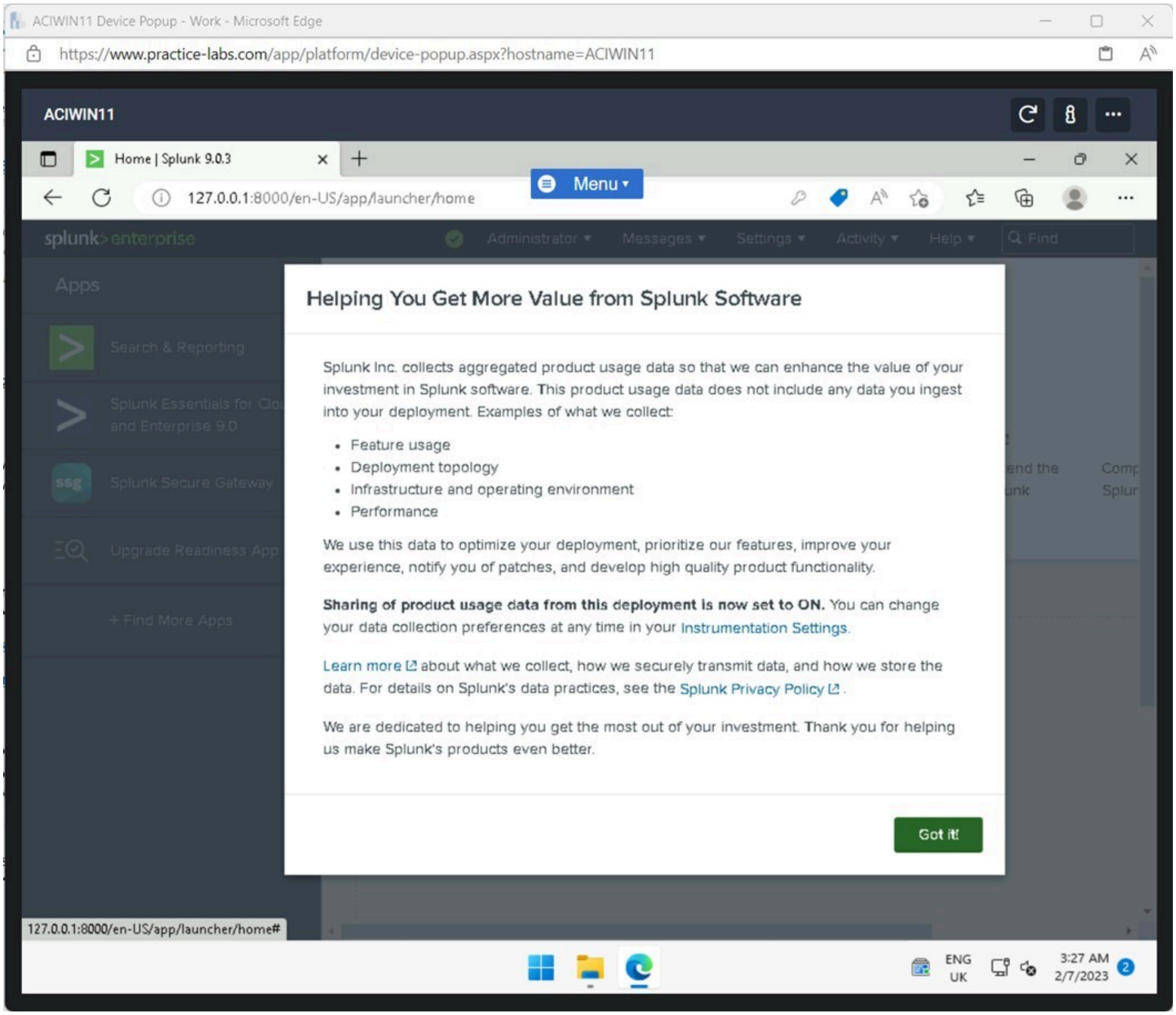


Figure 1.13 Screenshot of ACIWIN11: Displaying clicking Got it on the pop-up window.

Step 14

In the **Splunk Enterprise** web application, click **Settings**.

In the fly-out window, in the **DATA** pane, select **Forwarding and receiving**.

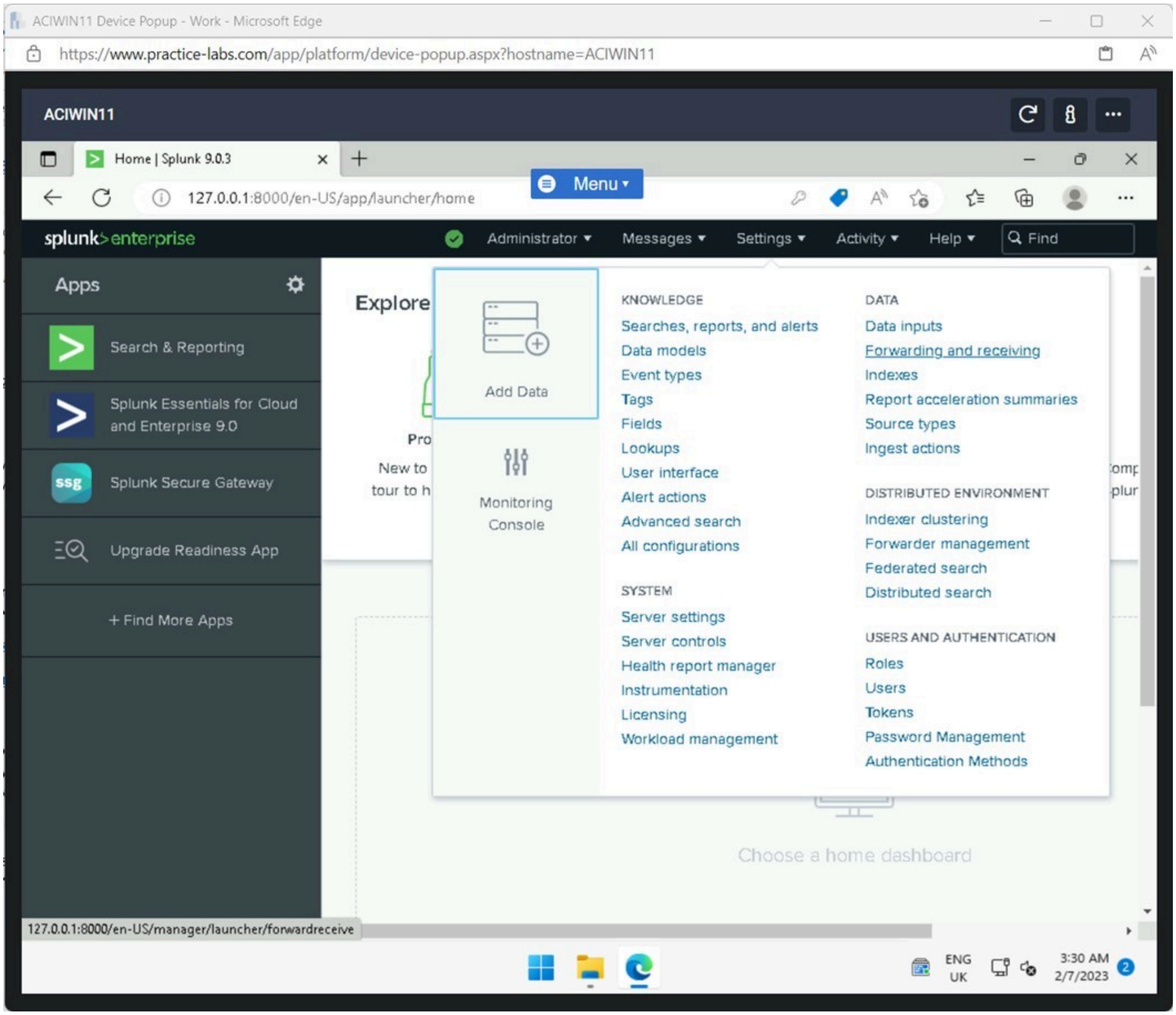


Figure 1.14 Screenshot of ACIWIN11: Displaying clicking Settings and selecting Forwarding and receiving in the Data pane.

Step 15

On the **Forwarding and receiving** window, click **+ADD new** in the **Receive data** pane.

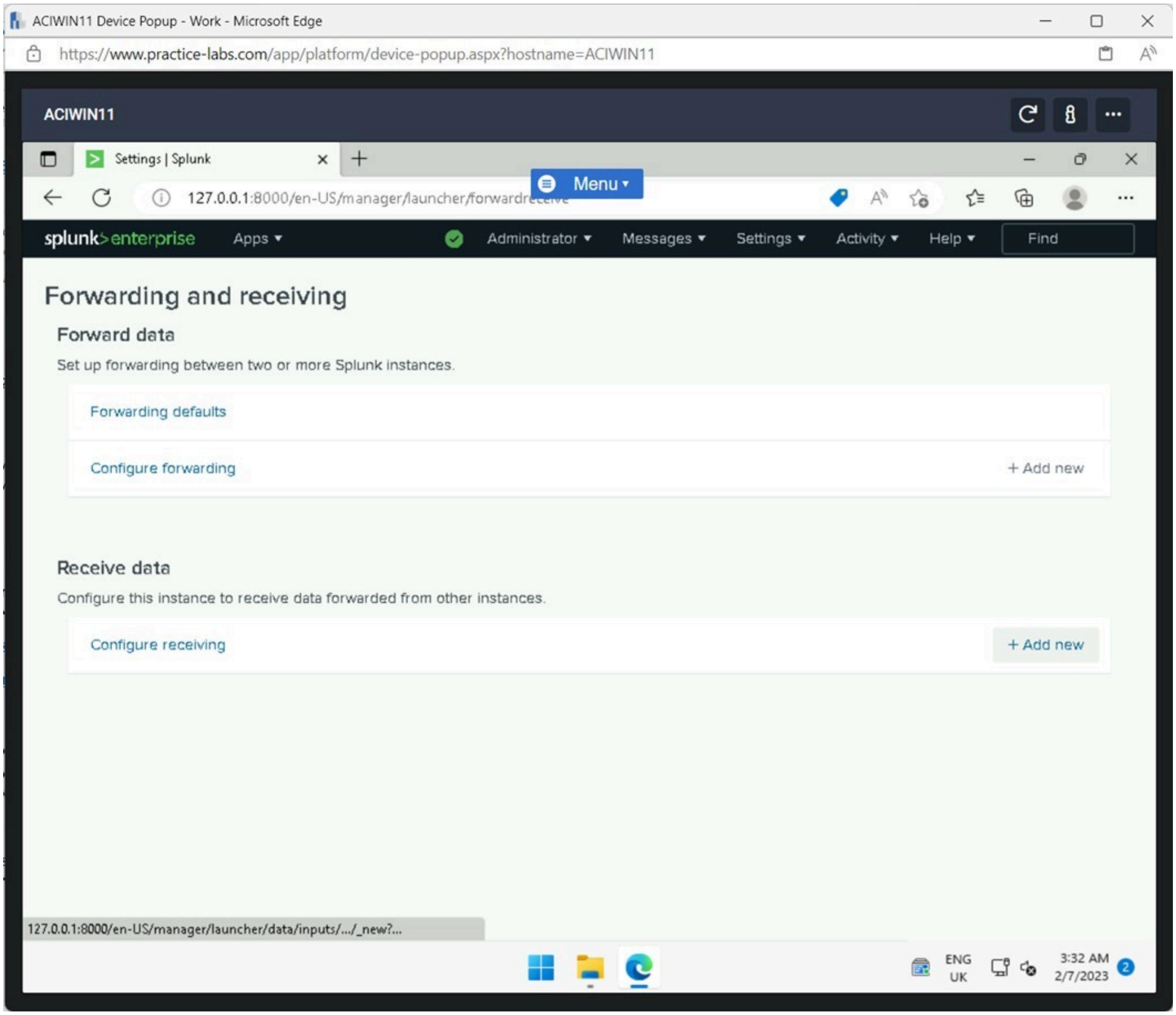


Figure 1.15 Screenshot of ACIWIN11: Displaying selecting Add new in the Receive data pane.

Step 16

On the **Add new** window, enter the following in the **Listen on this port** field:

```
9997
```

Click **Save**.

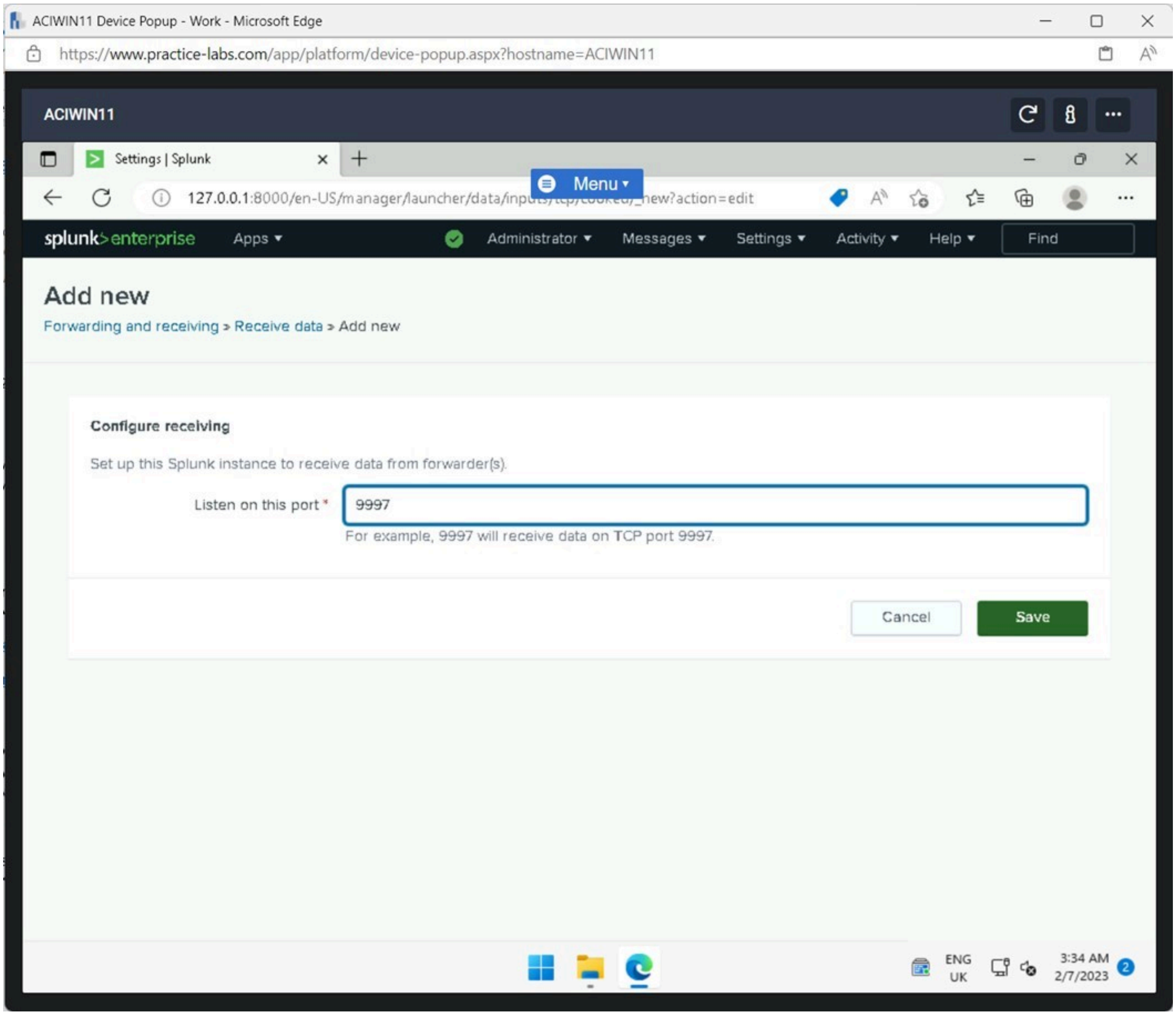


Figure 1.16 Screenshot of ACIWIN11: Displaying entering the listening port and clicking Save.

> ***Note:*** *The* ***Splunk Enterprise*** *application was configured to listen for network traffic. The Windows Firewall needs to be configured to ensure the traffic can be received on the port.*

Step 17

Close the **Microsoft Edge** browser.

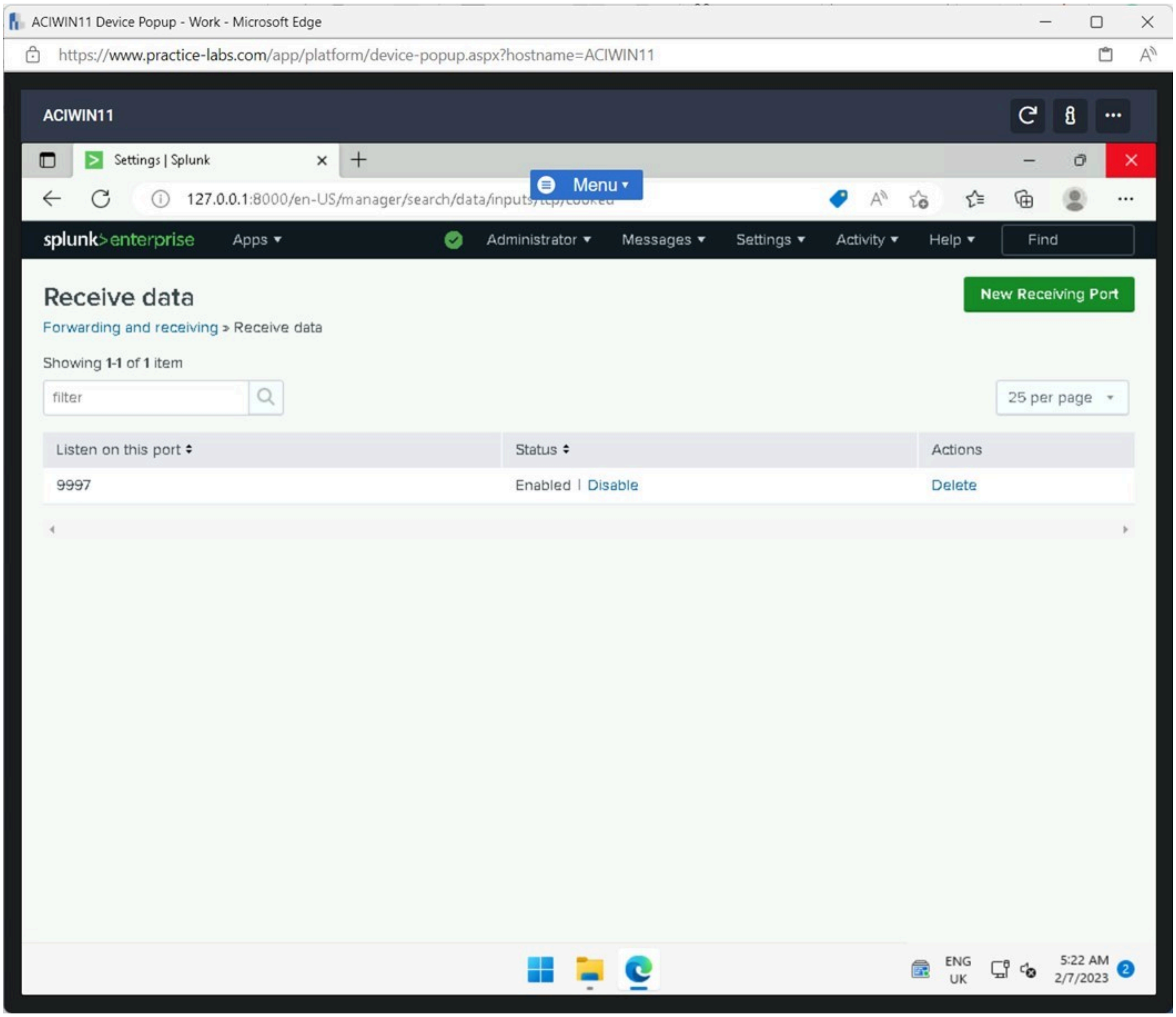


Figure 1.17 Screenshot of ACIWIN11: Displaying closing the Microsoft Edge browser.

Step 18

Click **Start**, type the following, and press **Enter**:

```
Windows Firewall
```

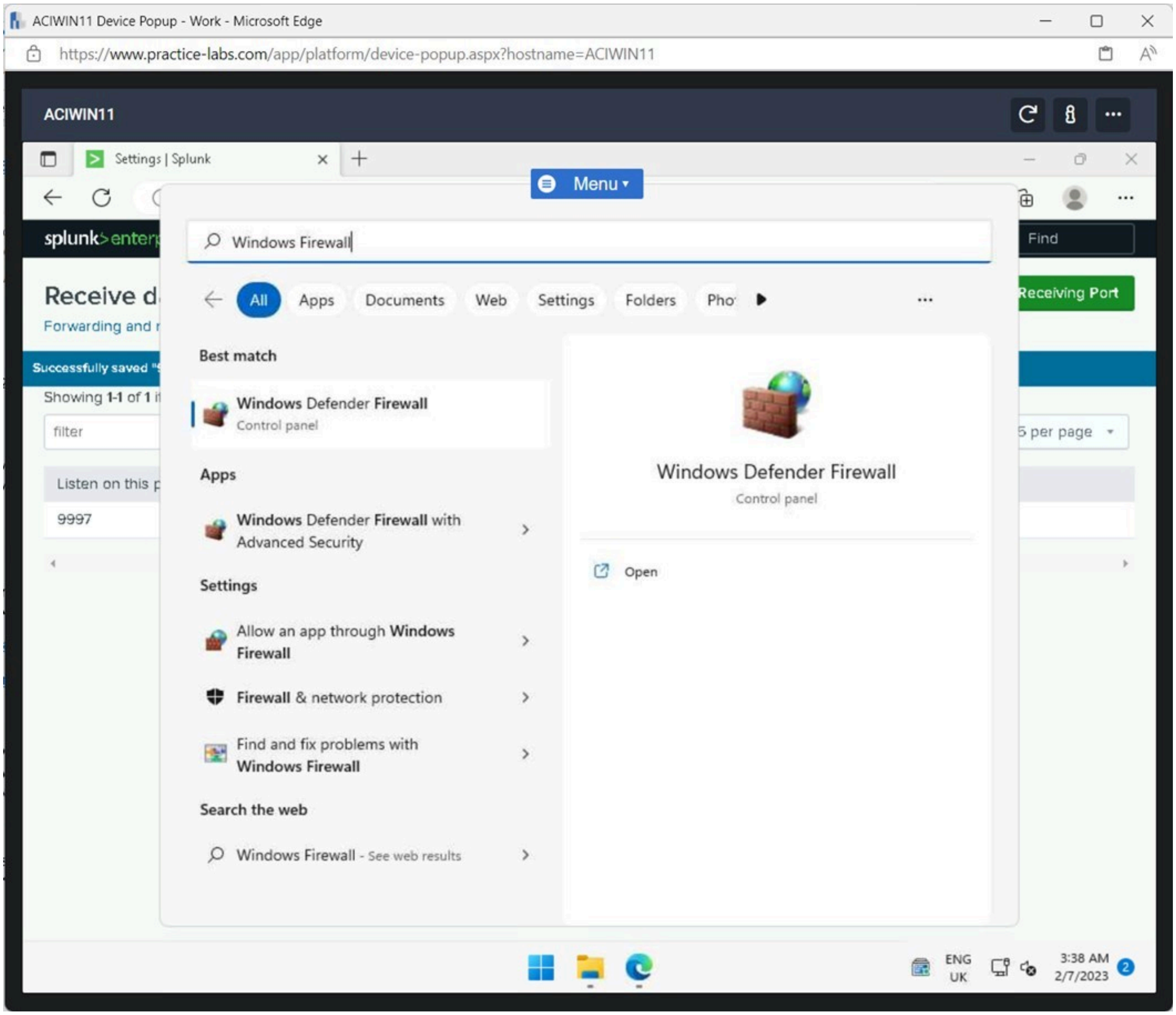


Figure 1.18 Screenshot of ACIWIN11: Displaying clicking Start, typing Windows Firewall and pressing enter.

Step 19

On the **Windows Defender Firewall** window, click **Advanced settings.**

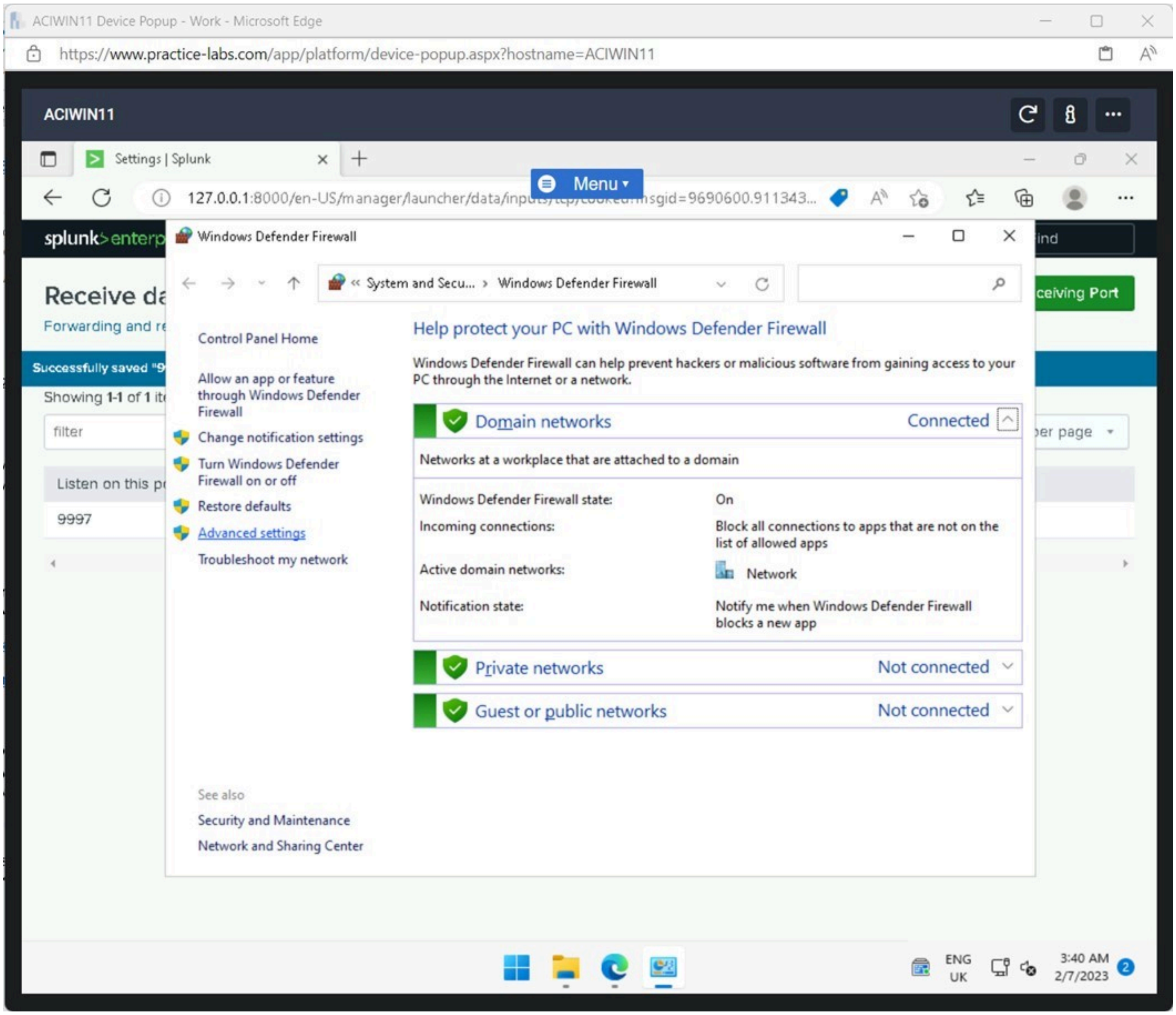


Figure 1.19 Screenshot of ACIWIN11: Displaying clicking Advanced settings in the Windows Defender Firewall window.

Step 20

On **Windows Defender Firewall with Advanced Security**, select **Inbound Rules** and click **New Rule** in the right pane.

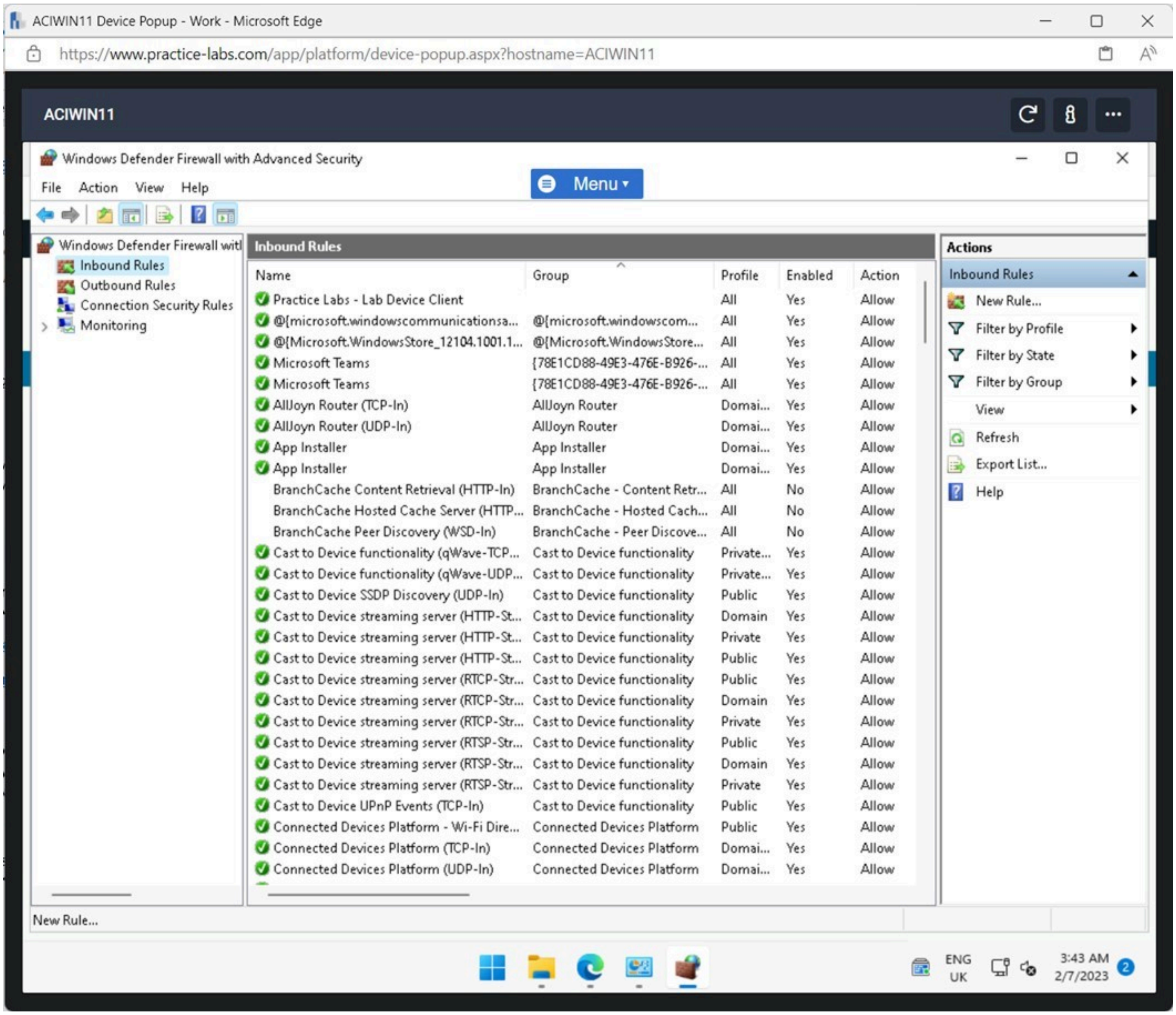


Figure 1.20 Screenshot of ACIWIN11: Displaying creating a new inbound rule.

Step 21

Select **Port** on the **Rule Type** window.

Click **Next**.

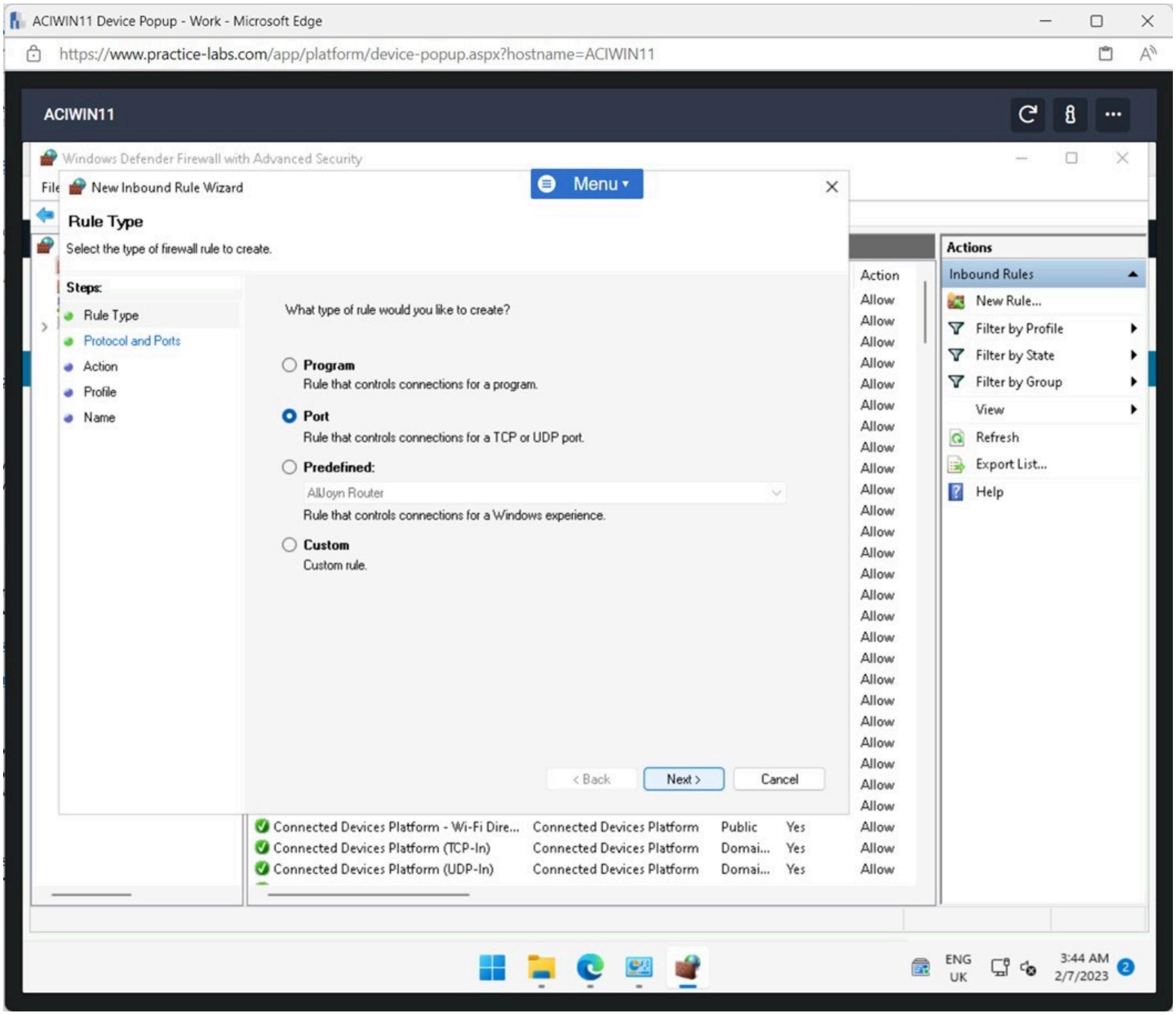


Figure 1.21 Screenshot of ACIWIN11: Displaying selecting Port and clicking Next.

Step 22

Enter the following in the **Specific local ports** field:

```
8089, 9997
```

Click **Next**.

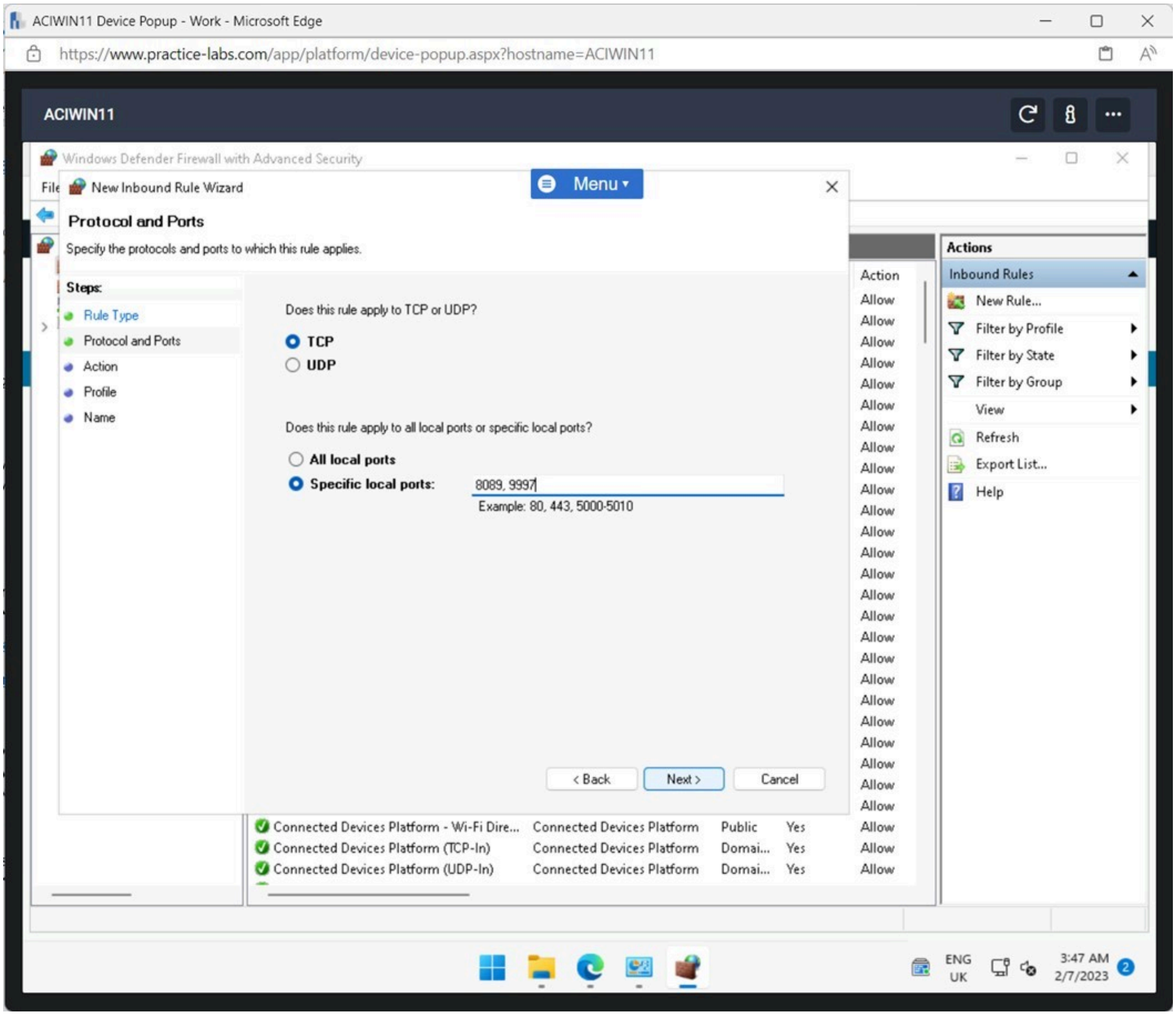


Figure 1.22 Screenshot of ACIWIN11: Displaying entering the port numbers and clicking Next.

Step 23

Click **Next** on the **Action** and **Profile** windows.

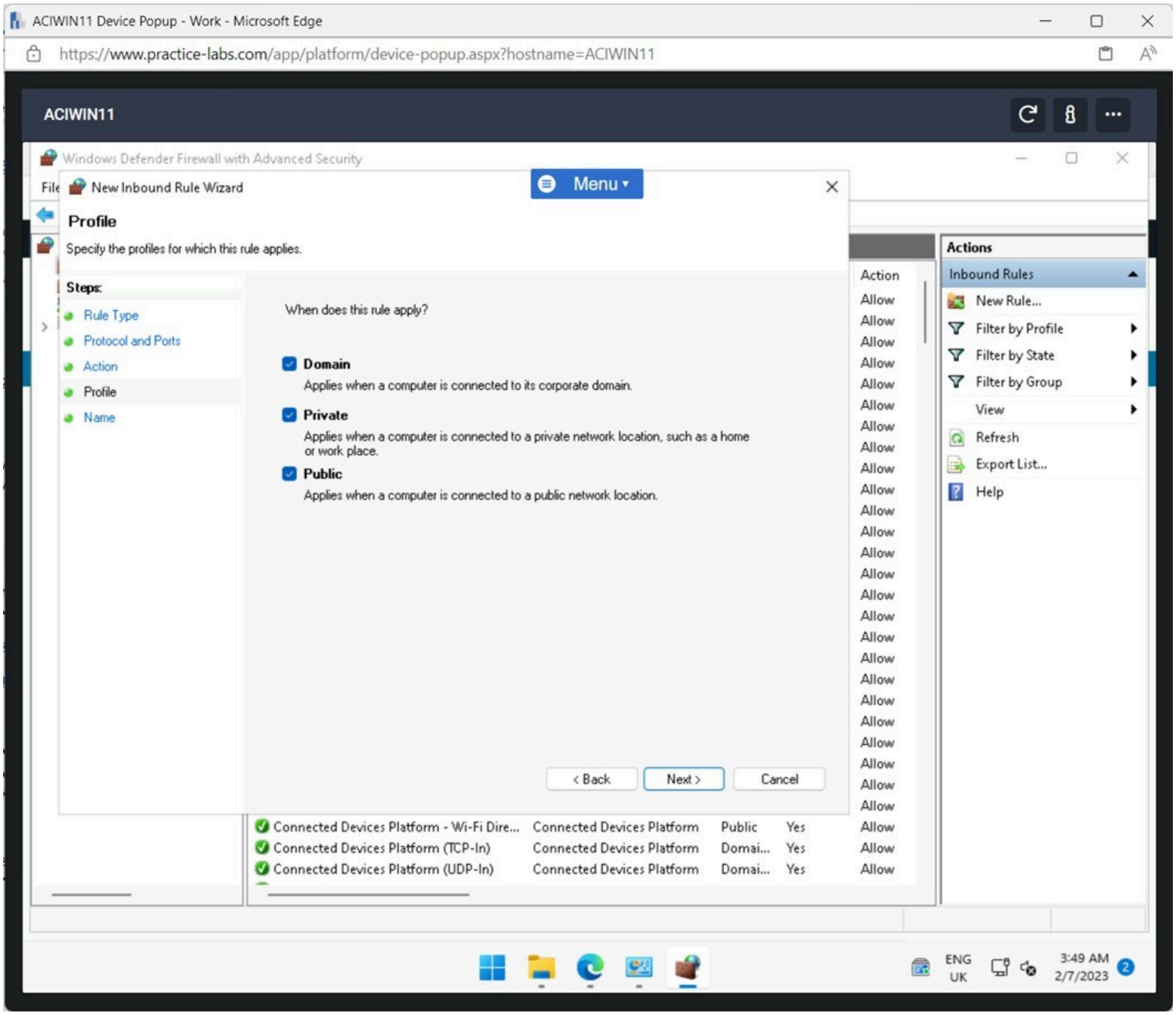


Figure 1.23 Screenshot of ACIWIN11: Displaying clicking Next on the Action and Profile windows.

Step 24

Enter the following in the **Name** field and click **Finish**.

```
Splunk
```

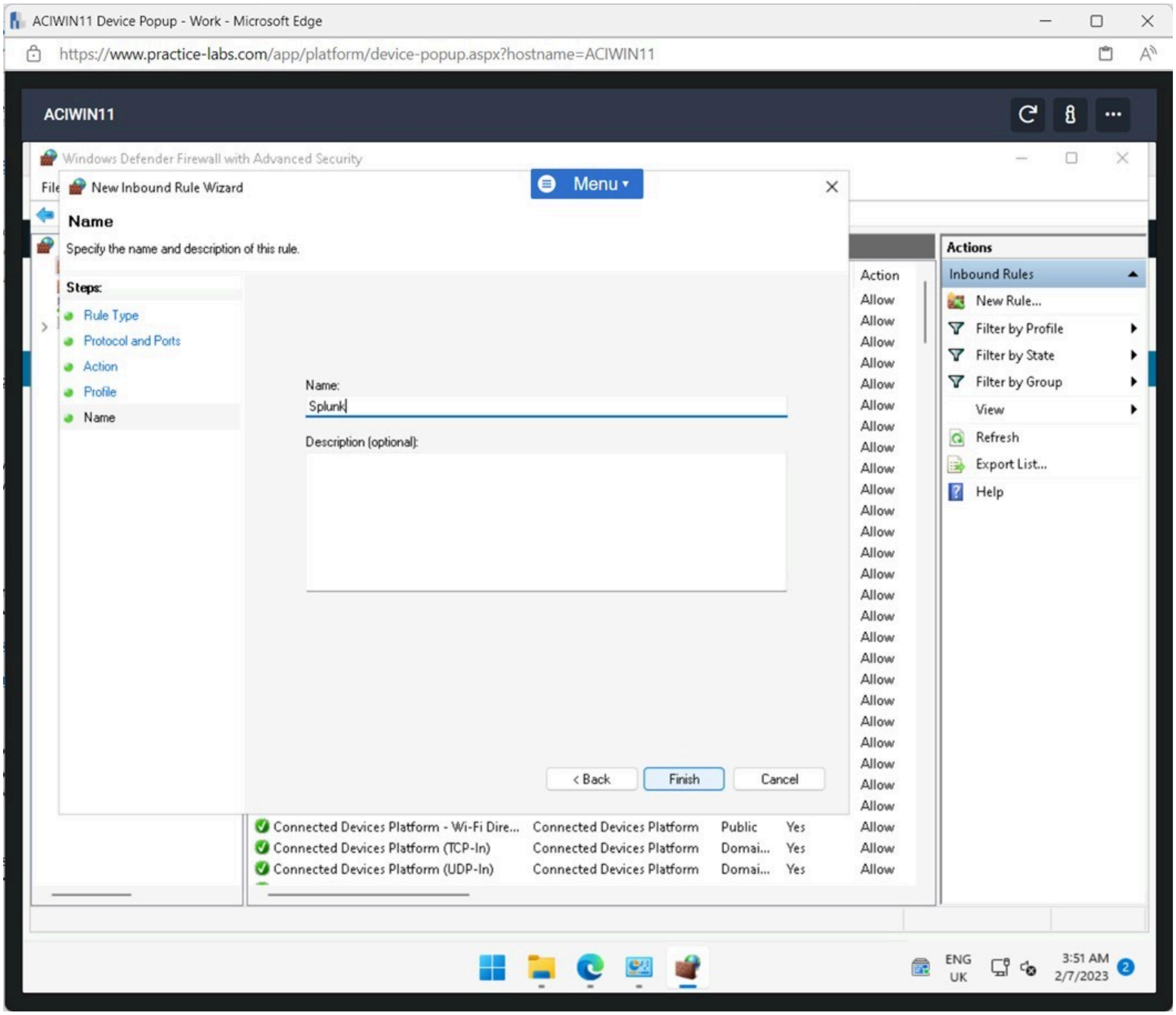


Figure 1.24 Screenshot of ACIWIN11: Displaying entering Splunk in the name field and clicking Finish.

Task 2 - Configuring a Splunk Client

To ensure the devices on the network can send log activity, the device needs to be configured to facilitate this.

In this task, a client device will be configured to forward its logs to the Splunk Enterprise application.

Step 1

Connect to **ACIDM01.**

Open **File Explorer** from the **Taskbar.**

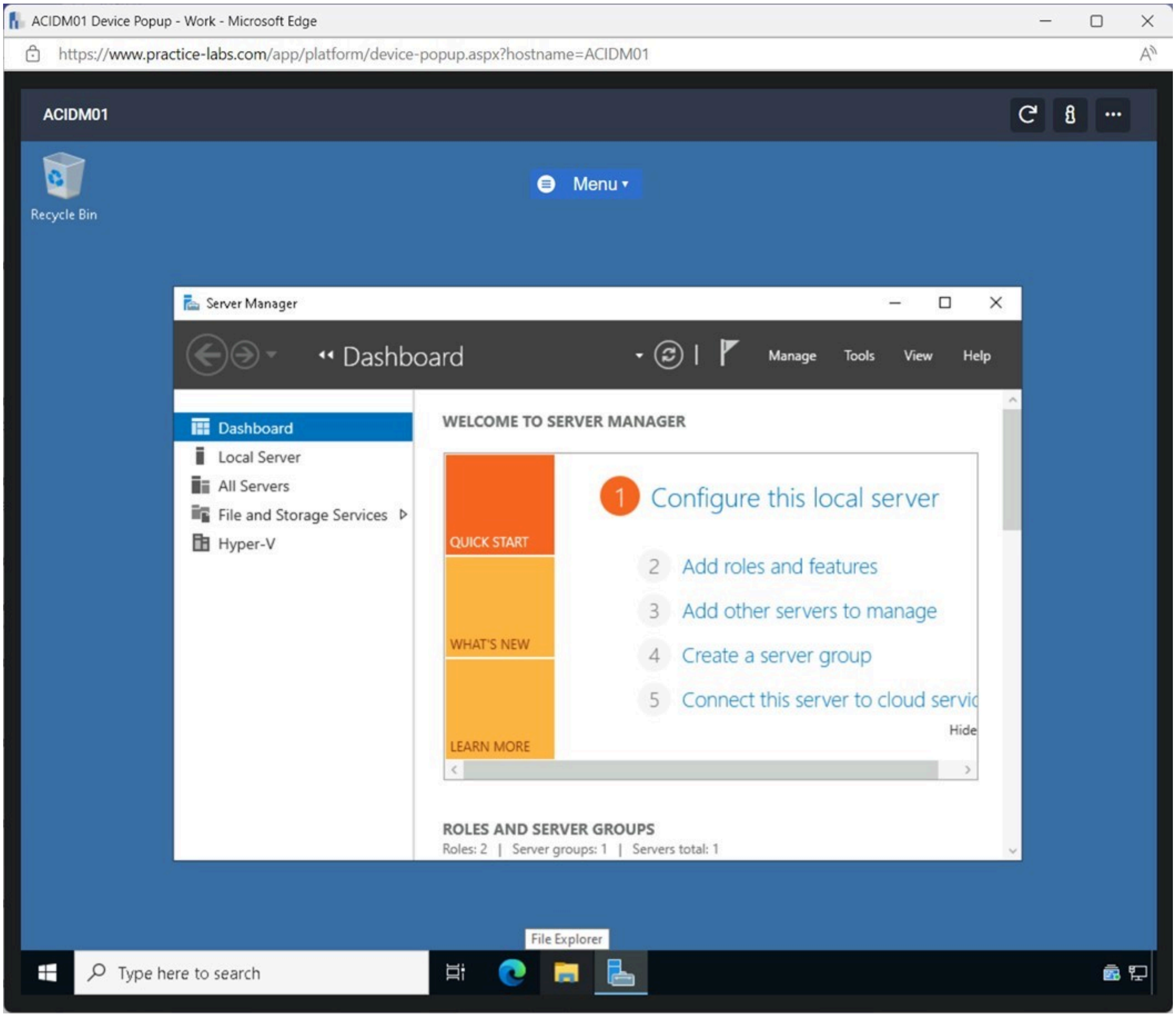


Figure 1.25 Screenshot of ACIDM01: Displaying opening File Explorer from the Taskbar.

Step 2

In **File Explorer**, select the **Downloads** folder and double click the **splunk-forwarder-9.0.3** installation file.

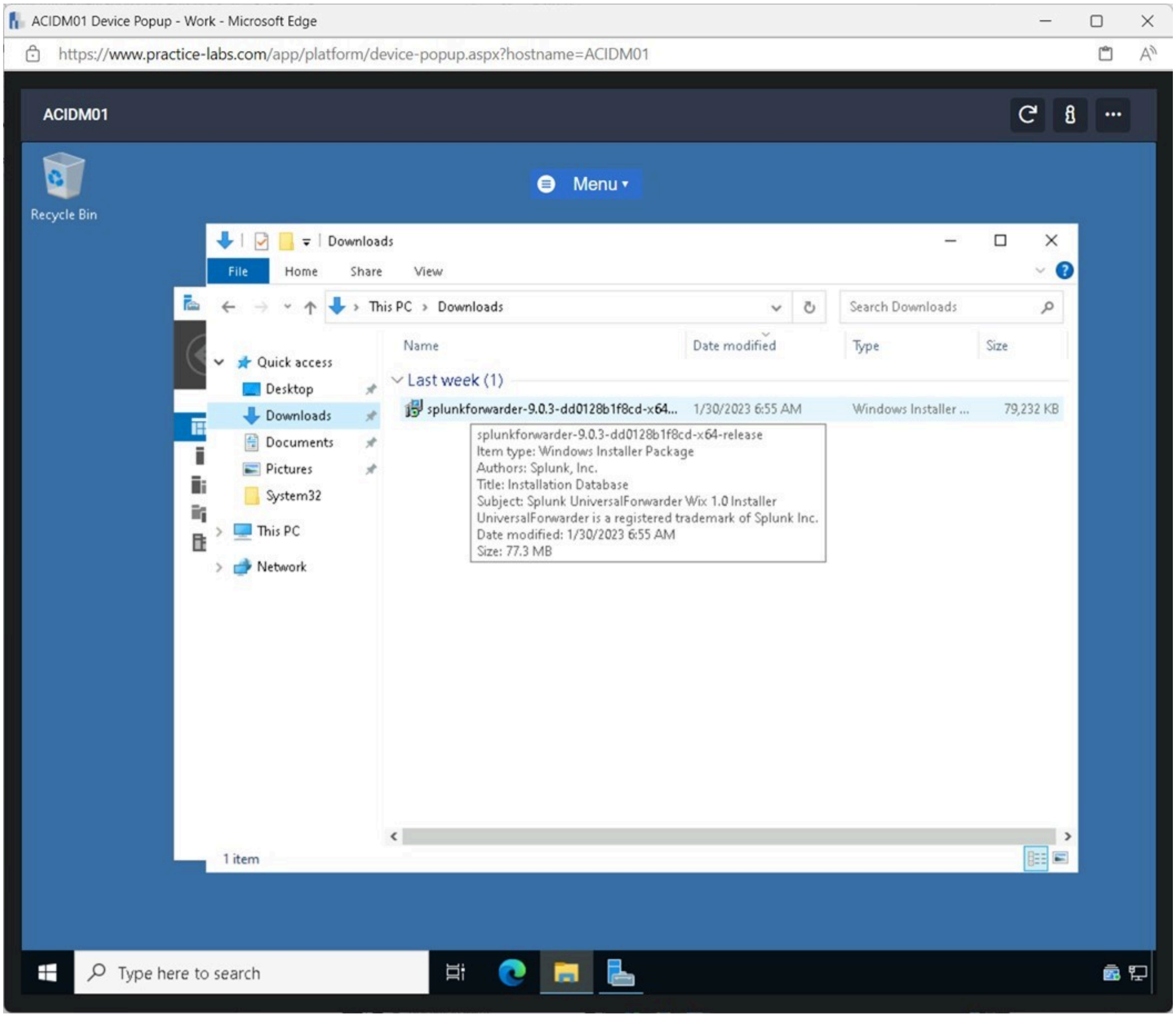


Figure 1.26 Screenshot of ACIDM01: Displaying double-clicking the splunkforwarder-9.0.3 installation file.

Step 3

Tick the **Check this box to accept the License Agreement** tick box and click **Customize Options**.

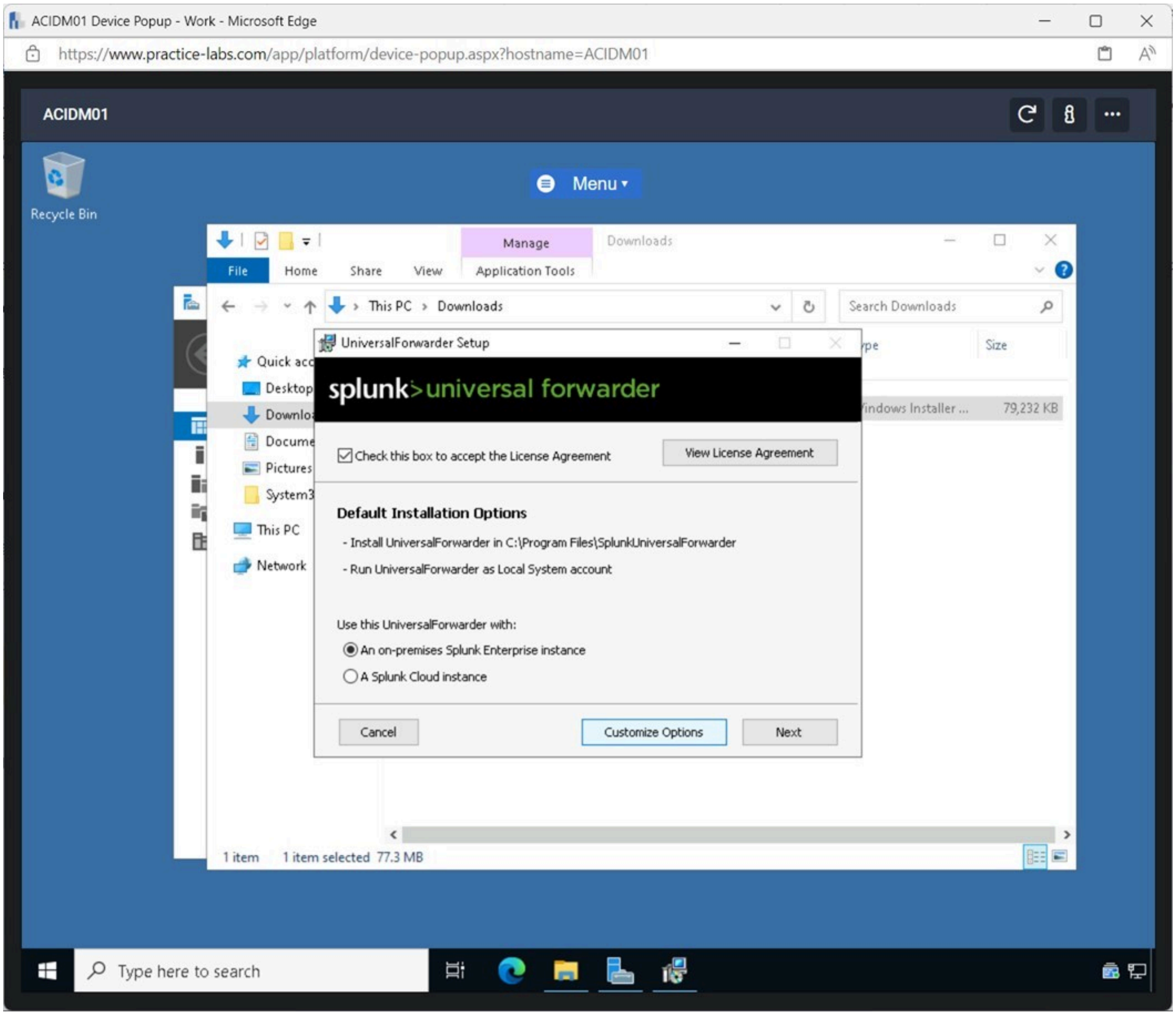


Figure 1.27 Screenshot of ACIDM01: Displaying ticking the tick box and clicking Customize Options.

Step 4

Click **Next** on the **UniversalForwarder Setup** window.

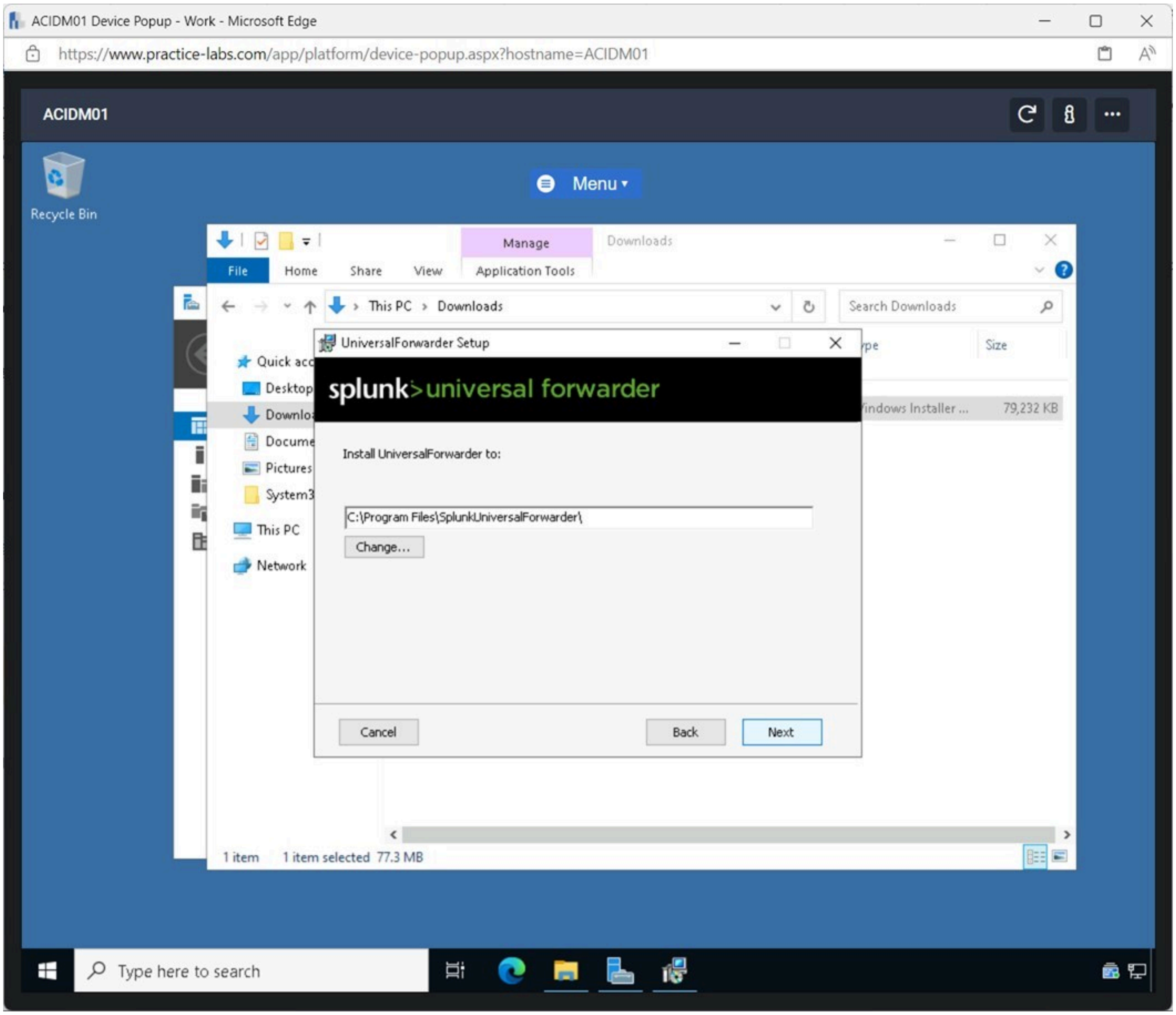


Figure 1.28 Screenshot of ACIDM01: Displaying clicking Next on the UniversalForwarder Setup screen.

Step 5

Click **Next** on the next installation screen.

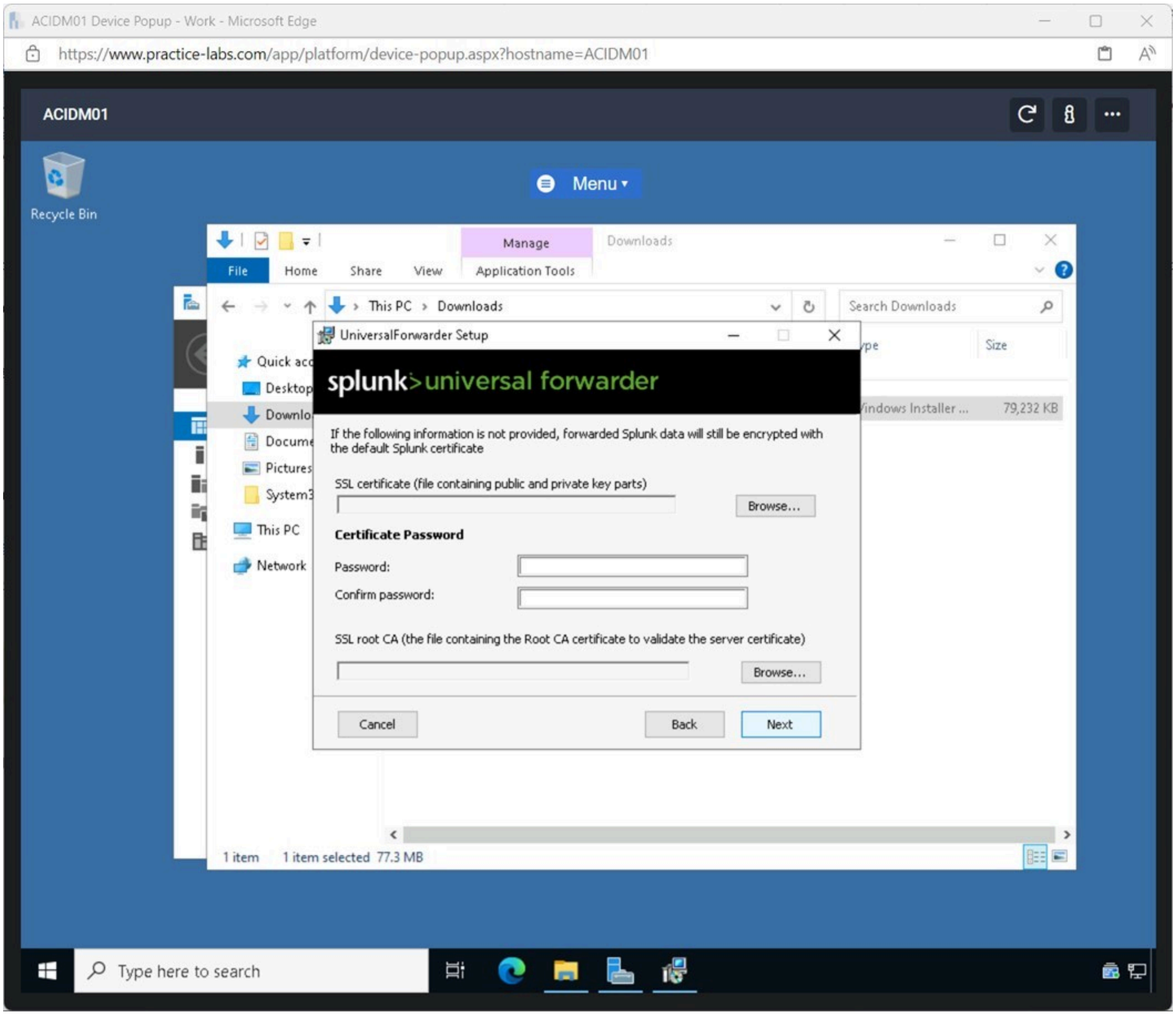


Figure 1.29 Screenshot of ACIWIN11: Displaying clicking Next on the installation screen.

Step 6

Select the **Domain Account** radio button and click **Next**.

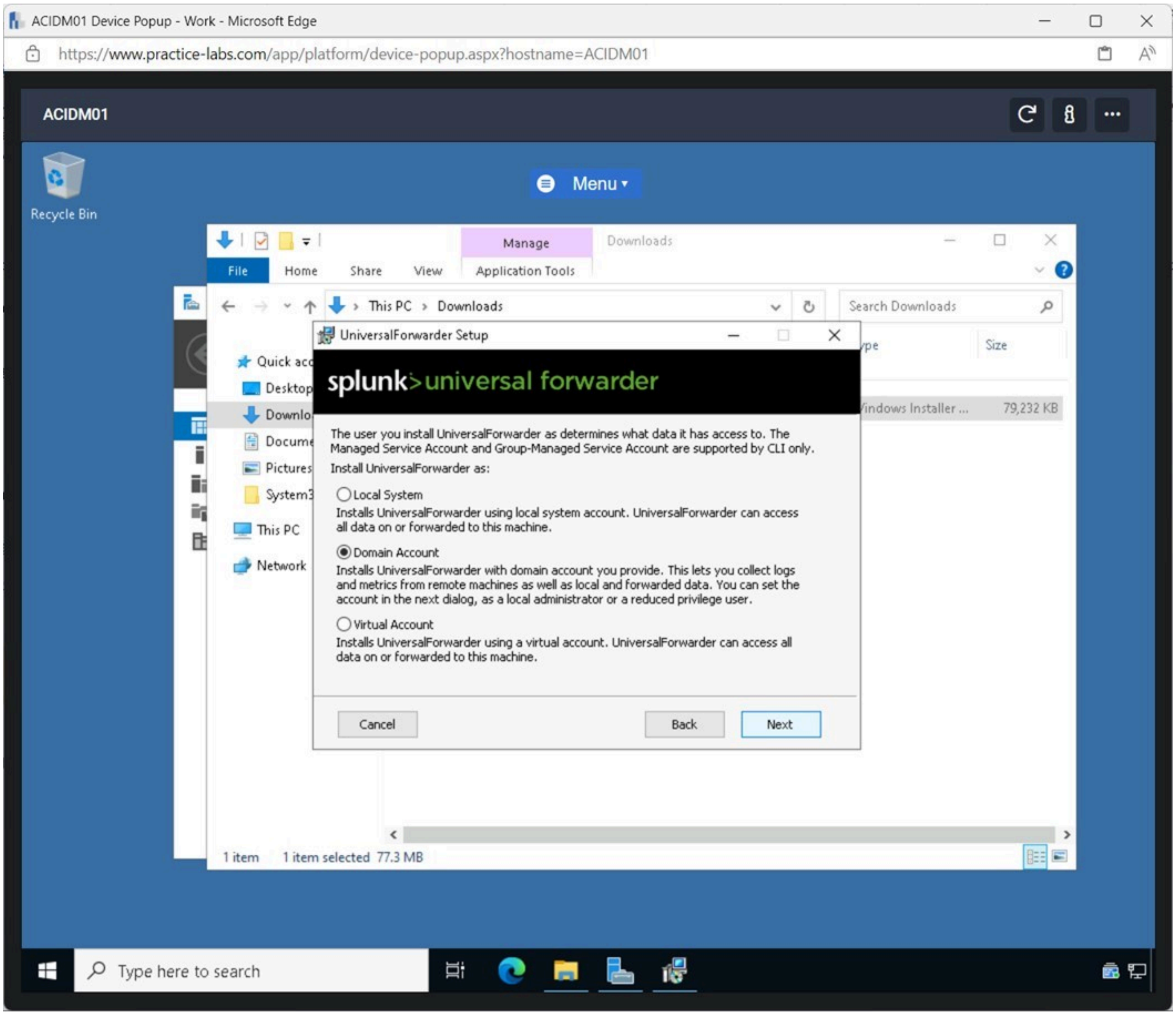


Figure 1.30 Screenshot of ACIDM01: Displaying selecting the Domain Account radio button and clicking Next.

Step 7

Enter the following details on the **UniversalForwader Setup** screen:

```
Username:       aciplab\administrator
Password:       Passw0rd
Confirm password:    Passw0rd
```

Click **Next**.

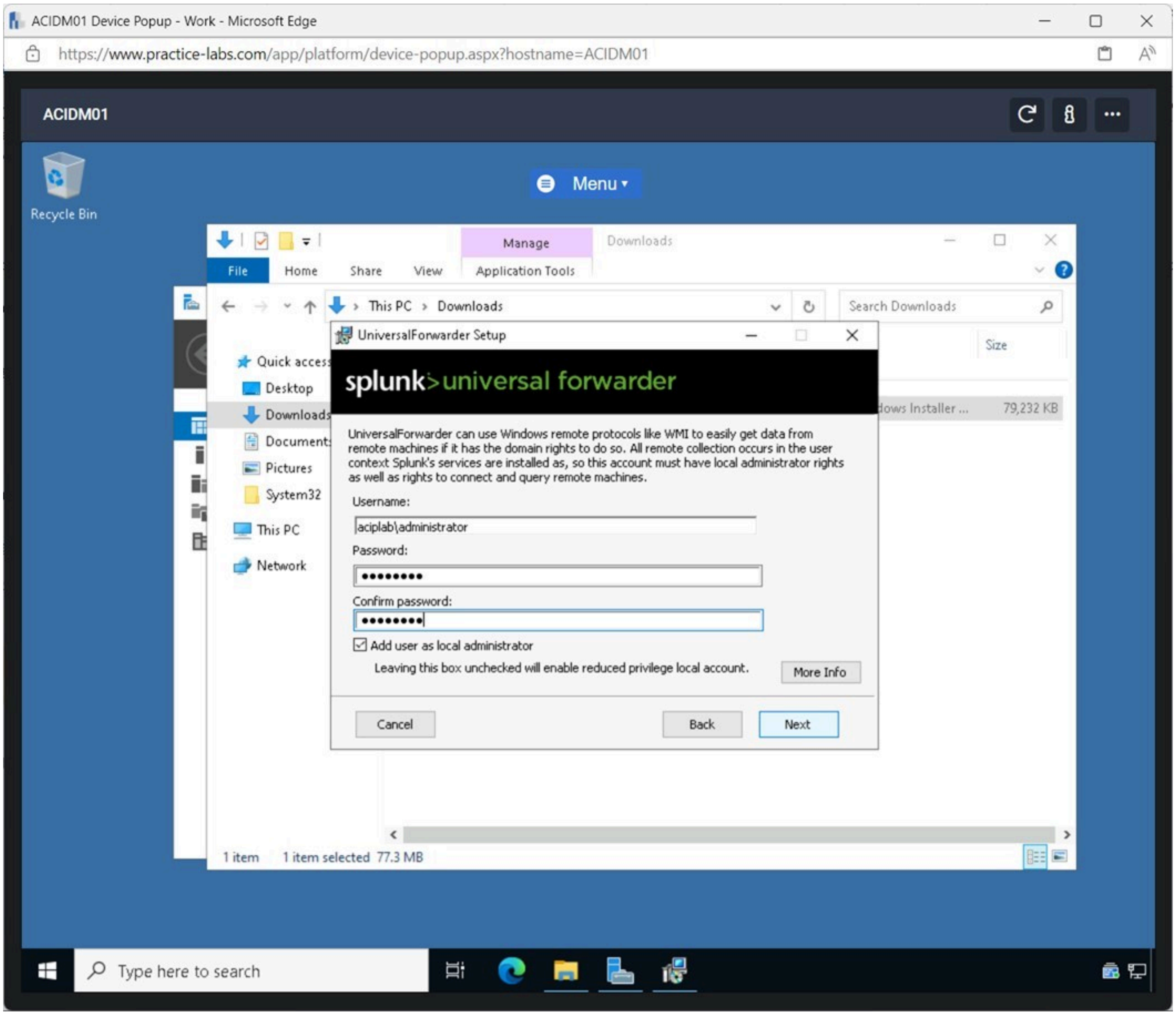


Figure 1.31 Screenshot of ACIDM01: Displaying entering the Domain credentials and clicking Next.

Step 8

On the **UniversalForwarder Setup** screen, tick all the boxes in the **Windows Event Logs** pane.

Click **Next**.

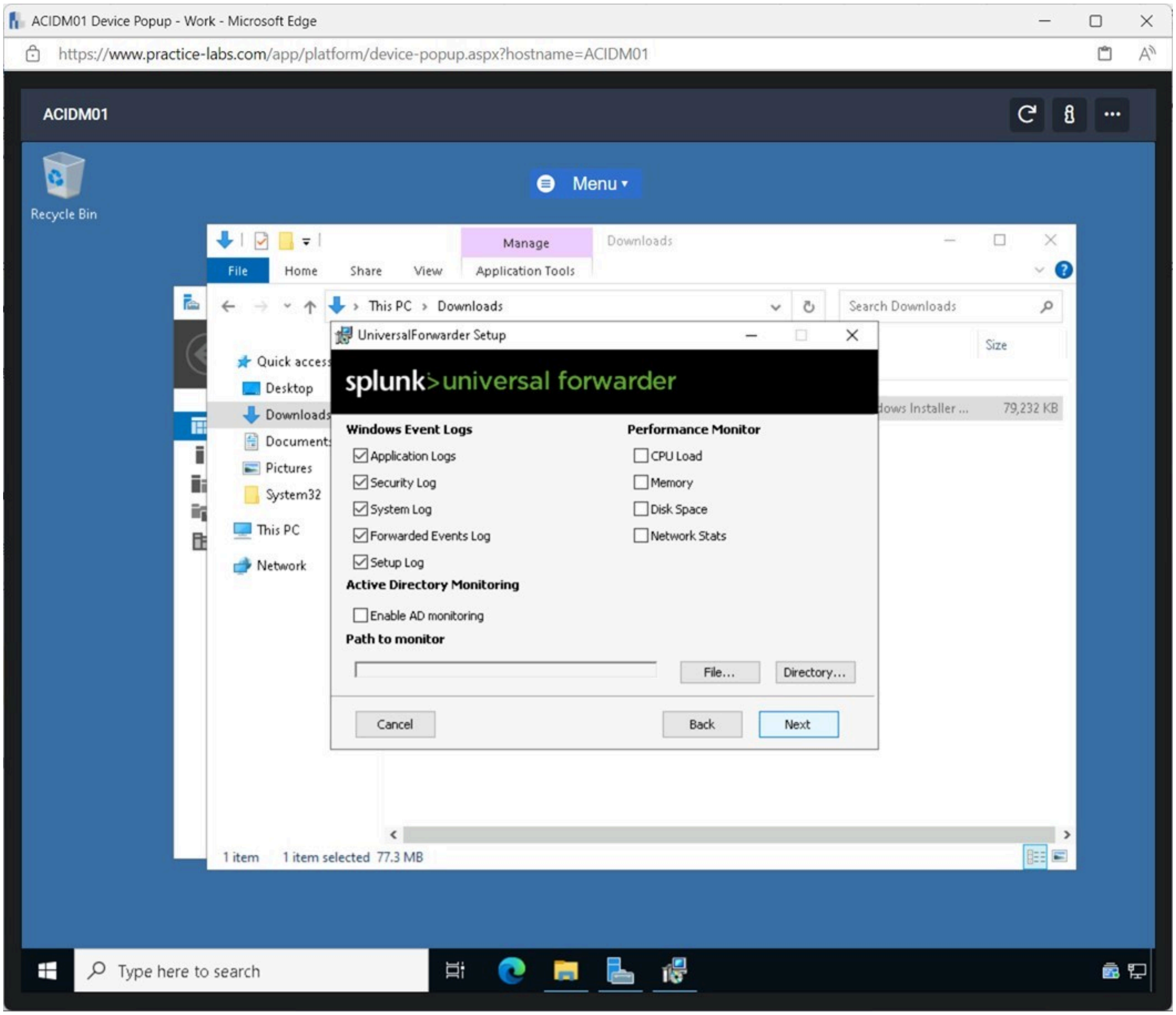


Figure 1.32 Screenshot of ACIDM01: Displaying ticking all the tick boxes in the Windows Event Logs pane and clicking Next.

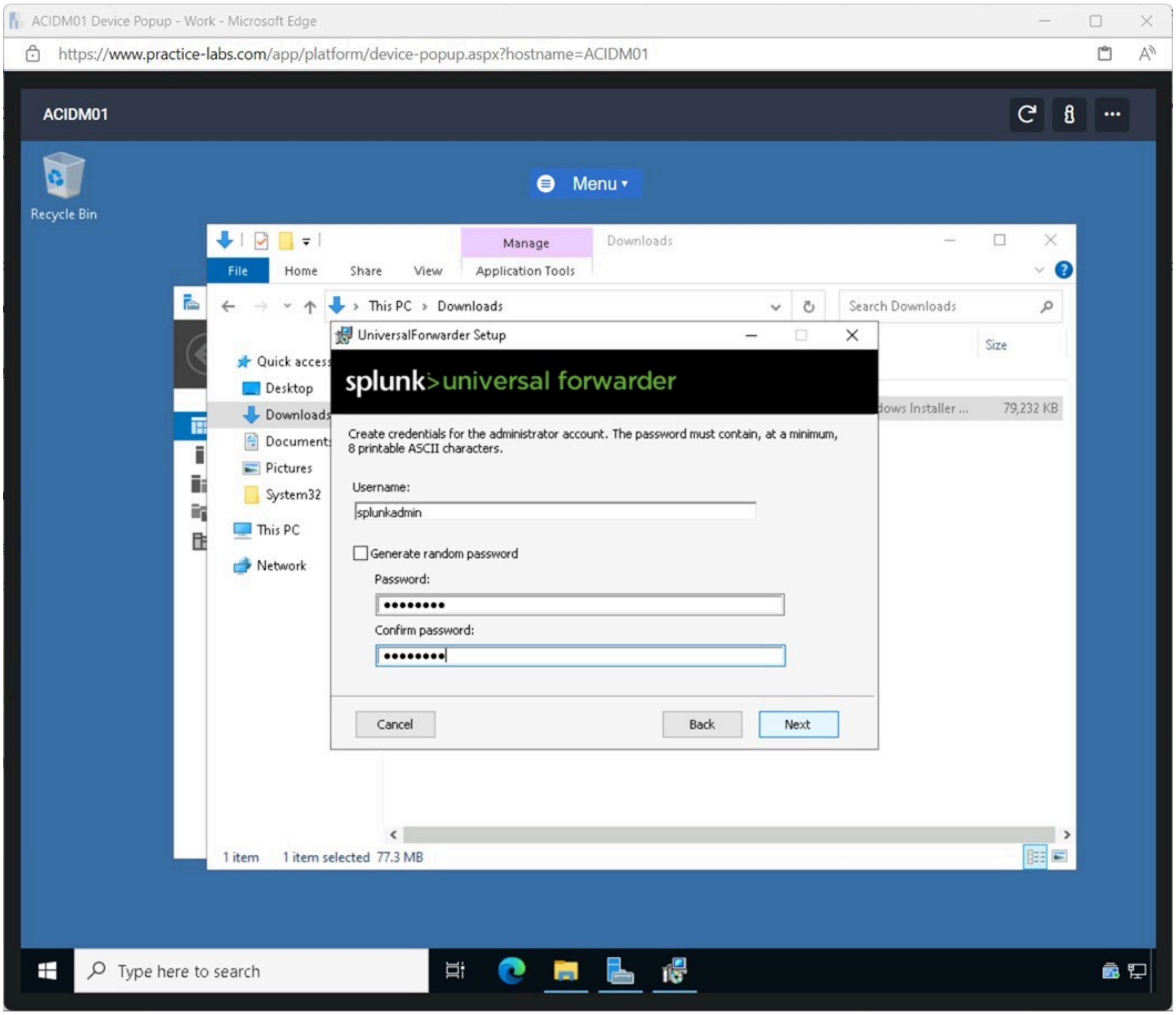
Step 9

Untick the **Generate random password** tickbox.

Enter the following details on the **UniversalForwader Setup** screen:

```
Username:        splunkadmin
Password:        Passw0rd
Confirm password:    Passw0rd
```

Click **Next**.



Step 1.33 Screenshot of ACIDM01: Displaying entering the credentials and clicking Next.

Step 10

In the **Hostname or IP** field, enter the following:

```
192.168.0.3
```

In the port field, enter the following:

```
8089
```

Click **Next**.

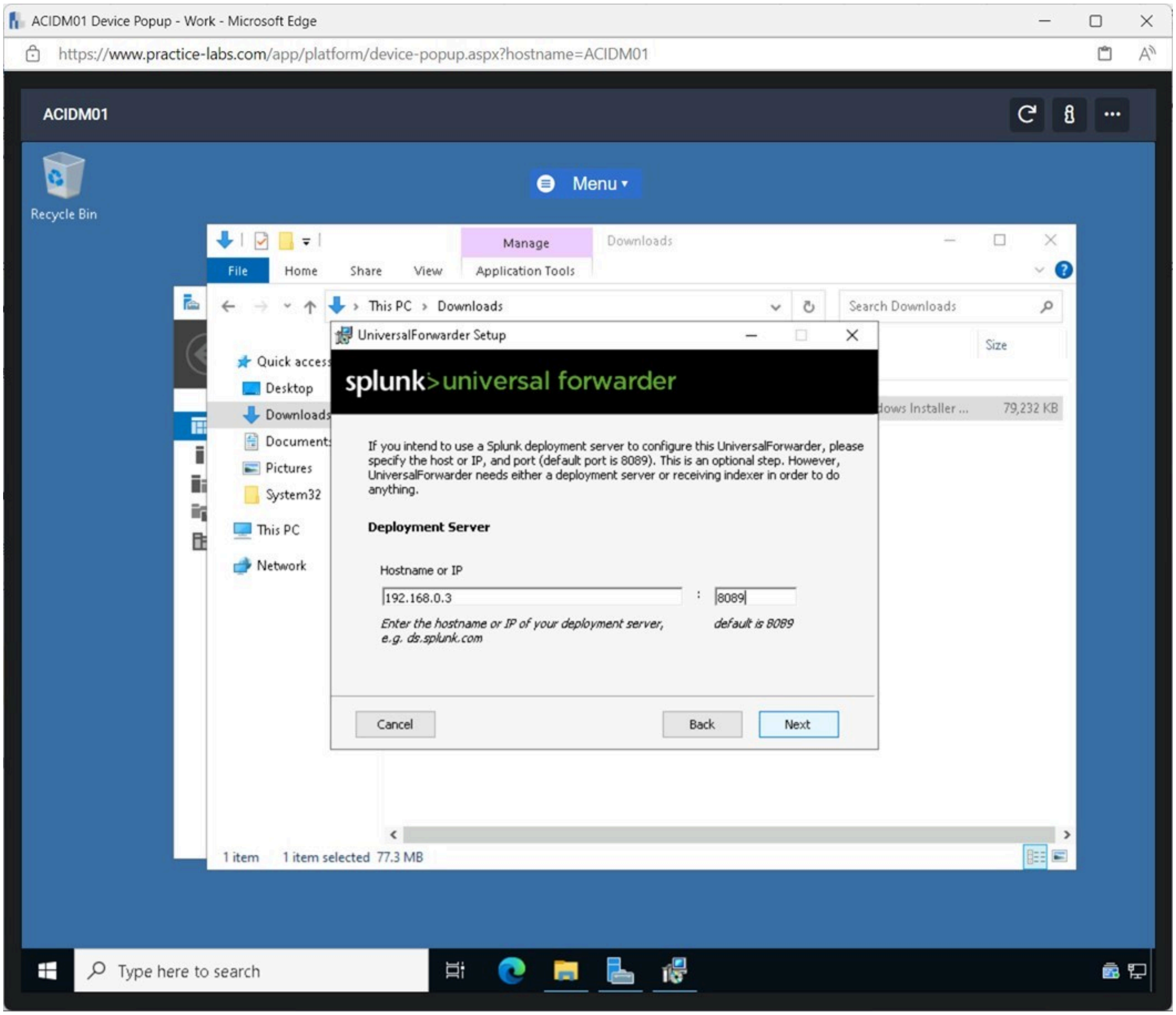


Figure 1.34 Screenshot of ACIDM01: Displaying entering the IP address and port number.

Step 11

In the **Hostname or IP** field, enter the following:

```
192.168.0.3
```

In the port field, enter the following:

```
9997
```

Click **Next**.

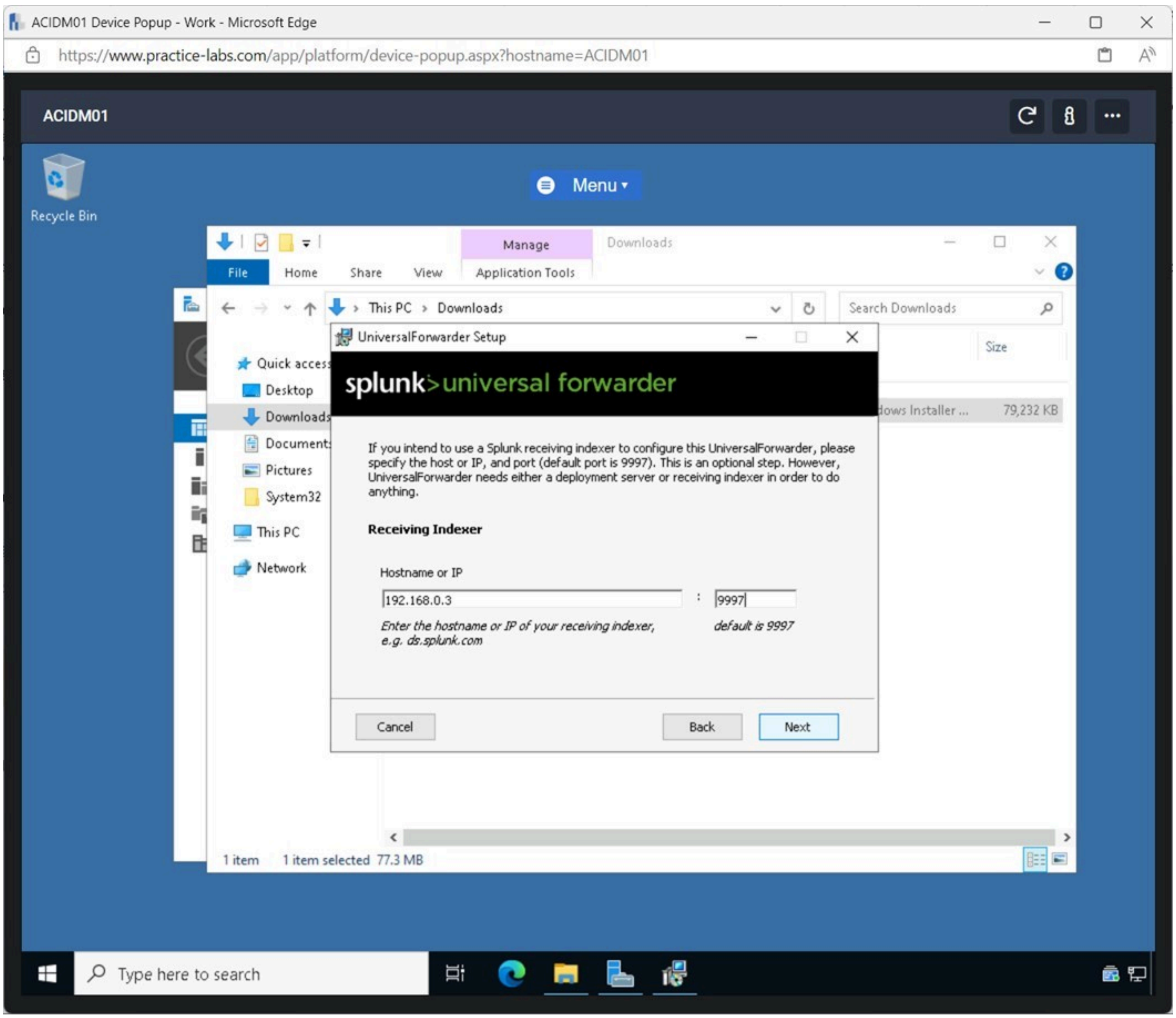


Figure 1.35 Screenshot of ACIDM01: Displaying entering the IP address and port number.

Step 12

Click **Install** on the **UniversalForwarder Setup** window.

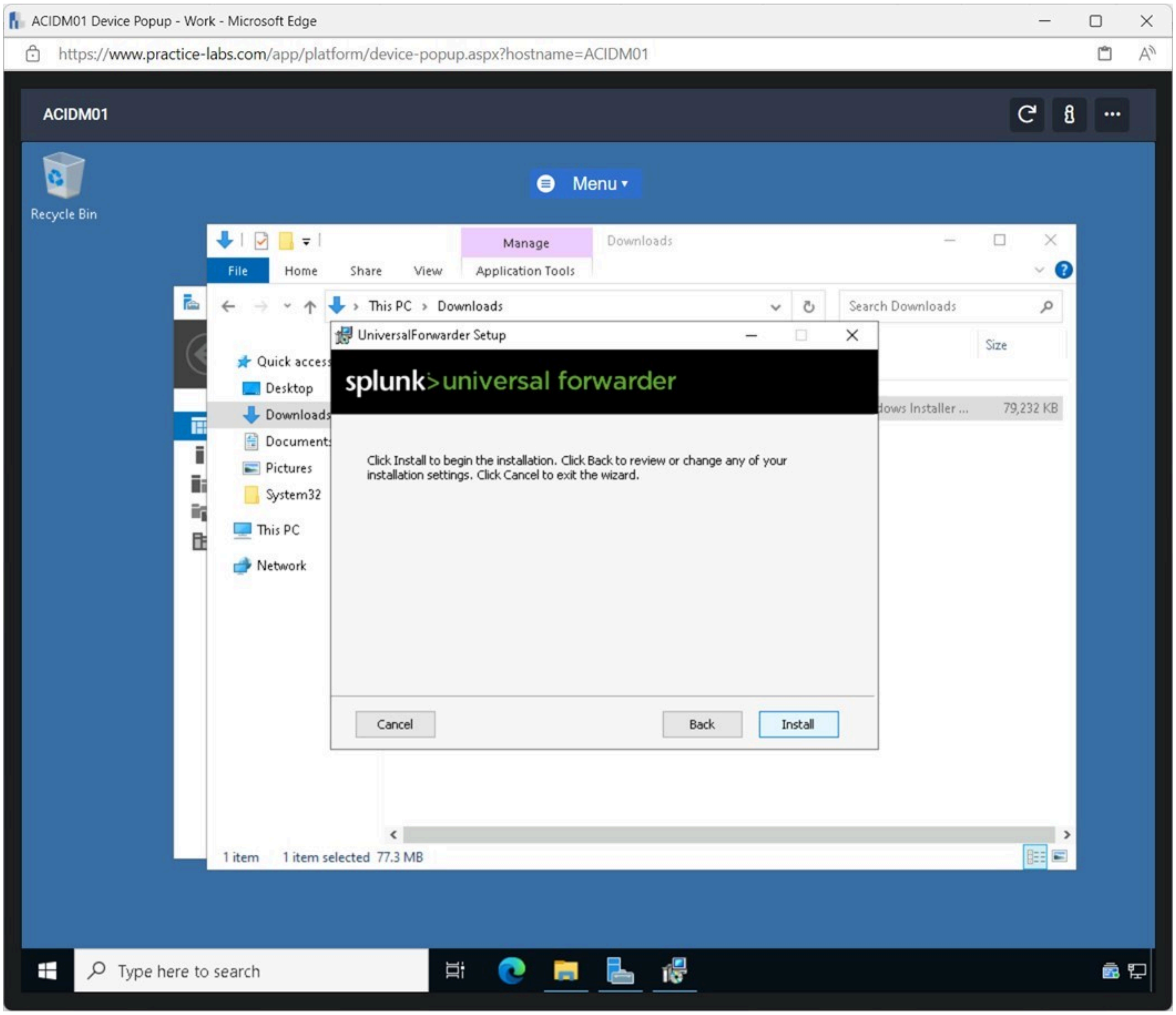


Figure 1.36 Screenshot of ACIDM01: Displaying clicking Install on the UniversalForwarder Setup window.

Step 13

After the installation has been completed, click **Finish**.

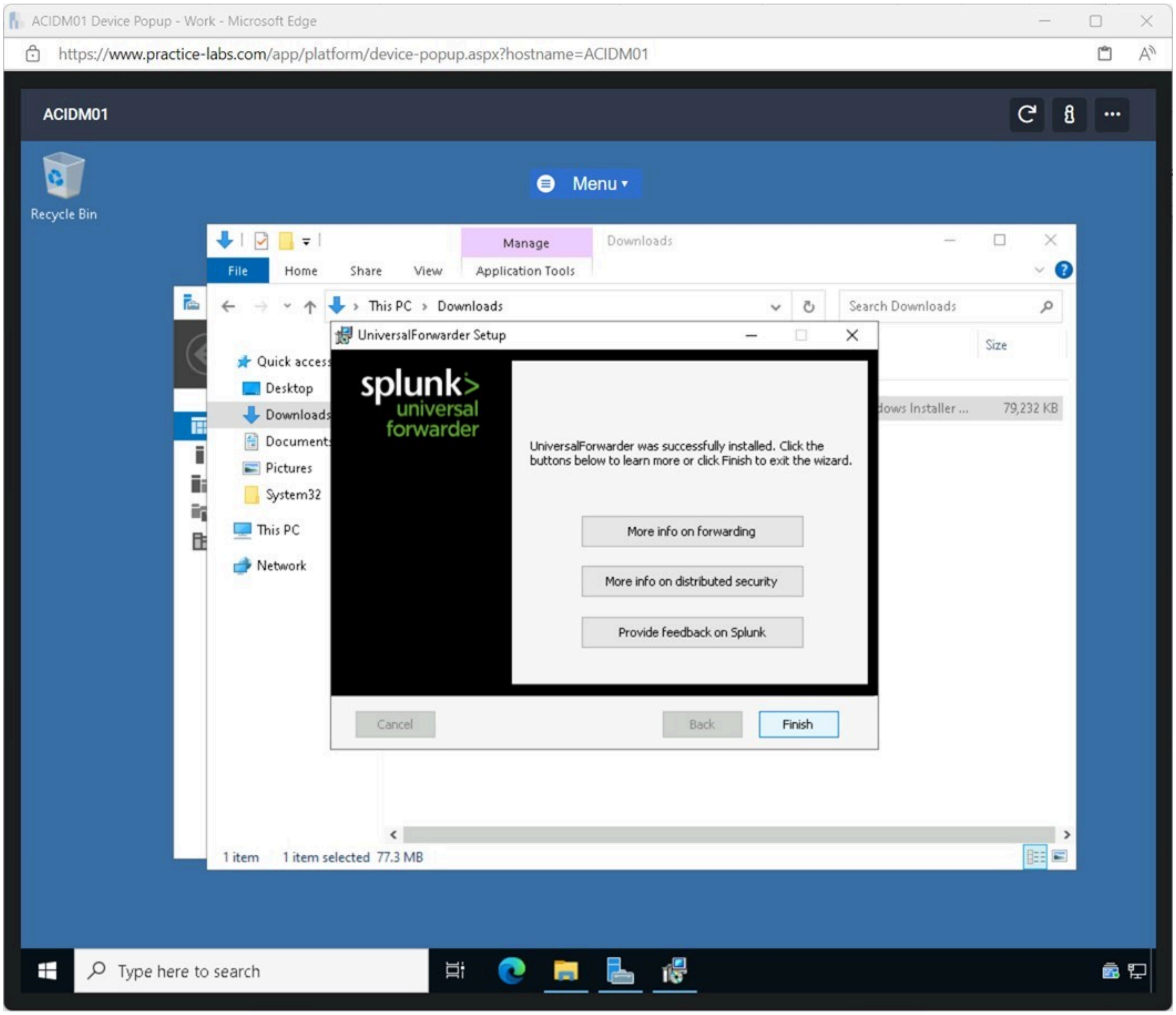


Figure 1.37 Screenshot of ACIDM01: Displaying clicking Finish after the application was installed.

Note: *The Splunk Universal forwarder was configured to forward the logs of the* ***ACIDM01*** *device to the Splunk Application, which was configured on* ***ACIWIN11****. The same steps can be performed on the ACIDC01 device to forward the logs to the Splunk Enterprise application.*

Task 3 - Collect Logs using Splunk Enterprise

After the Splunk Enterprise application is installed and configured, it can be used to collect logs from the devices on the network.

In this task, logs from the **ACIDM01** device will be collected using the Splunk Enterprise application.

Step 1

Connect to **ACIWIN11** and open **Microsoft Edge** from the **Taskbar**.

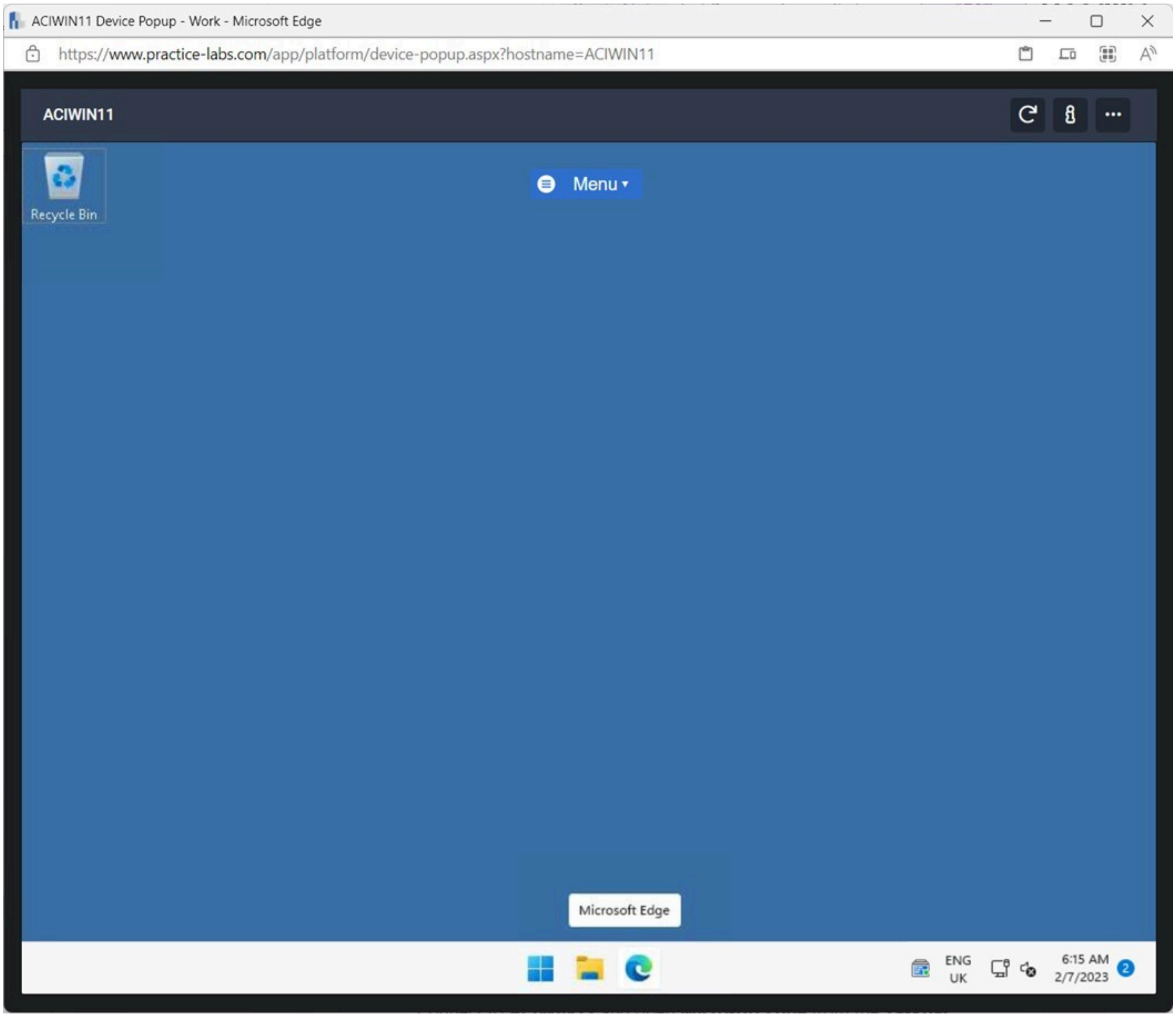


Figure 1.38 Screenshot of ACIWIN11: Displaying opening Microsoft Edge from the Taskbar.

Step 2

In the **Microsoft Edge** browser, browse to the following URL:

```
http://127.0.0.1:8000
```

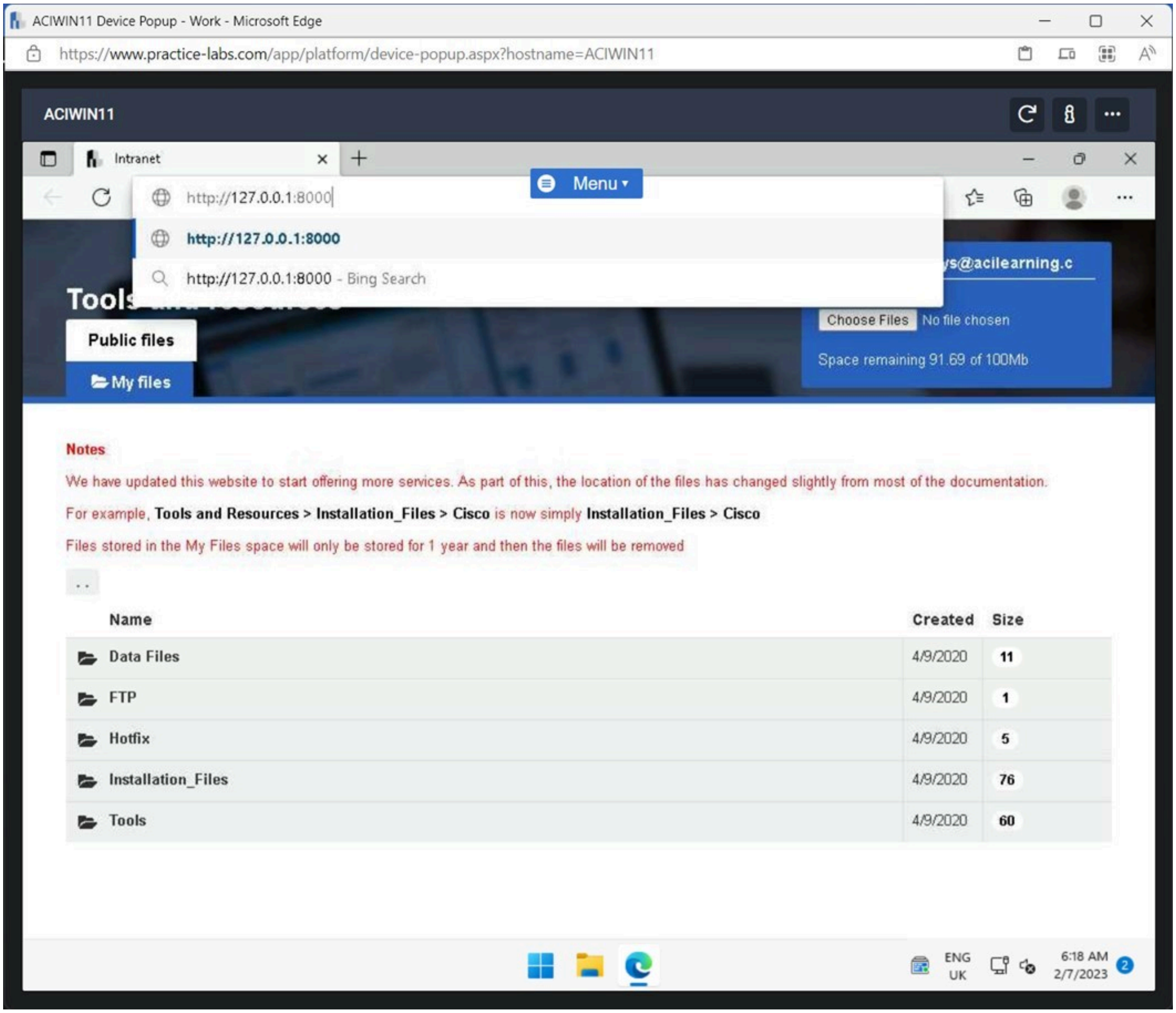


Figure 1.39 Screenshot of ACIWIN11: Displaying browsing to the http :// 127 . 0 . 0 . 1 : 8000 URL.

Step 3

Only if prompted, enter the following credentials:

```
Username: splunkadmin
Password: Passw0rd
```

Click **Sign In**.

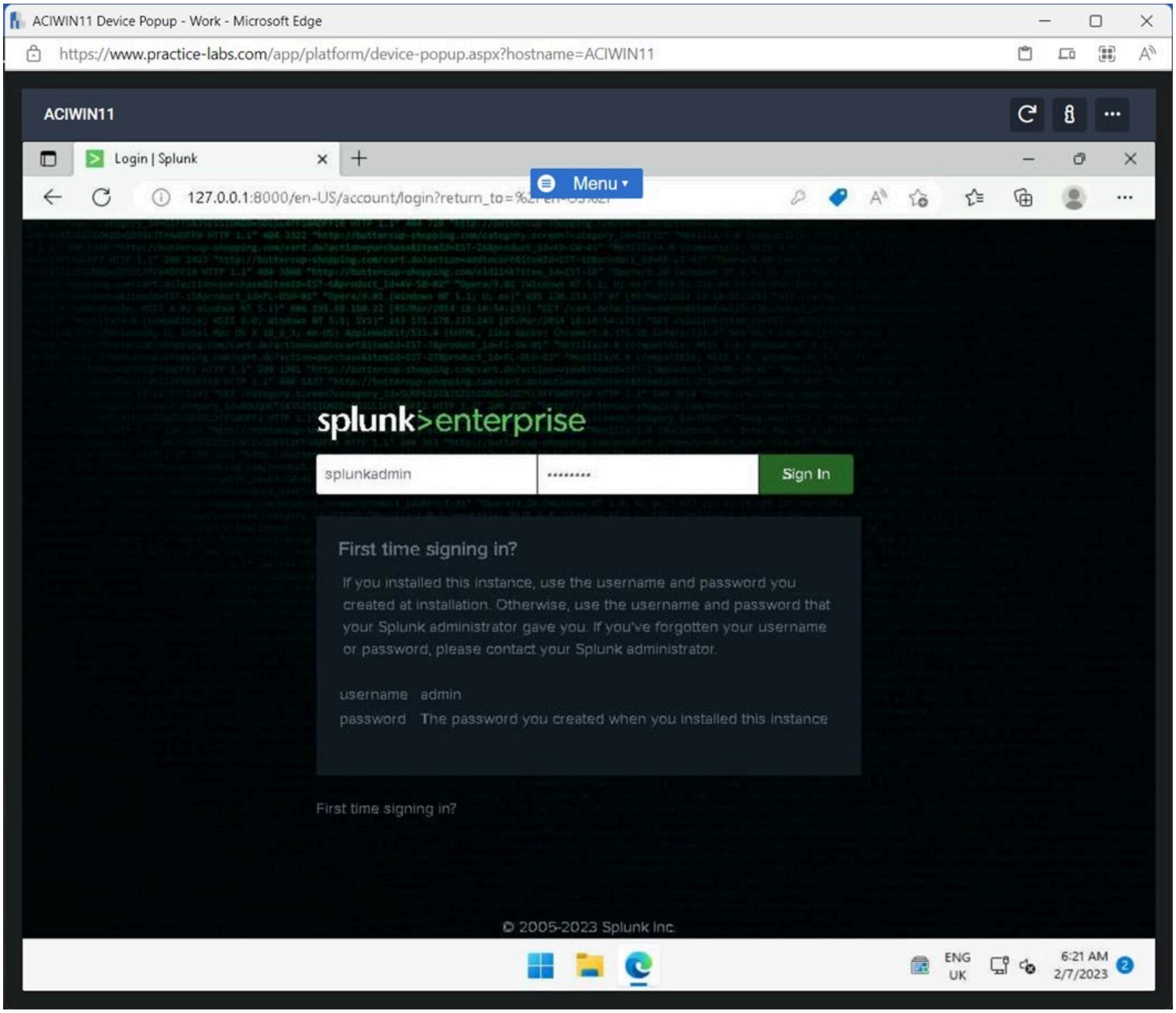


Figure 1.40 Screenshot of ACIWIN11: Displaying signing into the Splunk web application with the given credentials.

Step 4

Click **Add Data** in the **Explore Splunk Enterprise** pane.

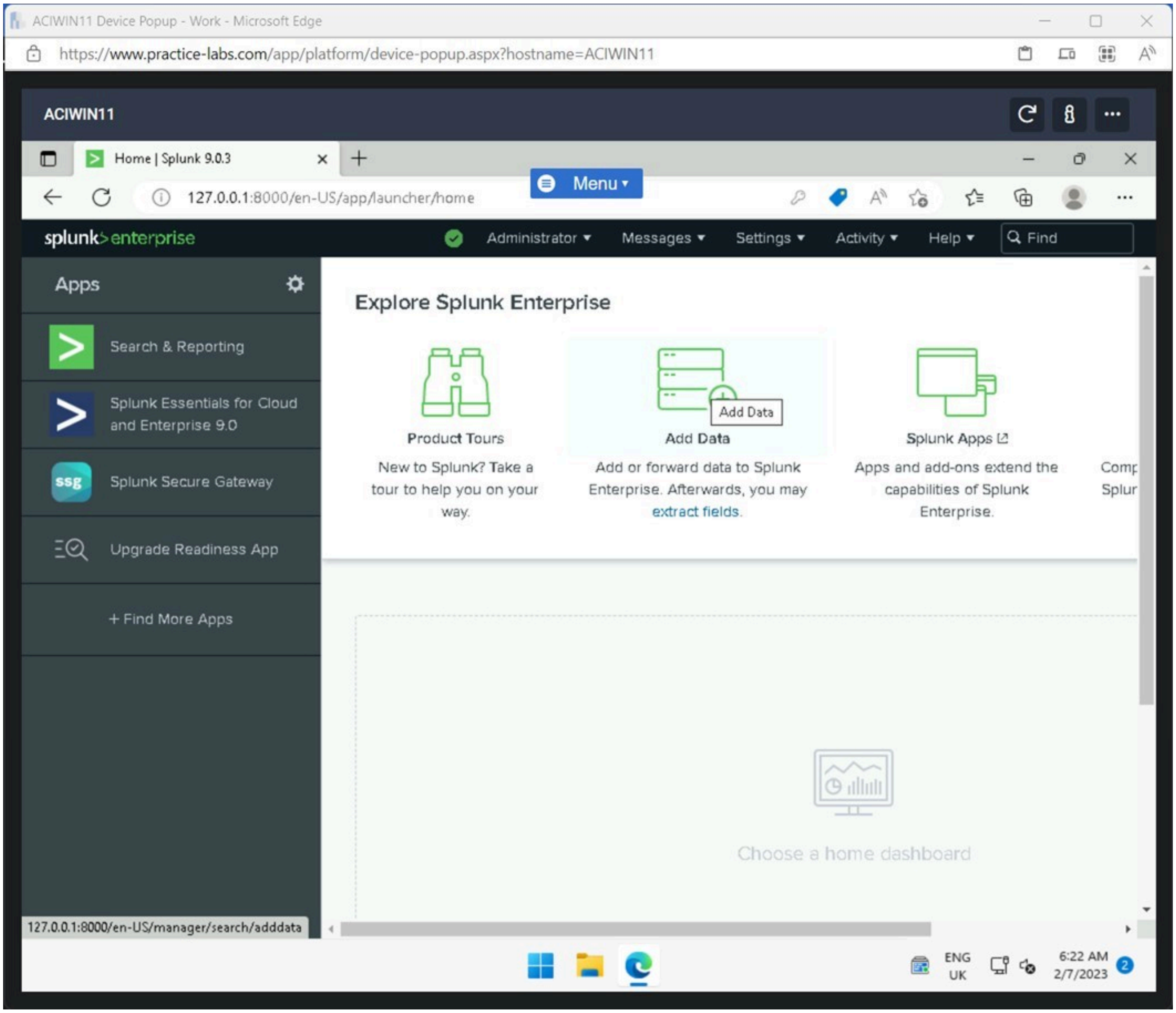


Figure 1.41 Screenshot of ACIWIN11: Displaying clicking Add Data in the Explore Splunk Enterprise pane.

Step 5

If a **Welcome, Administrator** window pops up, click **Skip**.

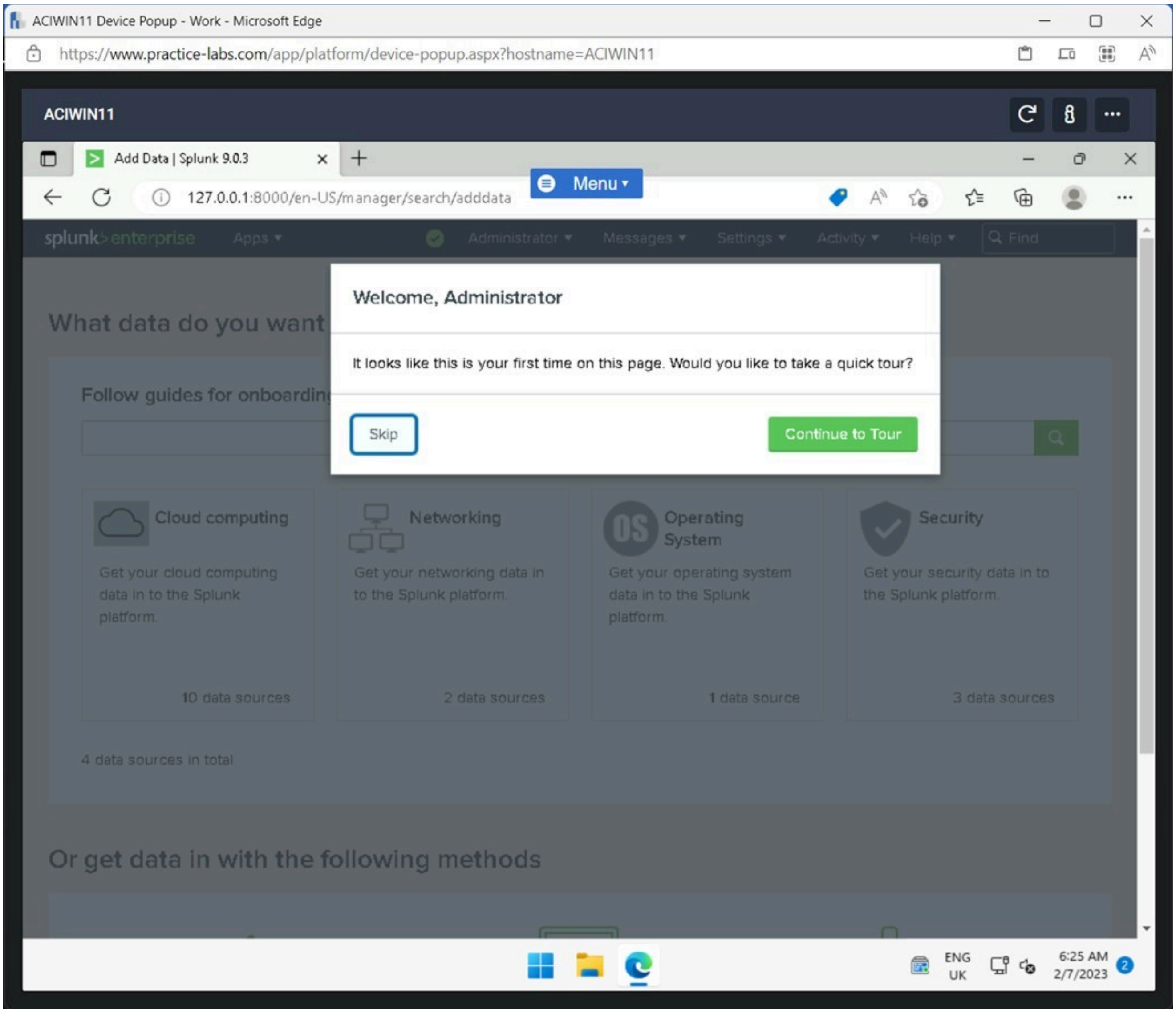


Figure 1.42 Screenshot of ACIWIN11: Displaying clicking Skip on the pop-up window.

Step 6

If needed, scroll down on the page and select **Forward data from a Splunk forwarder**.

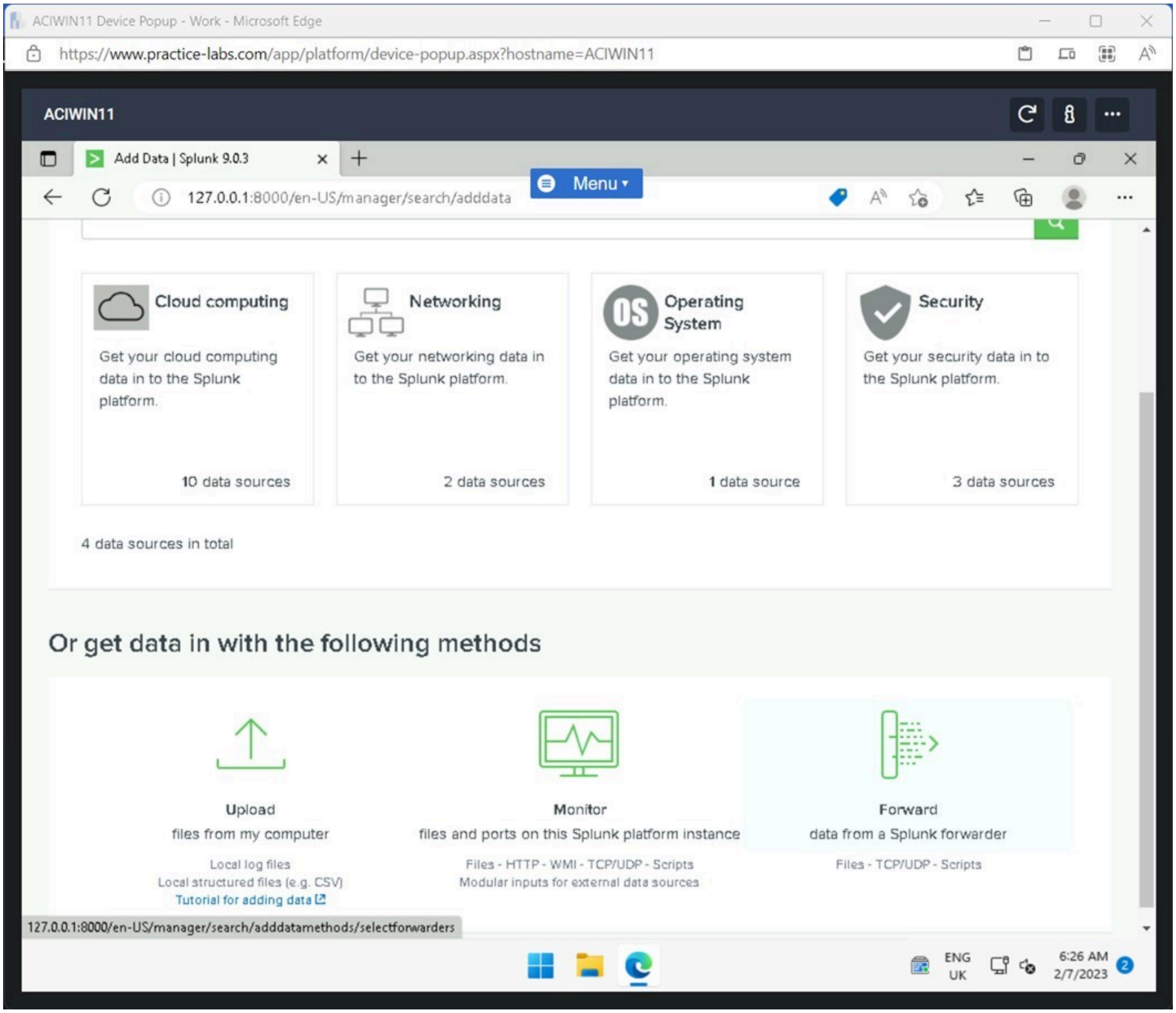


Figure 1.43 Screenshot of ACIWIN11: Displaying clicking Forward data from a Splunk forwarder.

Step 7

On the **Add Data** window, enter the following in the **New Server Class Name** field:

```
ACIDM01 Logs
```

Click **WINDOWS ACIDM01** in the **Available host(s)** pane.

Click **Next**.

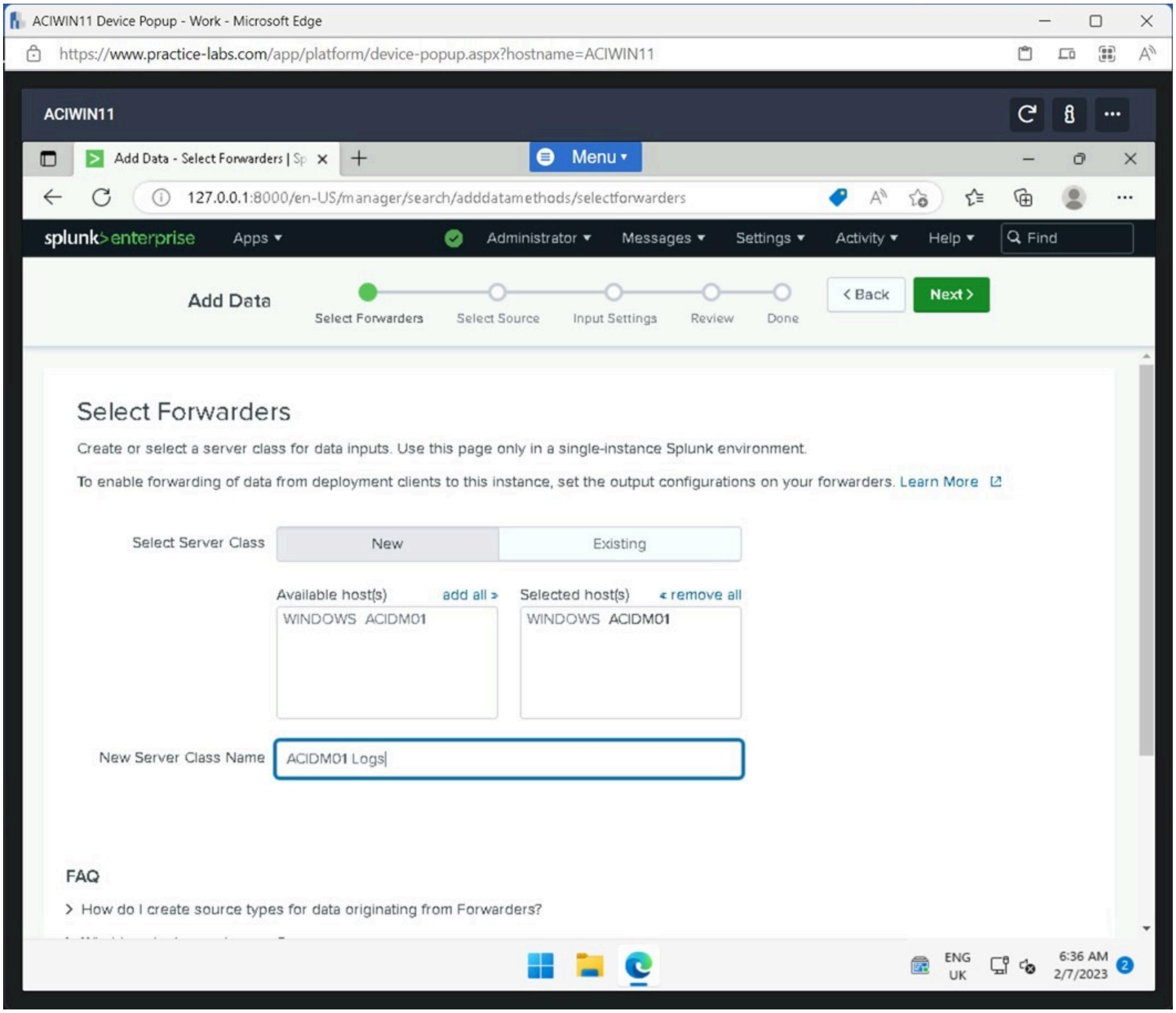


Figure 1.44 Screenshot of ACIWIN11: Displaying entering the New Server Class Name and clicking Next.

> ***Note:*** *If an error occurs, click back and select Forward again. There might be a delay before the Splunk Application detects the ACIDM01 device.*

Step 8

Click **Local Event Logs** on the **Select Source** window.

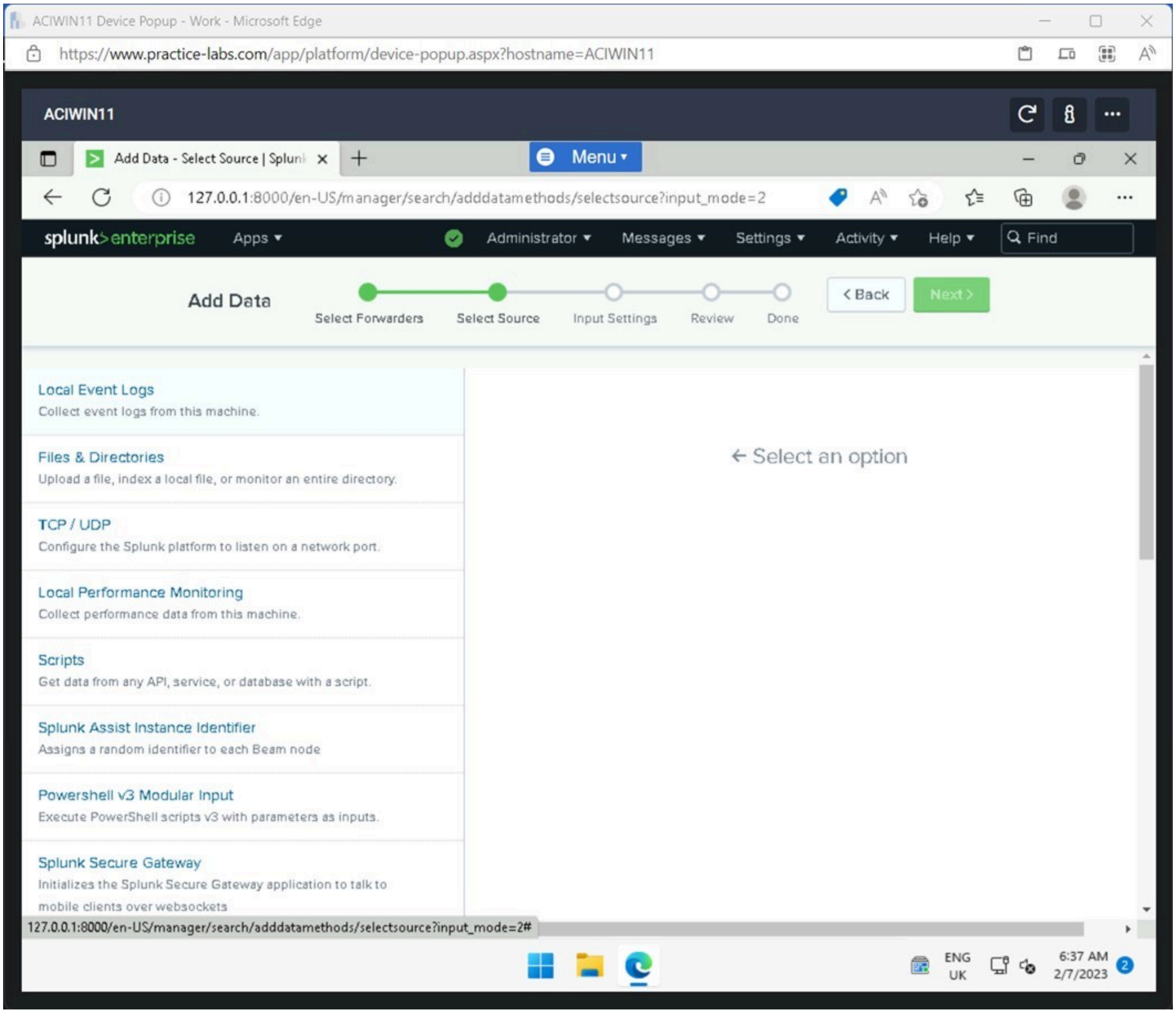


Figure 1.45 Screenshot of ACIWIN11: Displaying clicking Local Event Logs on the Select Source window.

Step 9

In **Select Event Logs** pane, click the following options:

```
Application
Security
System
```

Click **Next**.

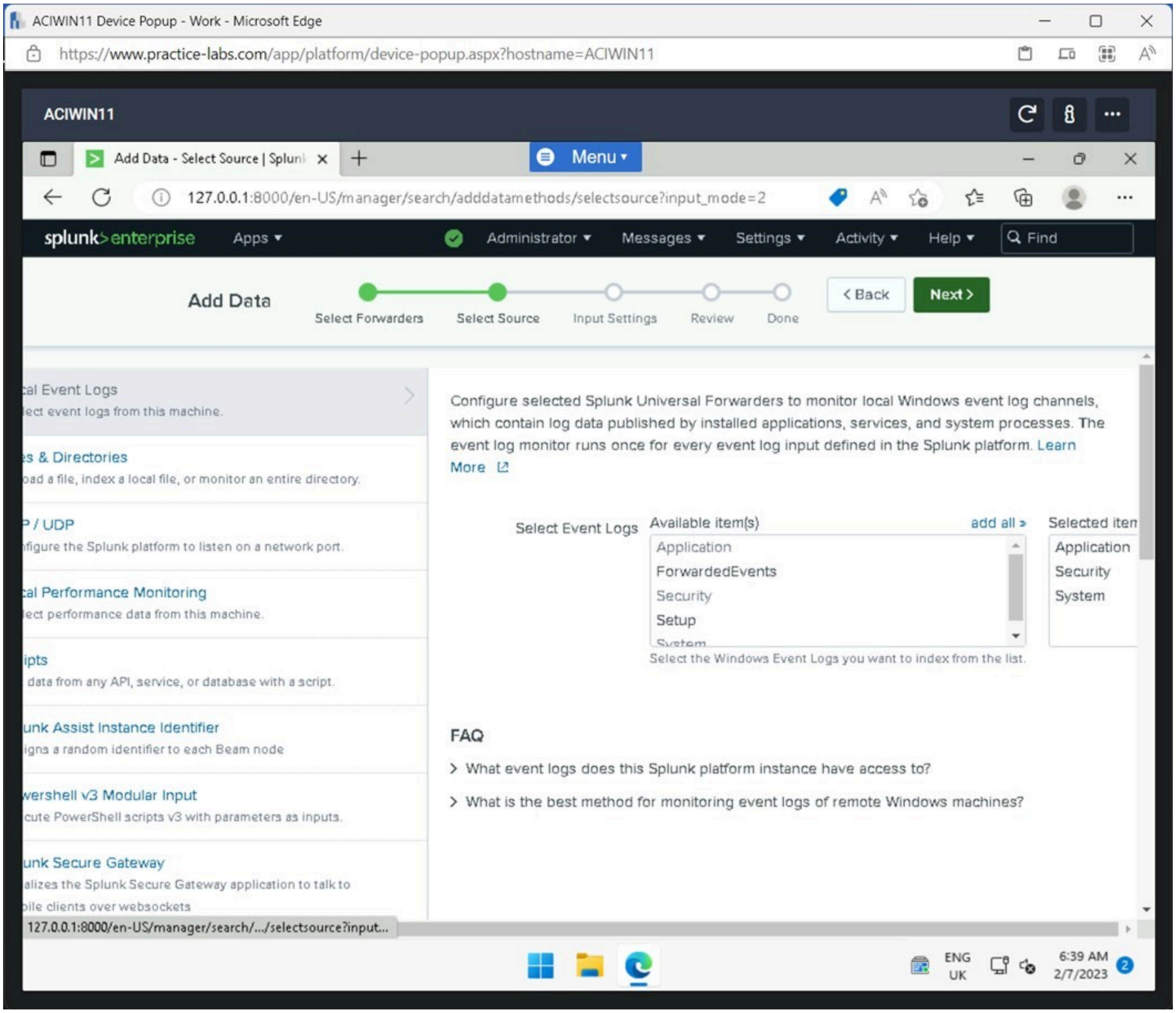


Figure 1.46 Screenshot of ACIWIN11: Displaying clicking the Available Items and selecting Next.

***Note:** By clicking the specified Lopg events, move it to the Selected items window.*

Step 10

Click **Review** on the **Input Settings** window.

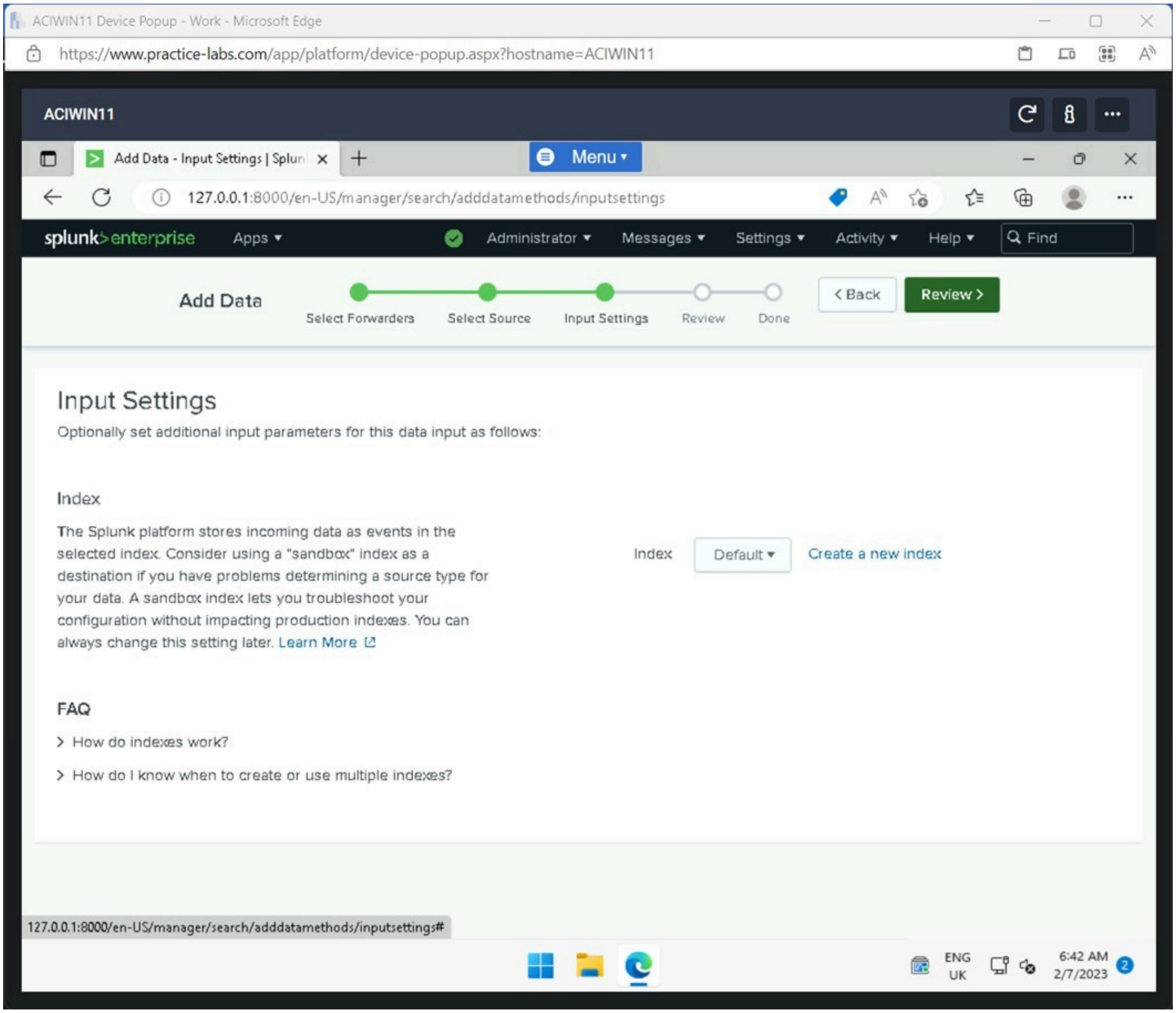


Figure 1.47 Screenshot of ACIWIN11: Displaying clicking Review on the Input Settings window.

Step 11

Click **Submit** on the **Review** window.

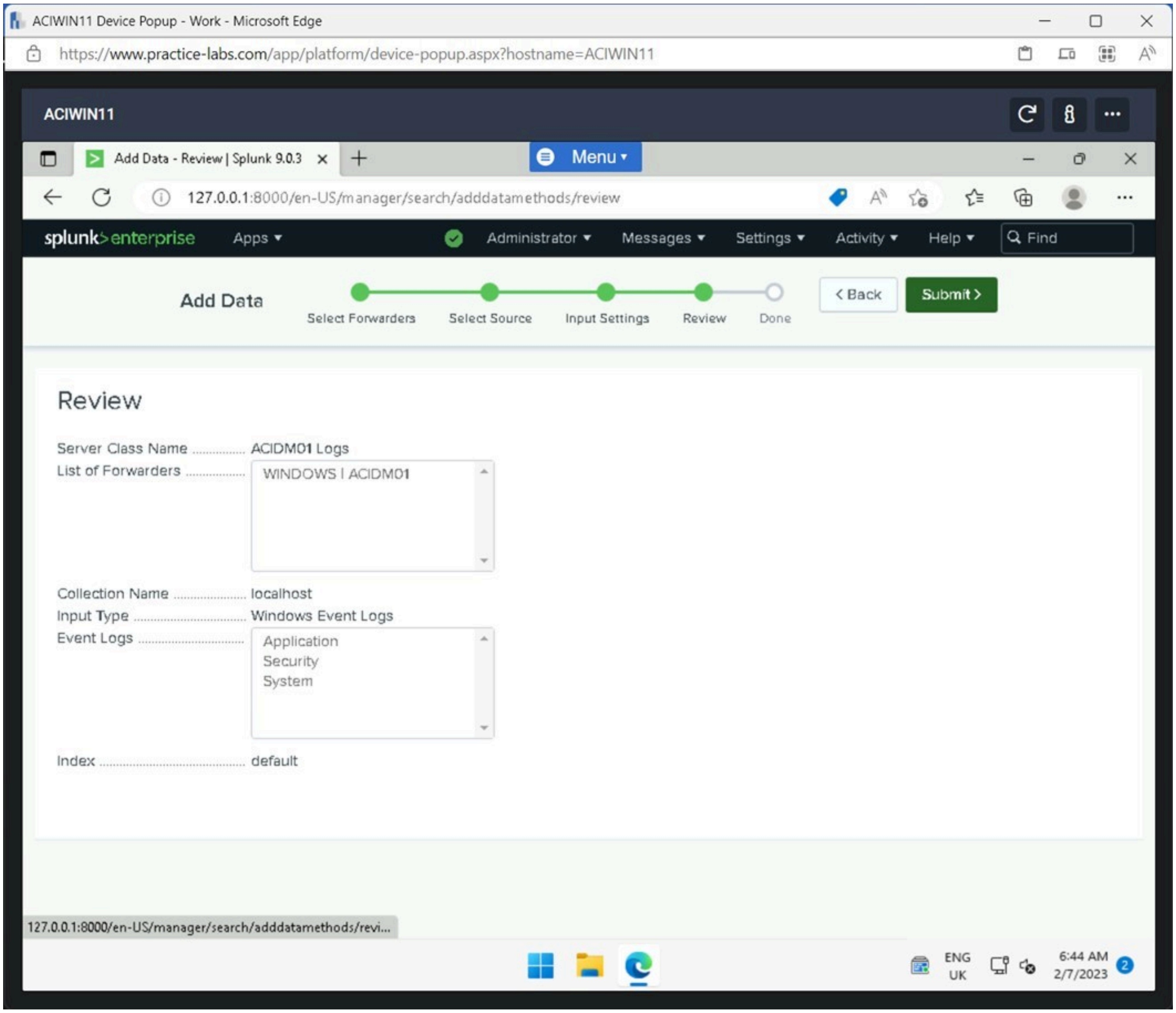


Figure 1.48 Screenshot of ACIWIN11: Displaying clicking Review on the Review window.

Step 12

Click **Start Searching** on the **Local event logs input has been created successfully** window.

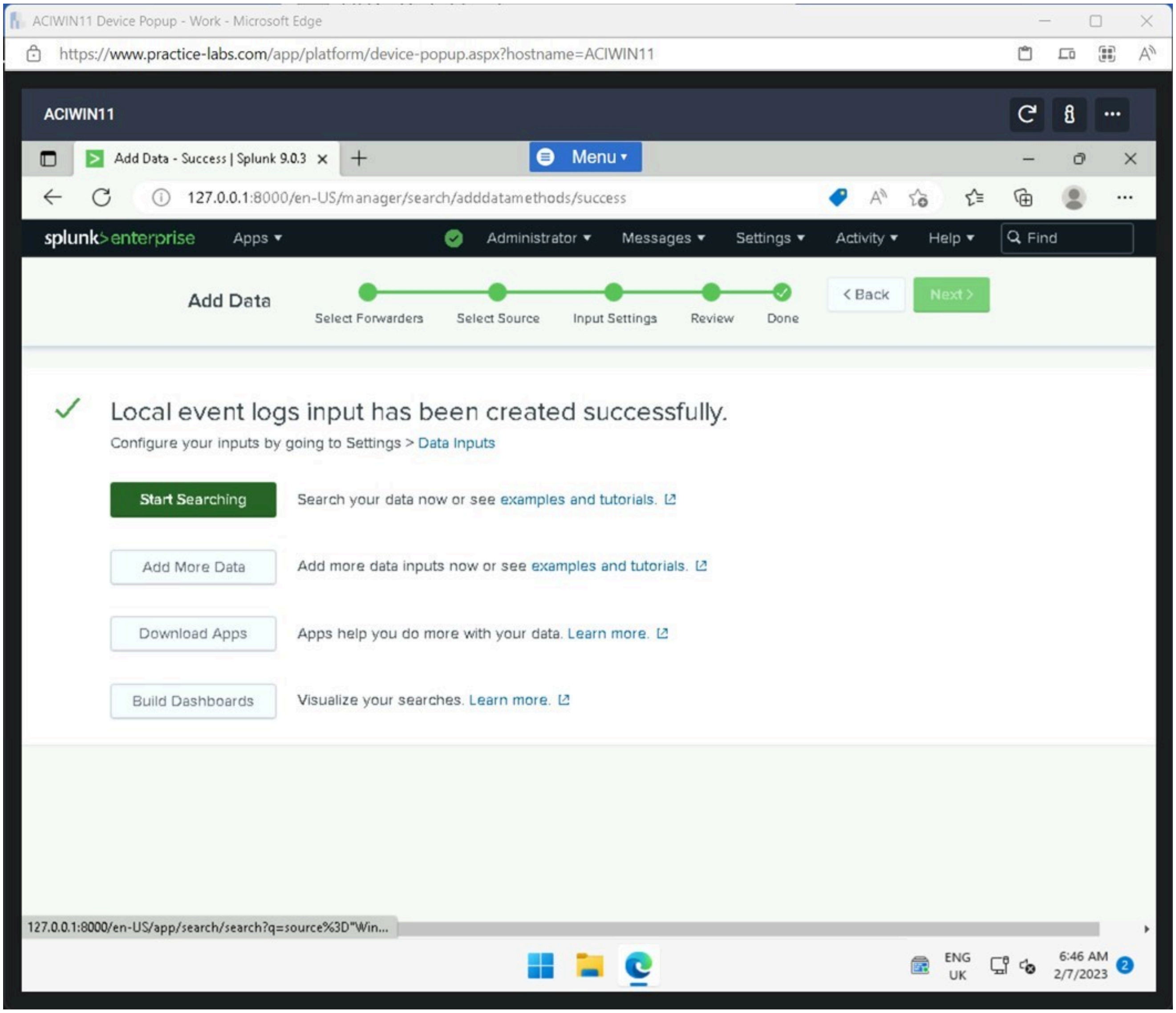


Figure 1.49 Screenshot of ACIWIN11: Displaying clicking Start Searching on the Local event logs input has been created successfully window.

Step 13

Close the **Microsoft Edge** browser window once you have viewed the event logs for the **ACIDM01**.

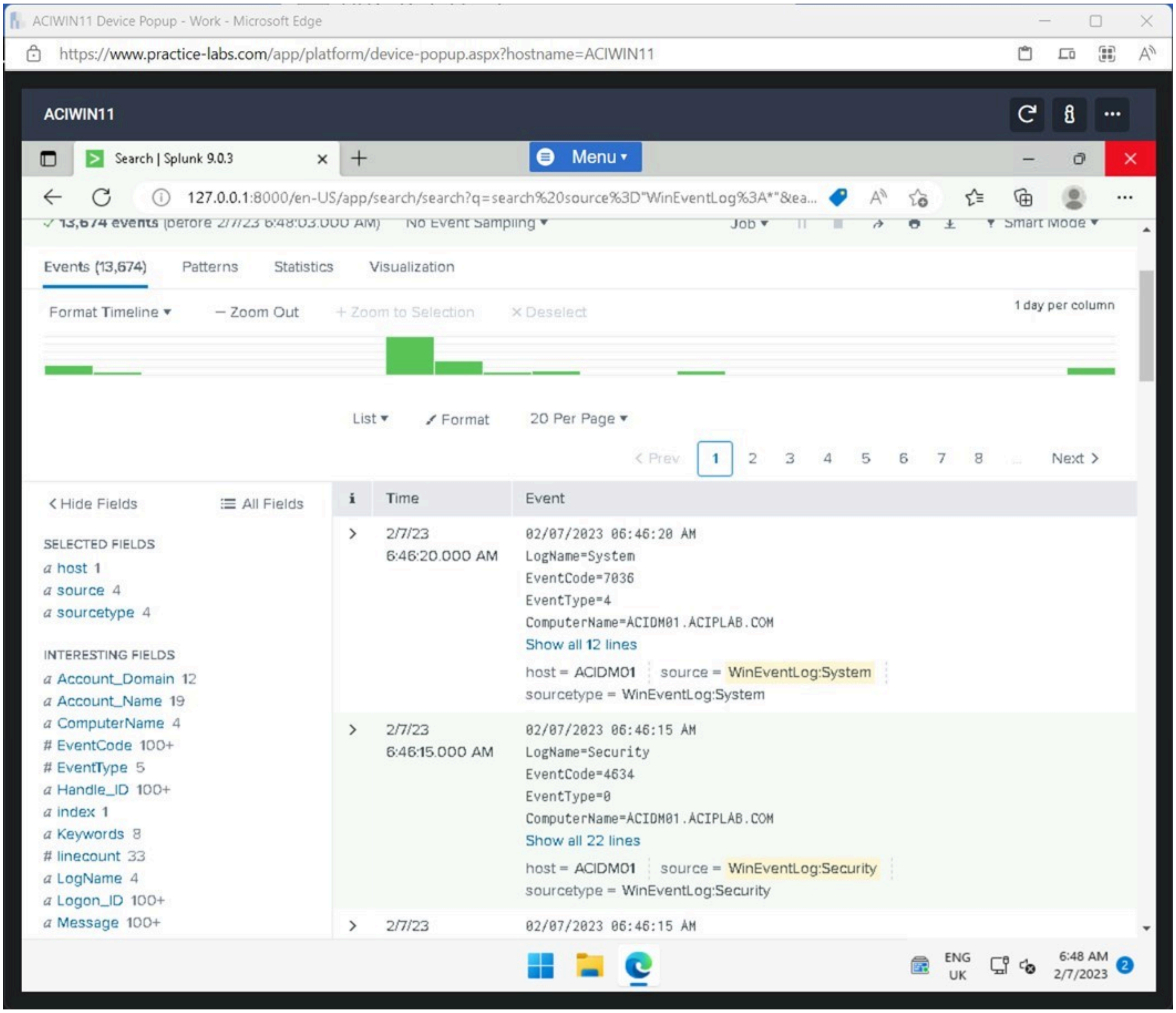


Figure 1.50 Screenshot of ACIWIN11: Displaying closing the Microsoft Edge browser.

> ***Note:*** *The Event logs for the ACIDM01 have been successfully imported into the Splunk Enterprise application. Additional devices can be added to have a central repository to view all the log files in one place.*

Exercise 2 - Encrypting Sensitive Data

As a Cyber Security Specialist, it is critical to protect sensitive data that is stored locally or in the cloud. Different types of sensitive data can be stored on the server, for example, Personally identifiable information (PII) or Cardholder Data (CHD)

To protect locally stored data, the server's hard drives can be encrypted to protect the data on the drive.

In this exercise, the Bitlocker Drive encryption feature will be installed, and a local drive will be encrypted.

Learning Outcomes

After completing this exercise, you should be able to:

- Install the BitLocker Drive Encryption Feature
- Encrypt a Local Drive on a Server

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01 -** Windows Server 2022 - Domain Controller
- **ACIDM01 -** Windows Server 2022 - Domain Member Server

Task 1 - Install the Bitlocker Drive Encryption Feature

To be able to encrypt a local disk on a Windows Server, the Bitlocker feature needs to be installed.

In this task, the Bitlocker Drive Encryption feature will be installed.

Step 1

Connect to **ACIDM01** in **Server Manager** and click **Add roles and features**.

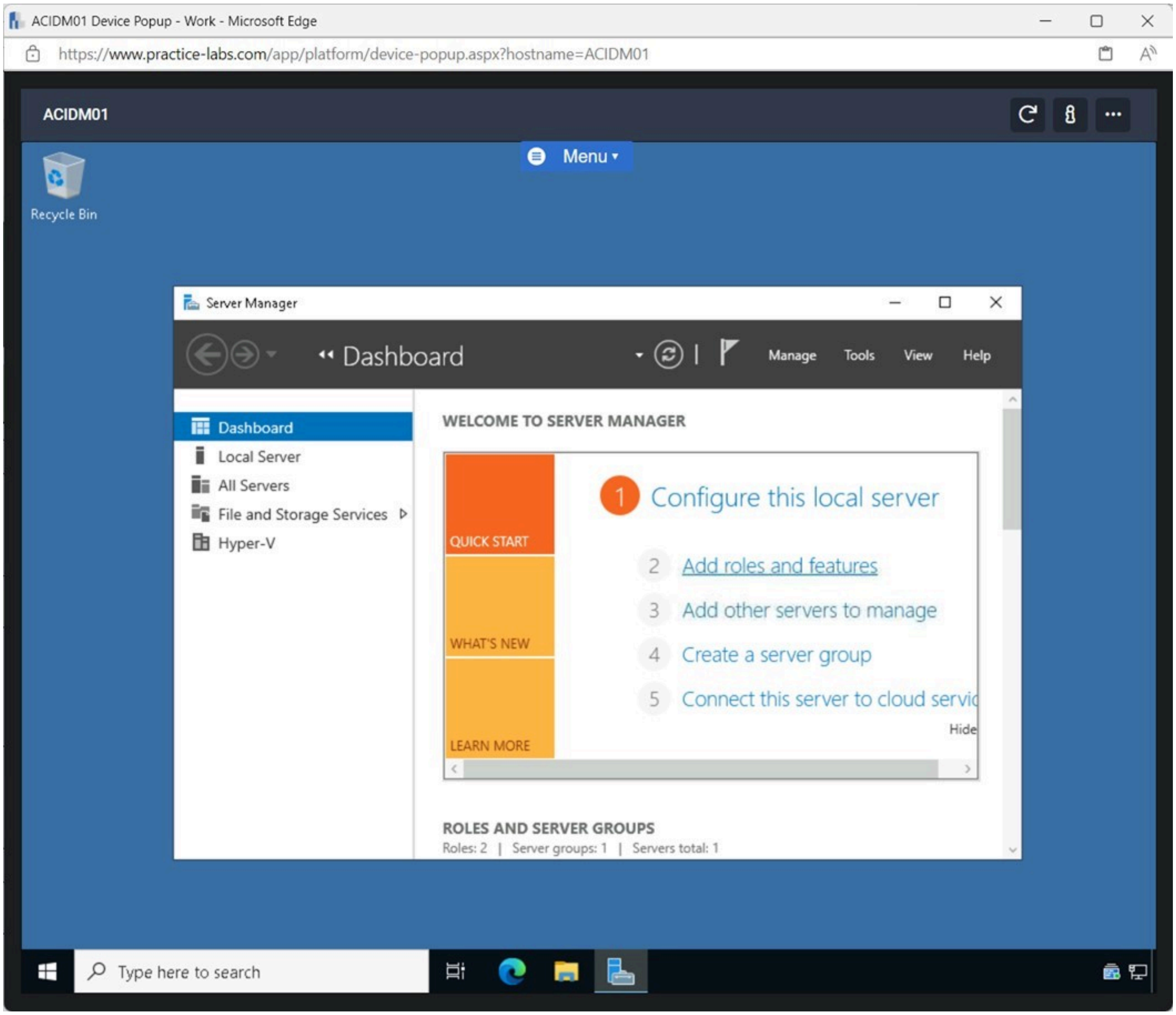


Figure 2.1 Screenshot of ACIDM01: Displaying clicking Add roles and features in Server Manager.

Step 2

In the **Add Roles and Features Wizard** window, click **Next**.

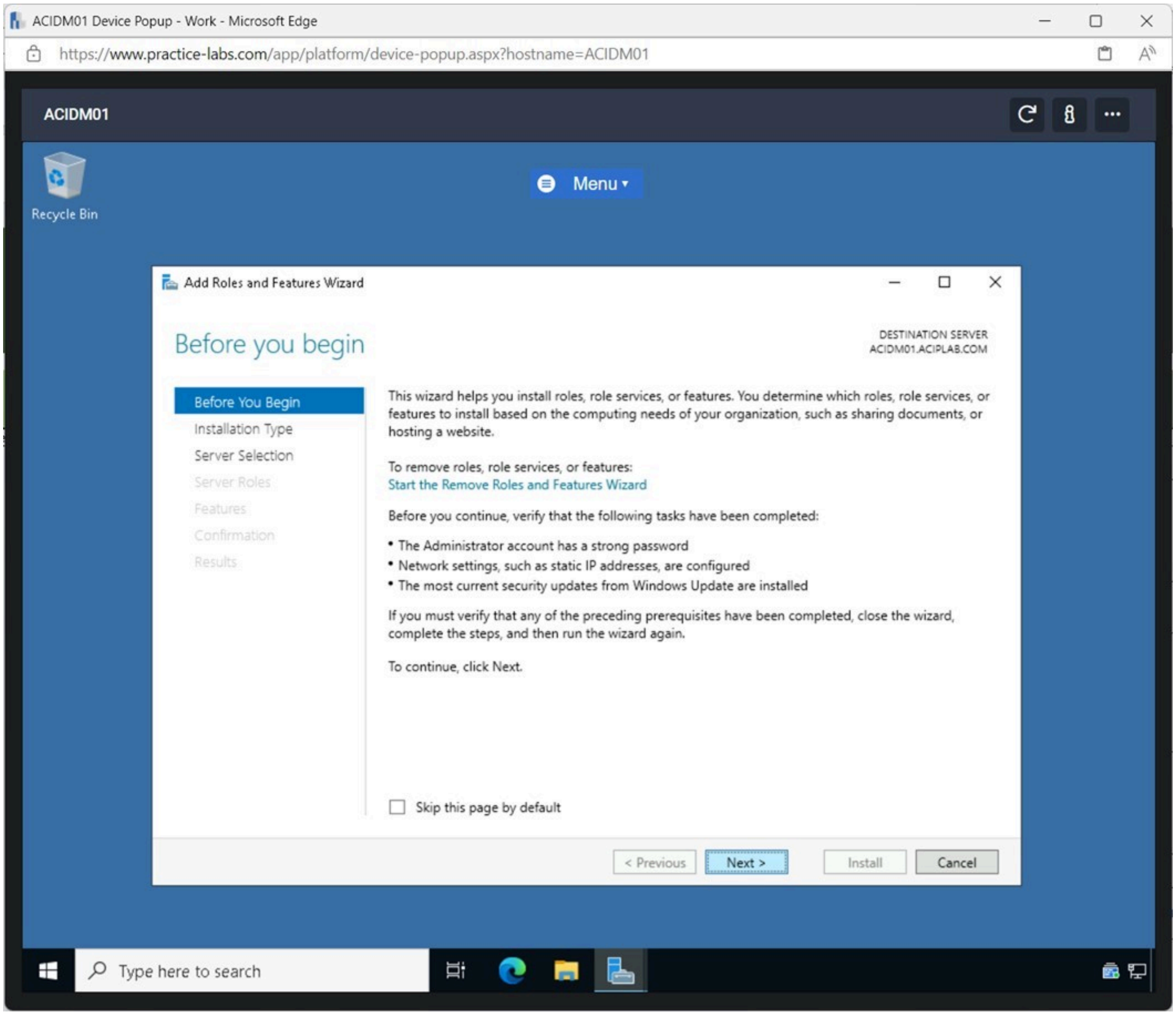


Figure 2.2 Screenshot of ACIDM01: Displaying clicking Next on the Add Roles and Features Wizard window.

Step 3

On the **Select installation type** pane, click **Next** until reaching the **Select features** pane.

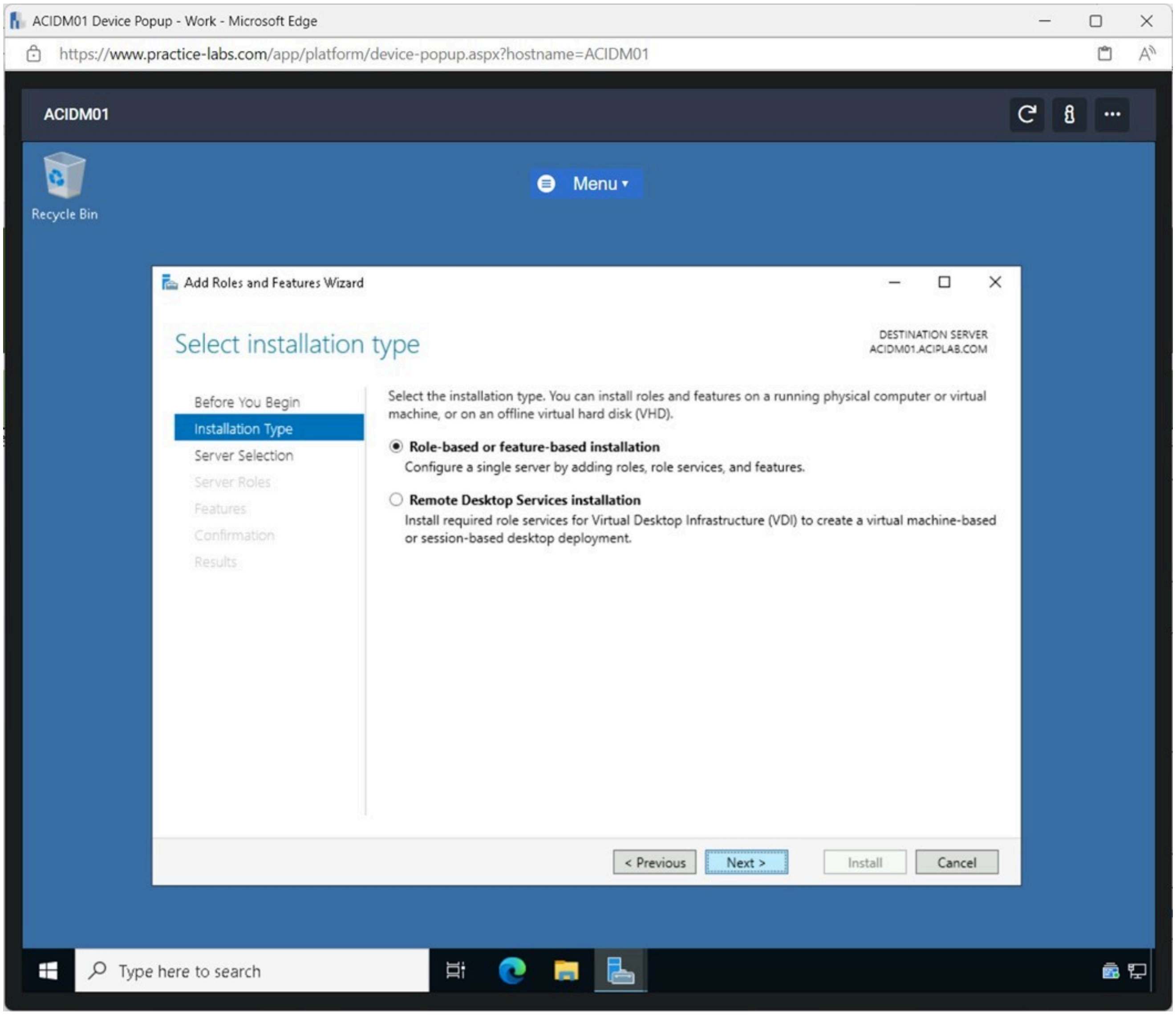


Figure 2.3 Screenshot of ACIDM01: Displaying clicking Next to reach the Select features pane.

Step 4

On the **Select features** pane, enable the **Bitlocker Drive Encryption** tickbox and click **Add Features** in the pop-up window.

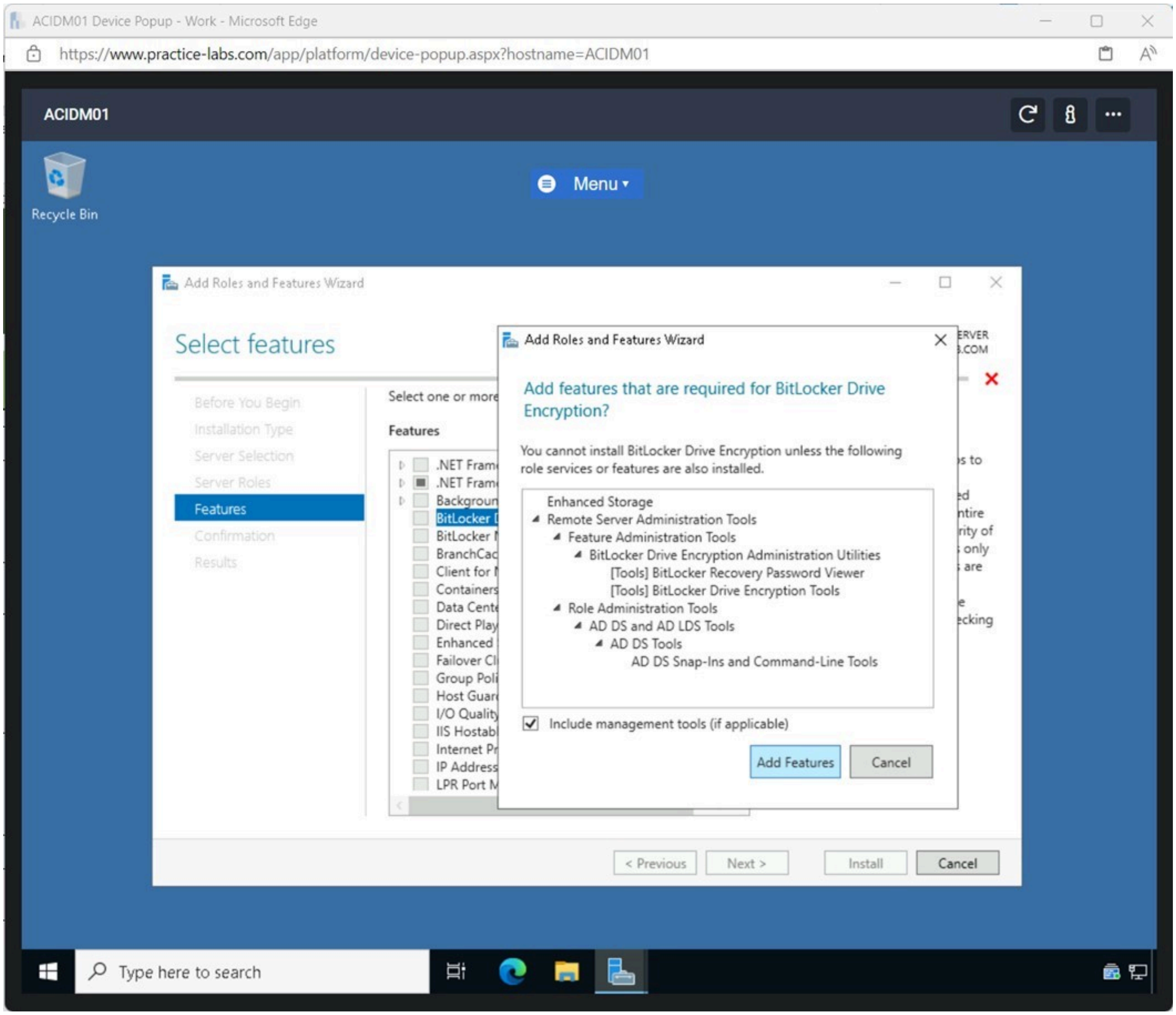


Figure 2.4 Screenshot of ACIDM01: Displaying enabling the Bitlocker Drive Encryption tickbox and clicking Add features on the pop-up window.

> ***Note:*** *When installing a specific feature on Windows Server, additional components are needed for the feature's functionality. Windows Server adds the additional needed components automatically.*

Step 5

Click **Next** on the **Select features** pane.

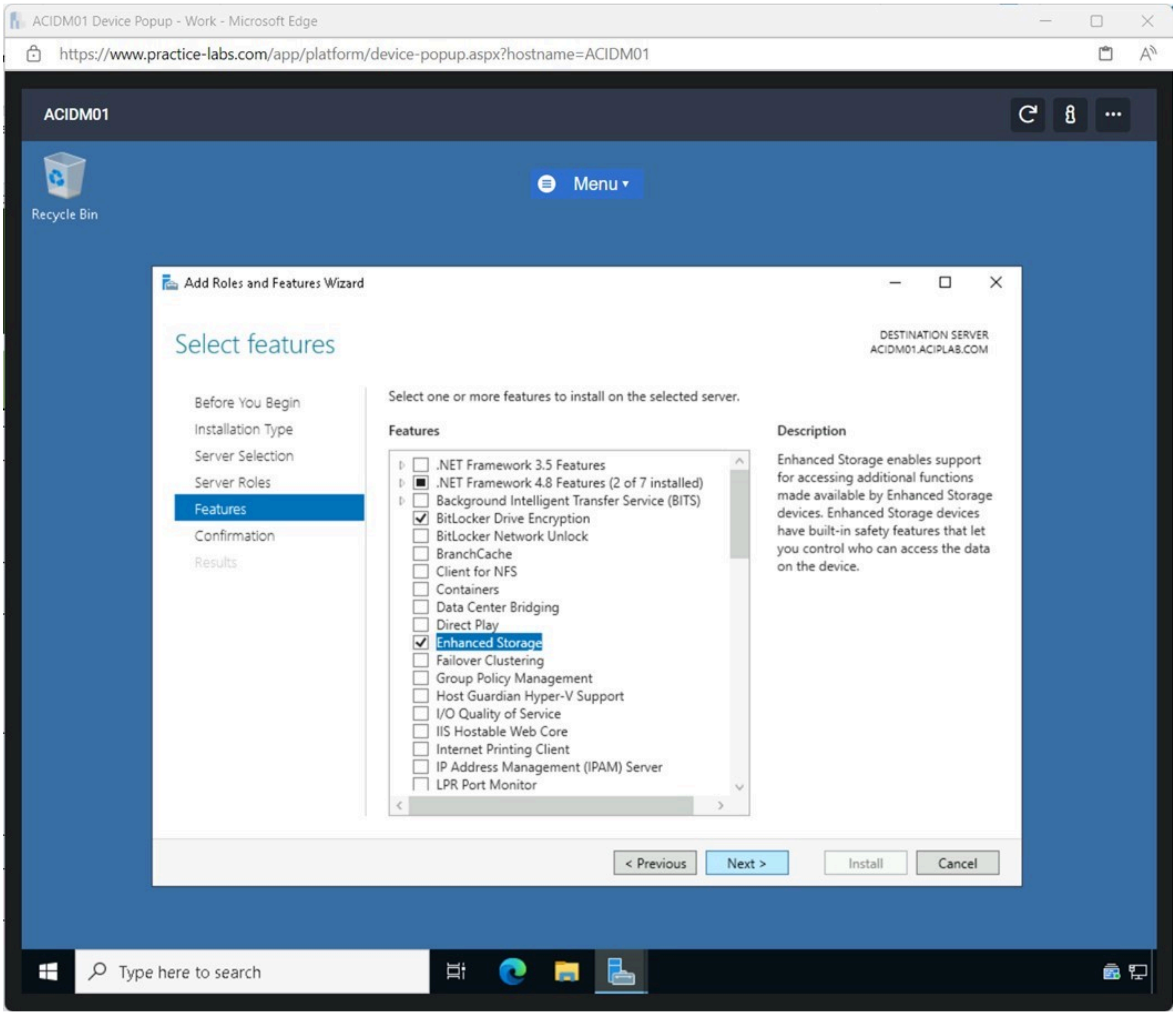


Figure 2.5 Screenshot of ACIDM01: Displaying clicking Next on the Select features pane.

Step 6

On the **Confirm installation selections** pane, click **Install**.

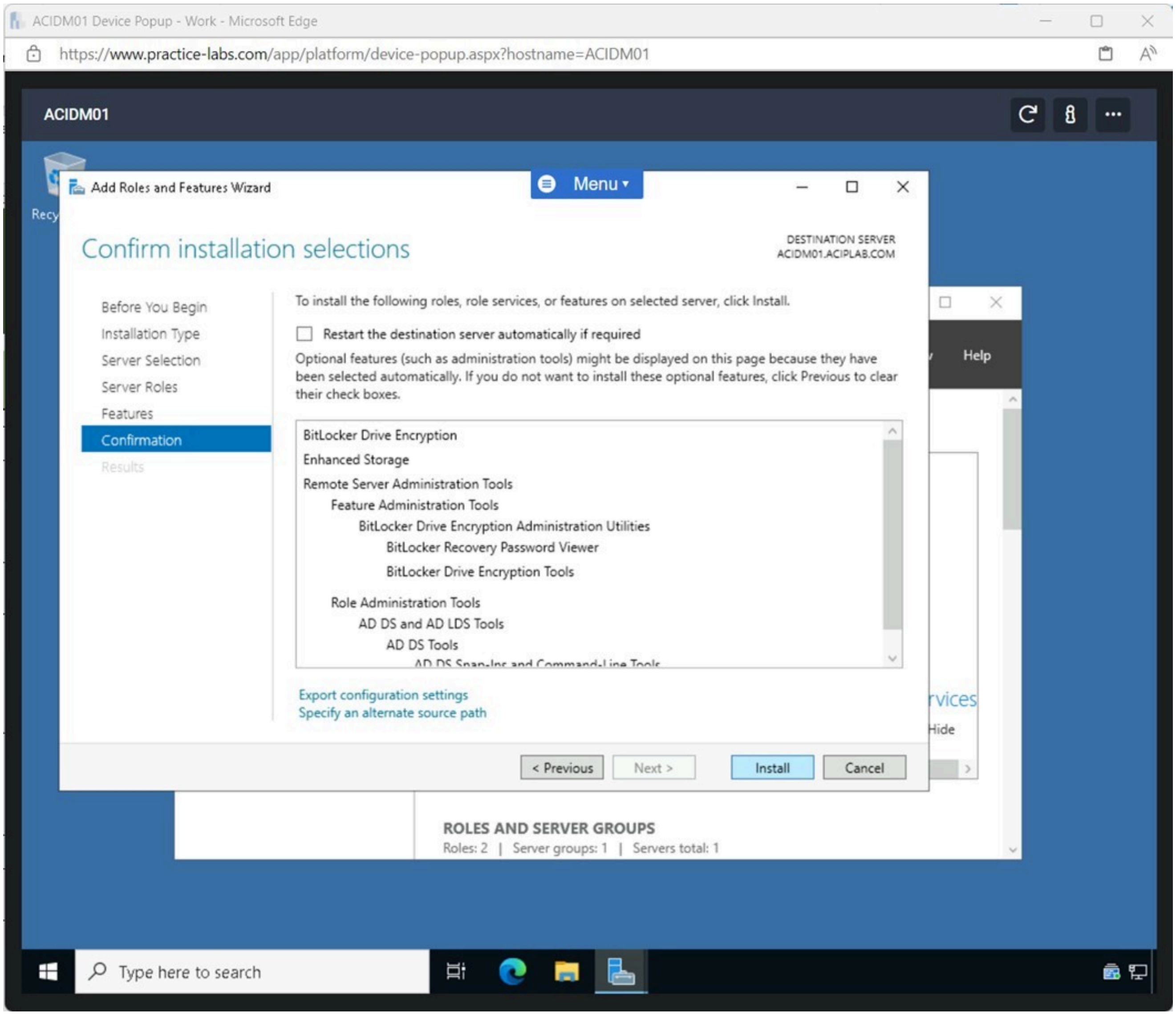


Figure 2.6 Screenshot of ACIDM01: Displaying clicking Install on the Confirm installation selections pane.

> ***Note:*** *The installation of the Bitlocker Drive Encryption feature will take a couple of minutes to complete.*

Step 7

Click **Close** after the installation has been completed.

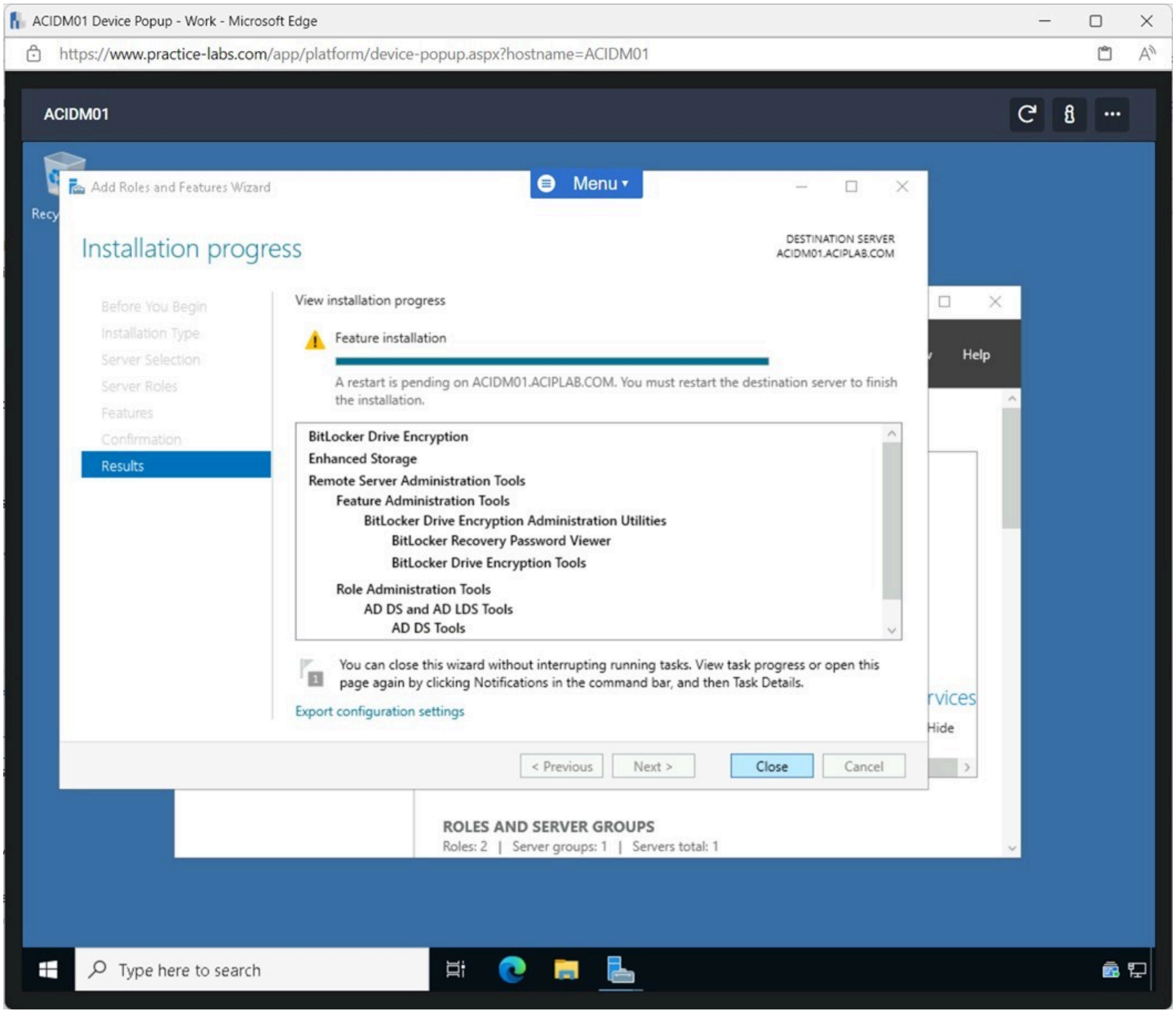


Figure 2.7 Screenshot of ACIDM01: Displaying clicking Close after the completion of the installation.

> ***Note:*** *After the feature has been installed, the* ***ACIDM01*** *device must be restarted to complete the installation.*

Step 8

Right-click **Start**, select **Shut down or sign out** and click **Restart**.

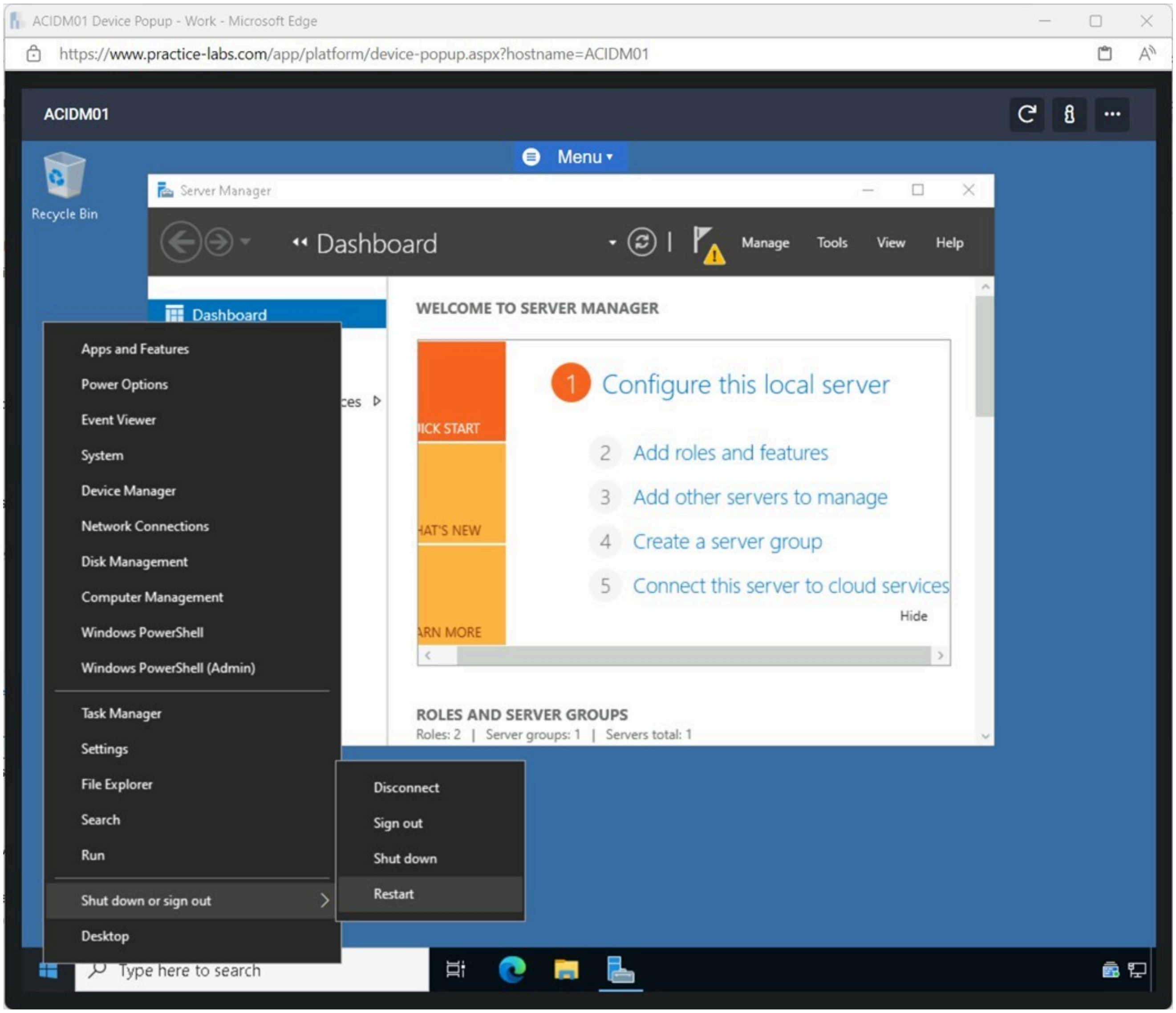


Figure 2.8 Screenshot of ACIDM01: Displaying right-clicking Start selecting Shut down or sign out and clicking Restart.

Step 9

On the **Choose a reason that best describes why you want to shut down this computer** pop-up window in the drop-down menu, select **Other (Planned)** and click **Continue**.

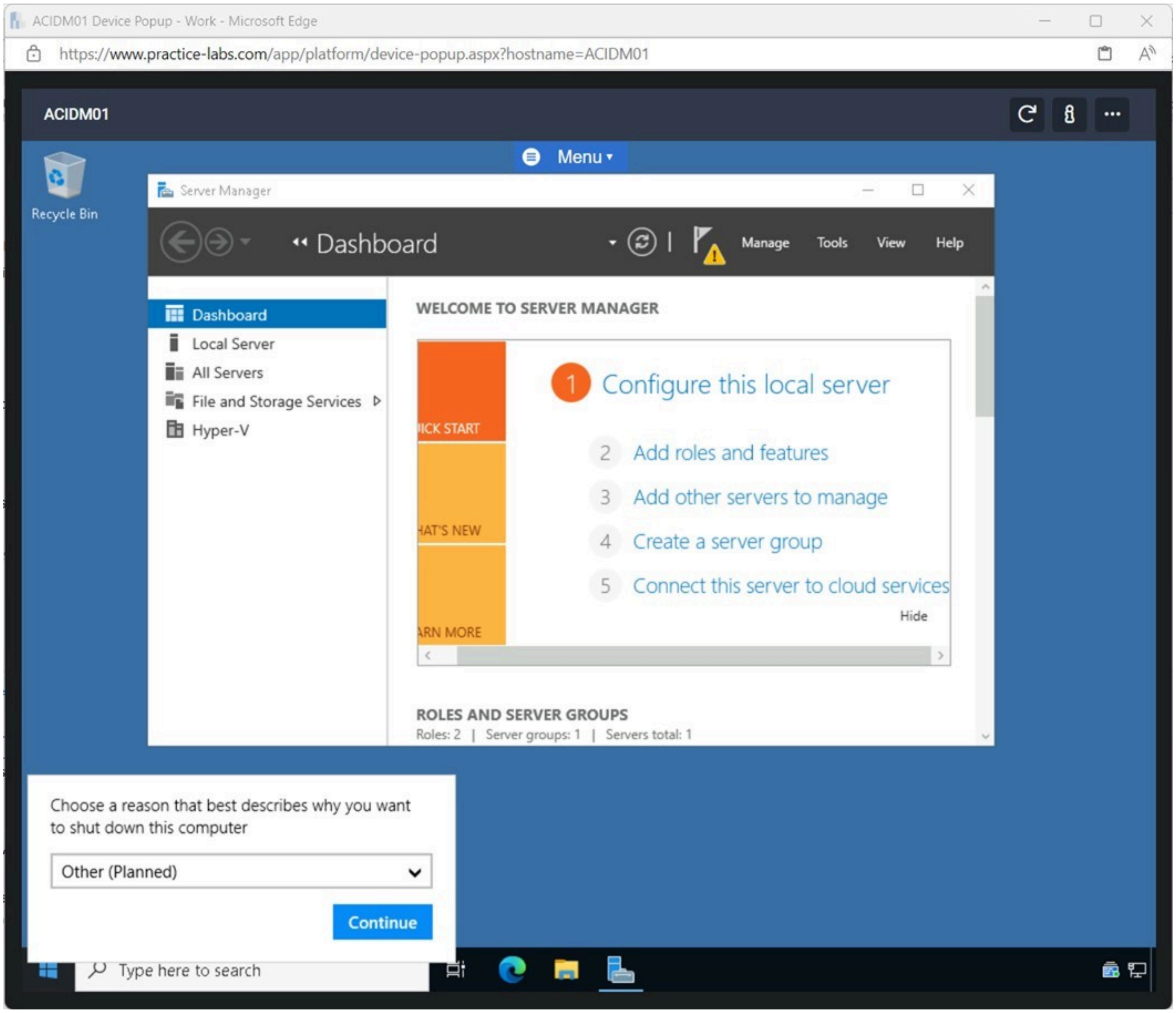


Figure 2.9 Screenshot of ACIDM01: Displaying selecting Other (Planned) in the pop-up drop-down menu and clicking Continue.

> ***Note:*** *The **ACIDM01** device will take a couple of minutes to restart to configure the installed feature.*

Task 2 - Encrypting a Local Drive on a Server

After installing the Bitlocker Drive Encryption feature, it can encrypt and secure a local drive.

In this task, a local drive will be encrypted.

Step 1

Reconnect to **ACIDM01** and open **File Explorer** from the **Taskbar**.

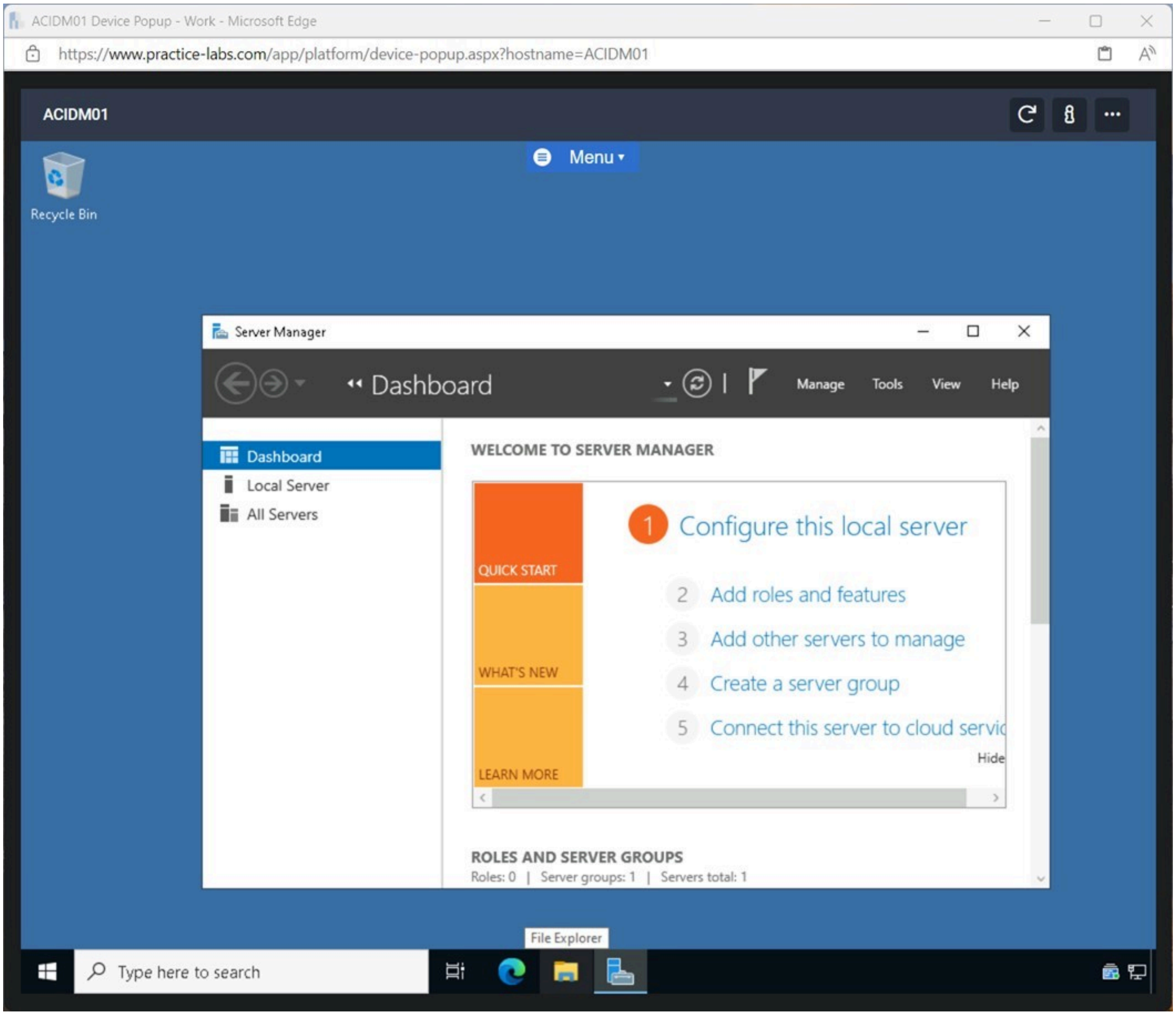


Figure 2.10 Screenshot of ACIDM01: Displaying opening File Explorer from the Taskbar.

Step 2

In **File Explorer**, expand **This PC**

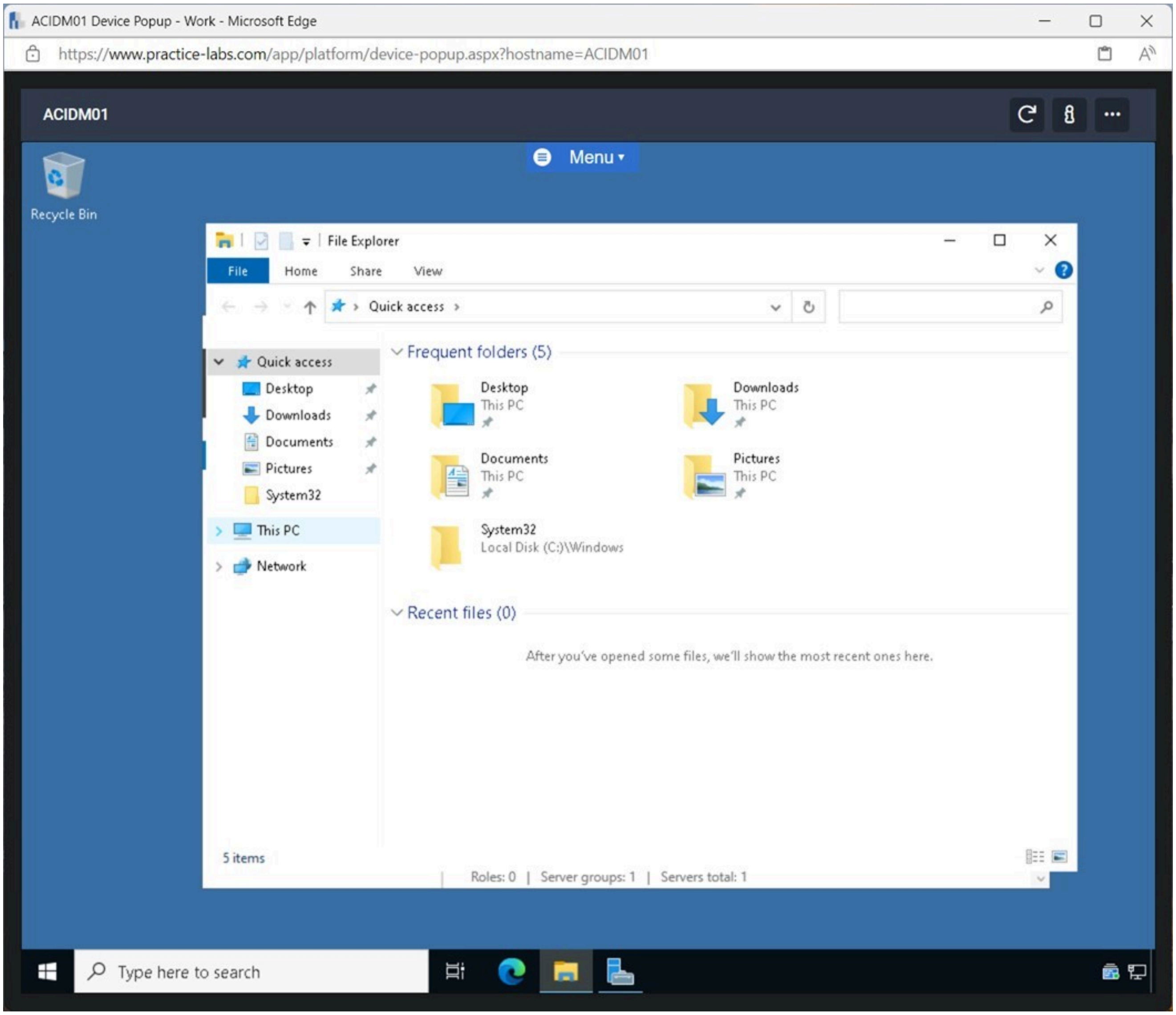


Figure 2.11 Screenshot of ACIDM01: Displaying expanding This PC in File Explorer.

Step 3

In the **This PC** pane, right-click **New Volume (D:)** and select **Turn on BitLocker**.

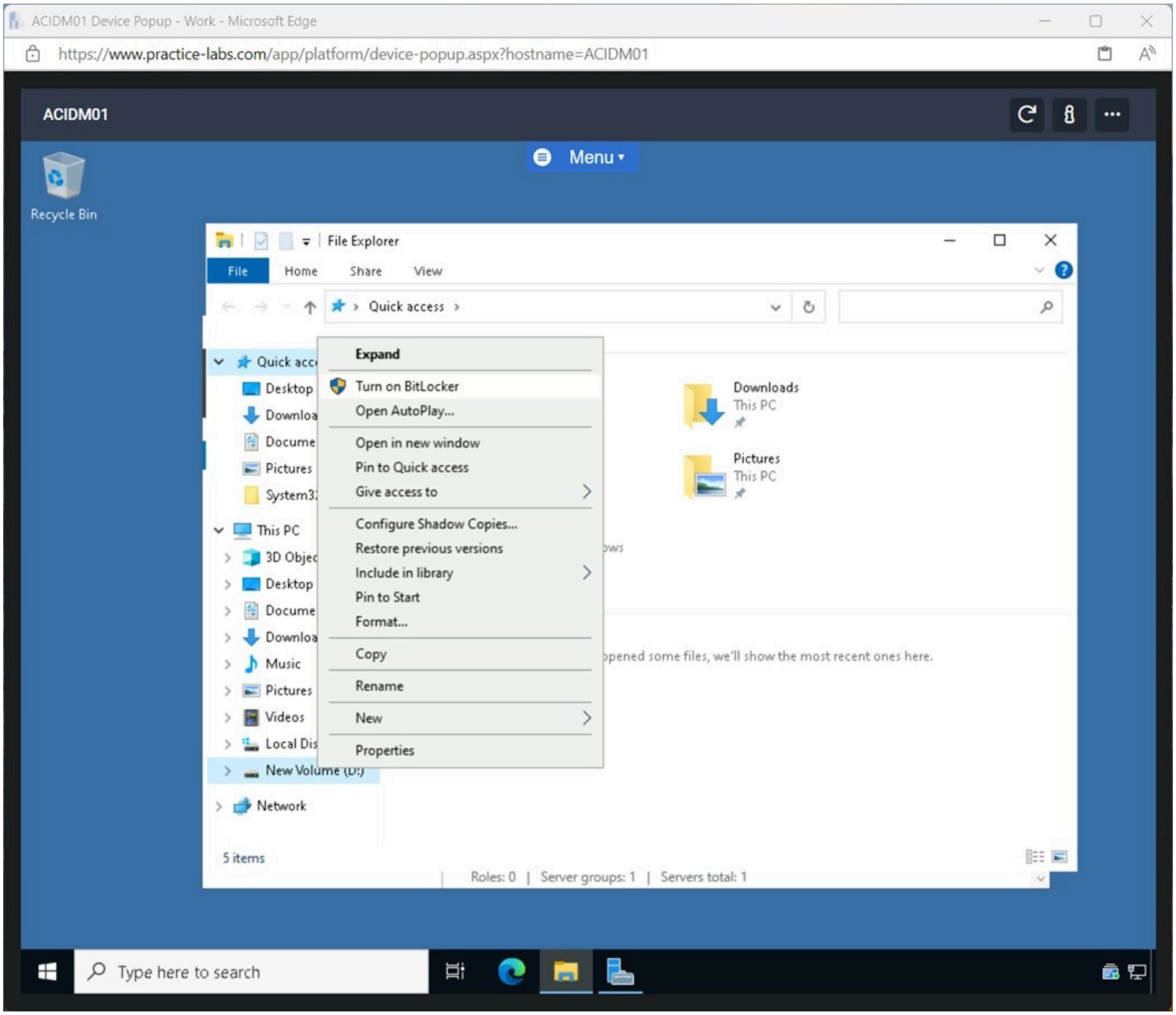


Figure 2.12 Screenshot of ACIDM01: Displaying right-clicking New Volume (D:) and selecting Turn on BitLocker.

Step 4

On the **BitLocker Drive Encryption** window, tick the **Use a password to unlock this drive** tickbox and enter the following:

```
Enter your password:    Passw0rd
Reenter your password:  Passw0rd
```

Click **Next**.

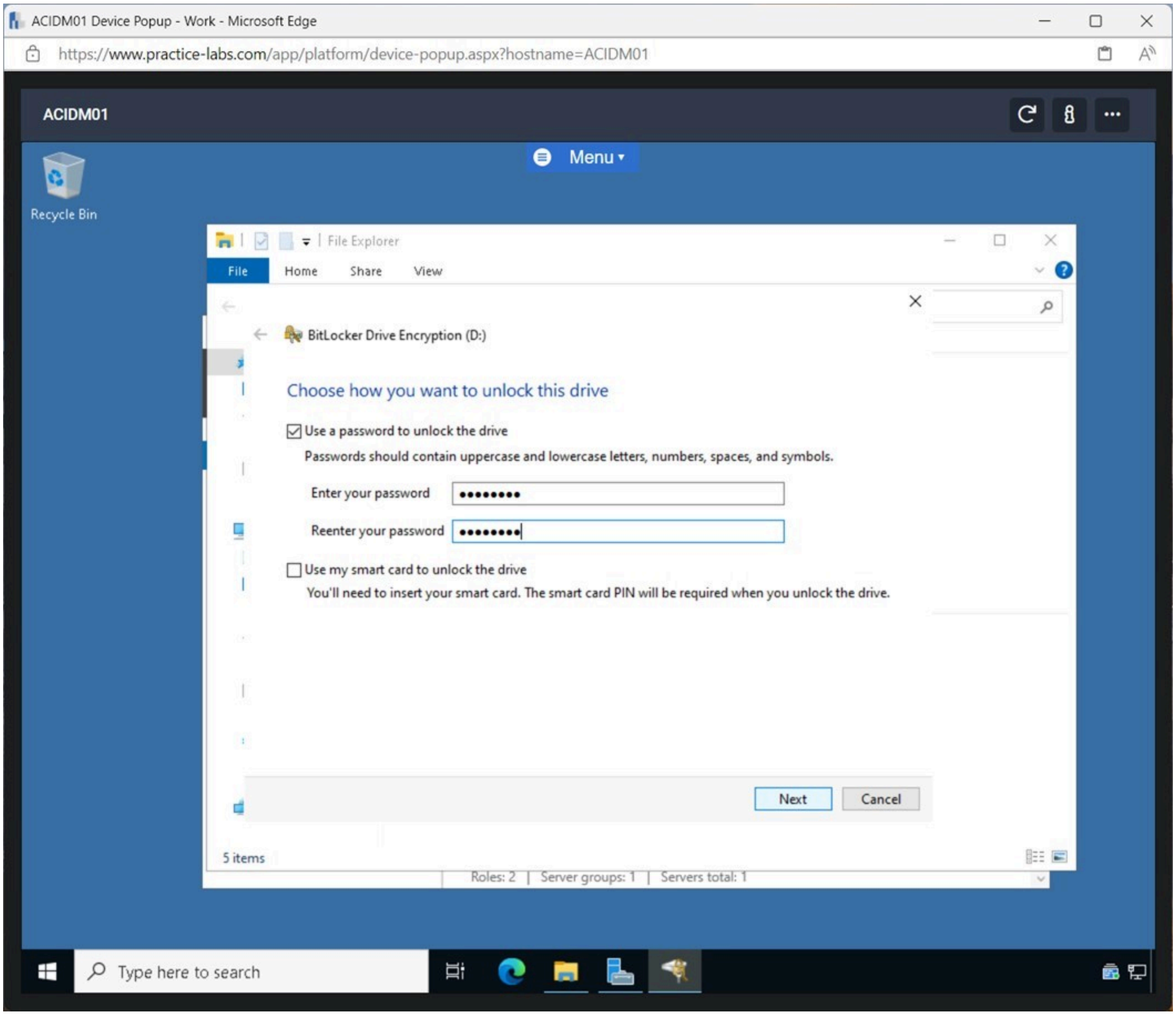


Figure 2.13 Screenshot of ACIDM01: Displaying ticking the Use a password to unlock the drive, entering the password in the required fields and clicking Next.

Step 5

On the **How do you want to back up your recovery key?** window, select **Save to a file**.

In the **Save BitLocker recovery key as** pop-up window, select **Documents** and click **Save**.

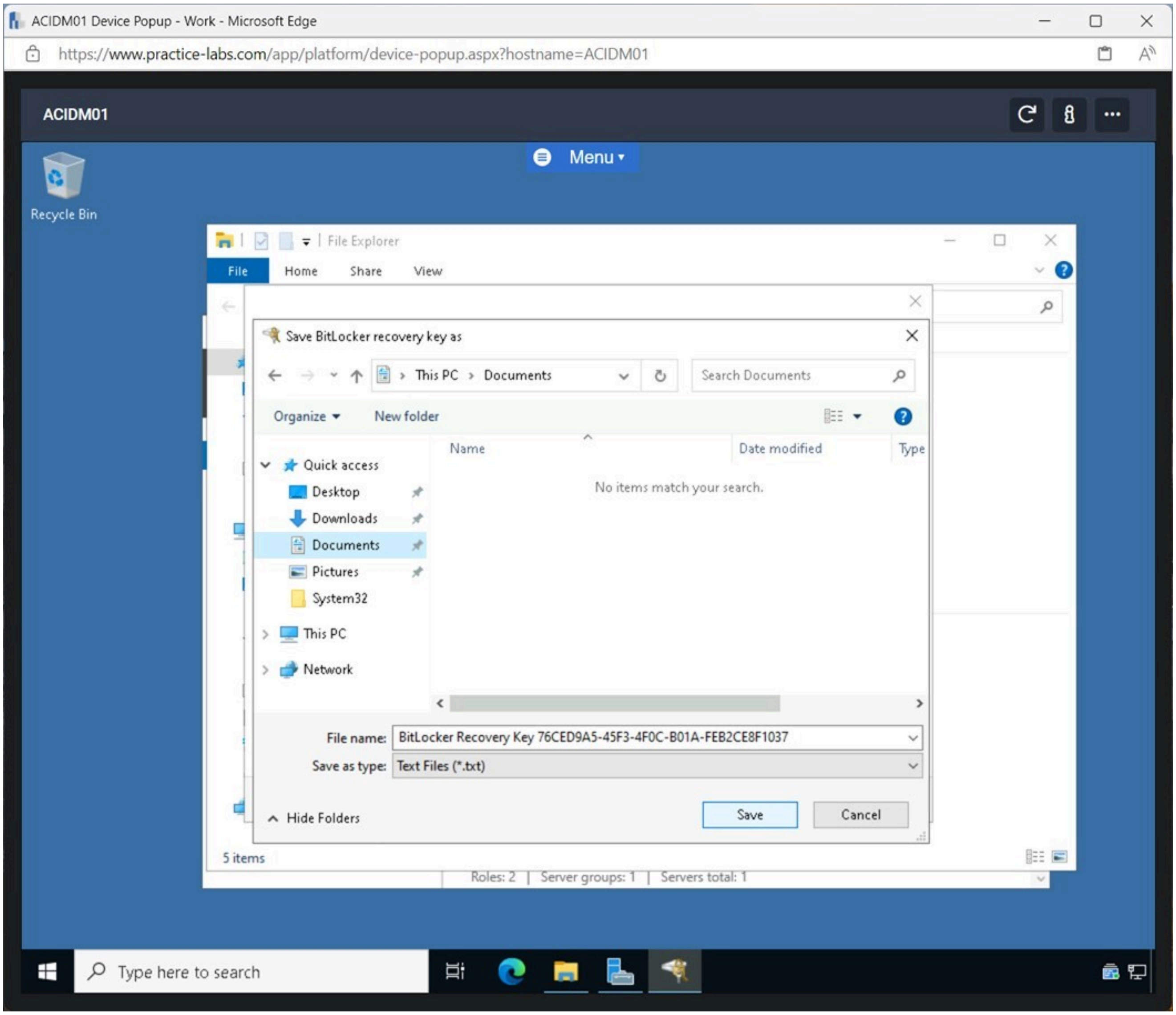


Figure 2.14 Screenshot of ACIDM01: Displaying clicking Save to a file, selecting Documents in the pop-up window and clicking Save.

Step 6

Click **Next** on the **How do you want to back up your recovery key?** pane.

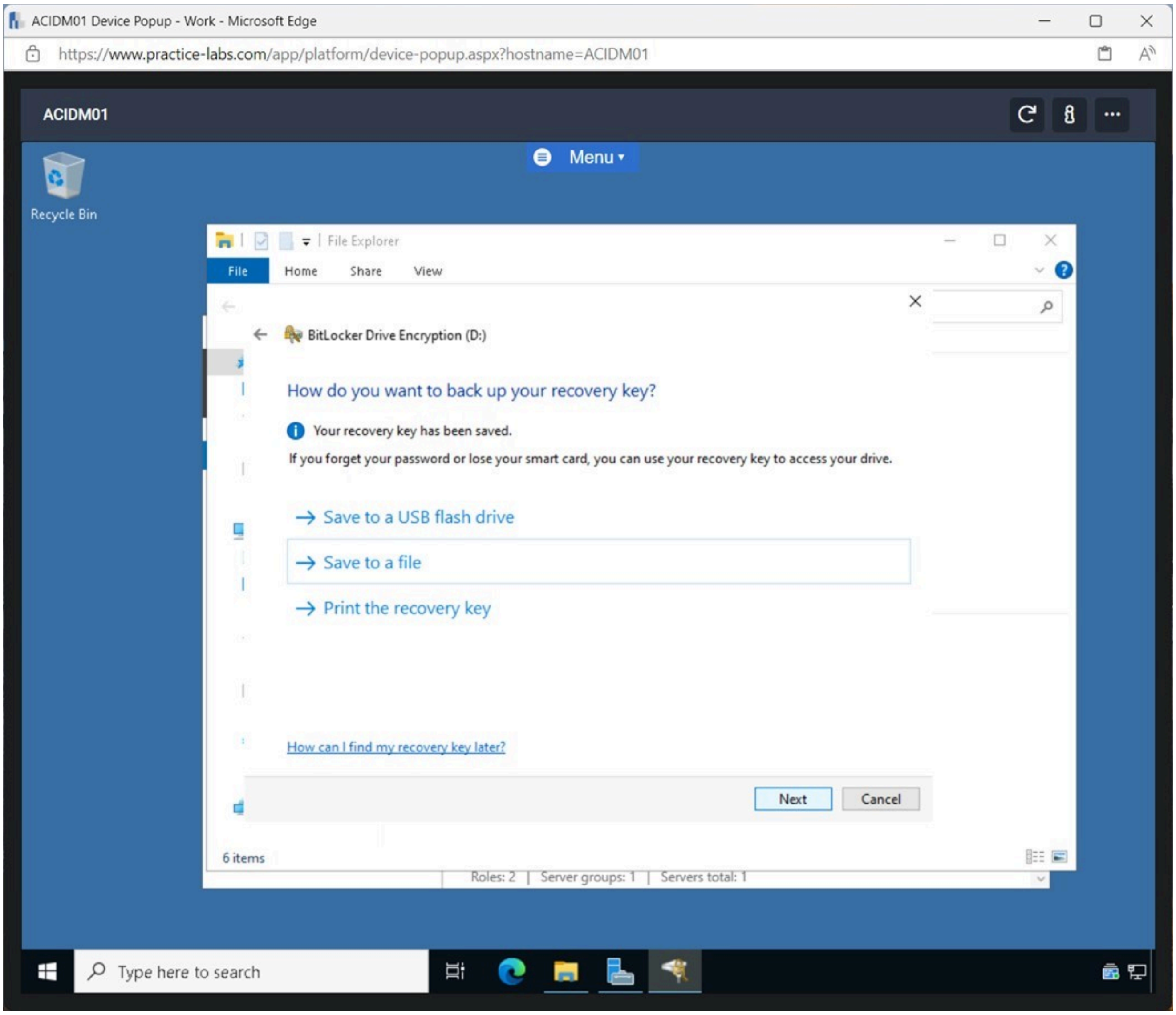


Figure 2.15 Screenshot of ACIDM01: Displaying clicking Next on the How do you want to back up your recovery key? pane.

> ***Note:*** *In a real-world scenario, the recovery key will not be stored on the local machine but would rather be saved to a secure network location or to a USB flash drive. This will ensure that the recovery key will be secure and accessible.*

Step 7

Click **Next** on the **Choose how much of your drive to encrypt** pane.

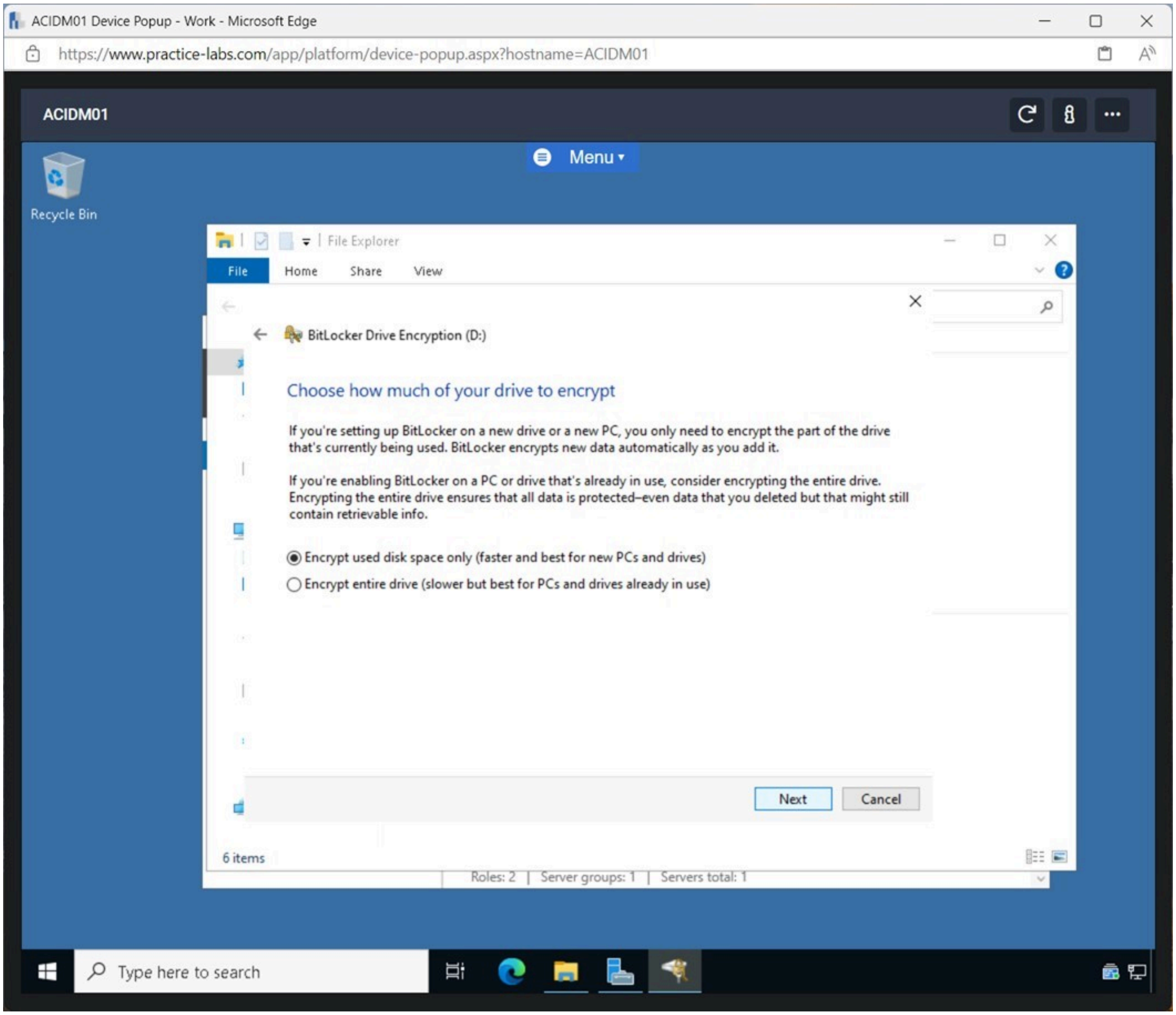


Figure 2.16 Screenshot of ACIDM01: Displaying clicking Next on the Choose how much of your drive to encrypt pane.

> ***Note:*** *As this is an empty drive, the default selection is used. In a real-world scenario, if the drive already had data on it that needs to be encrypted, the second option needs to be selected.*

Step 8

Click **Next** on the **Choose which encryption mode to use** pane.

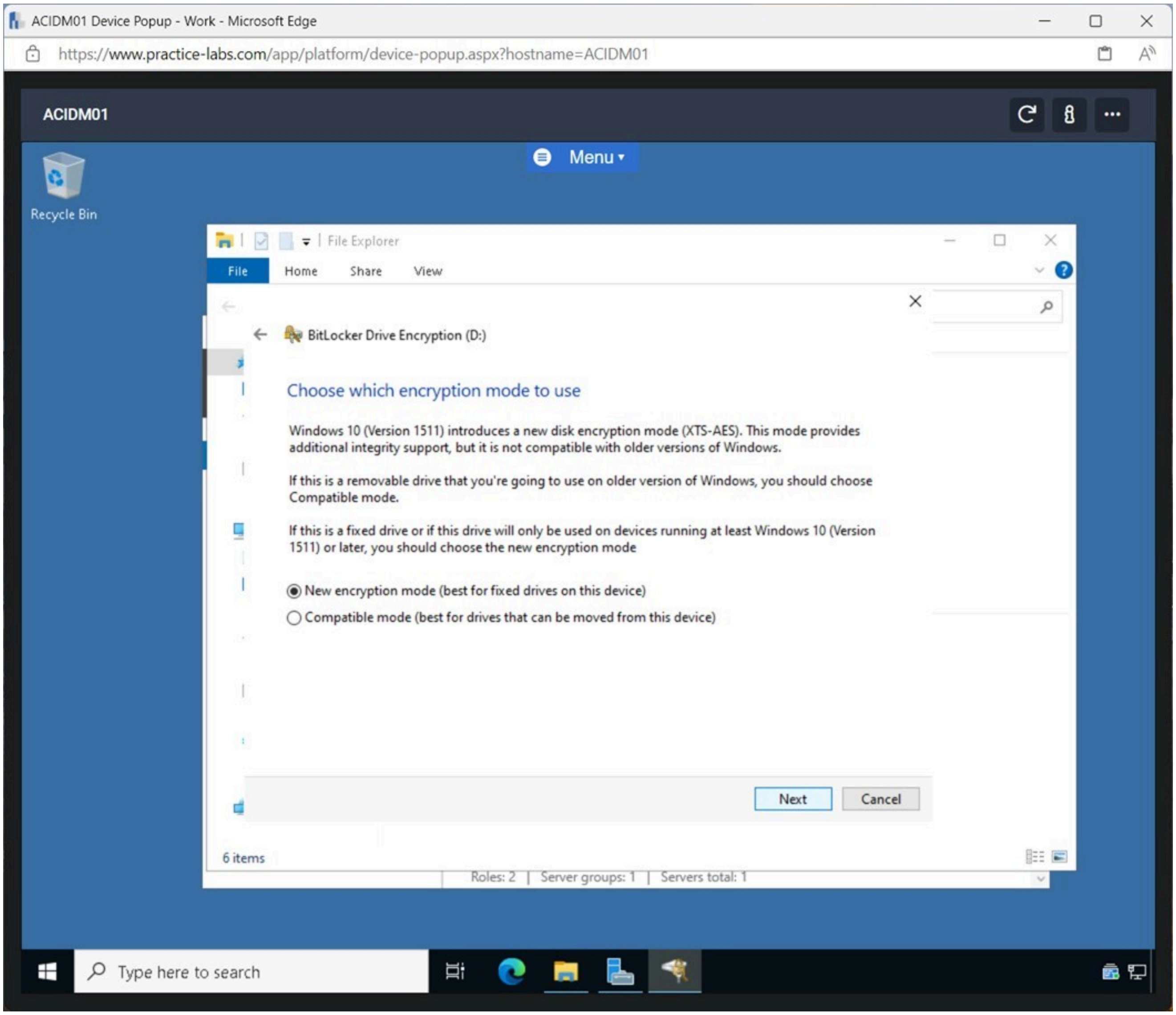


Figure 2.17 Screenshot of ACIDM01: Displaying clicking Next on the Choose which encryption method to use pane.

> ***Note:*** *Depending on the type of drive that will be encrypted, a different option can be selected. In this case, a fixed drive will be encrypted.*

Step 9

On the **Are you ready to encrypt this drive?** pane click **Start encrypting**.

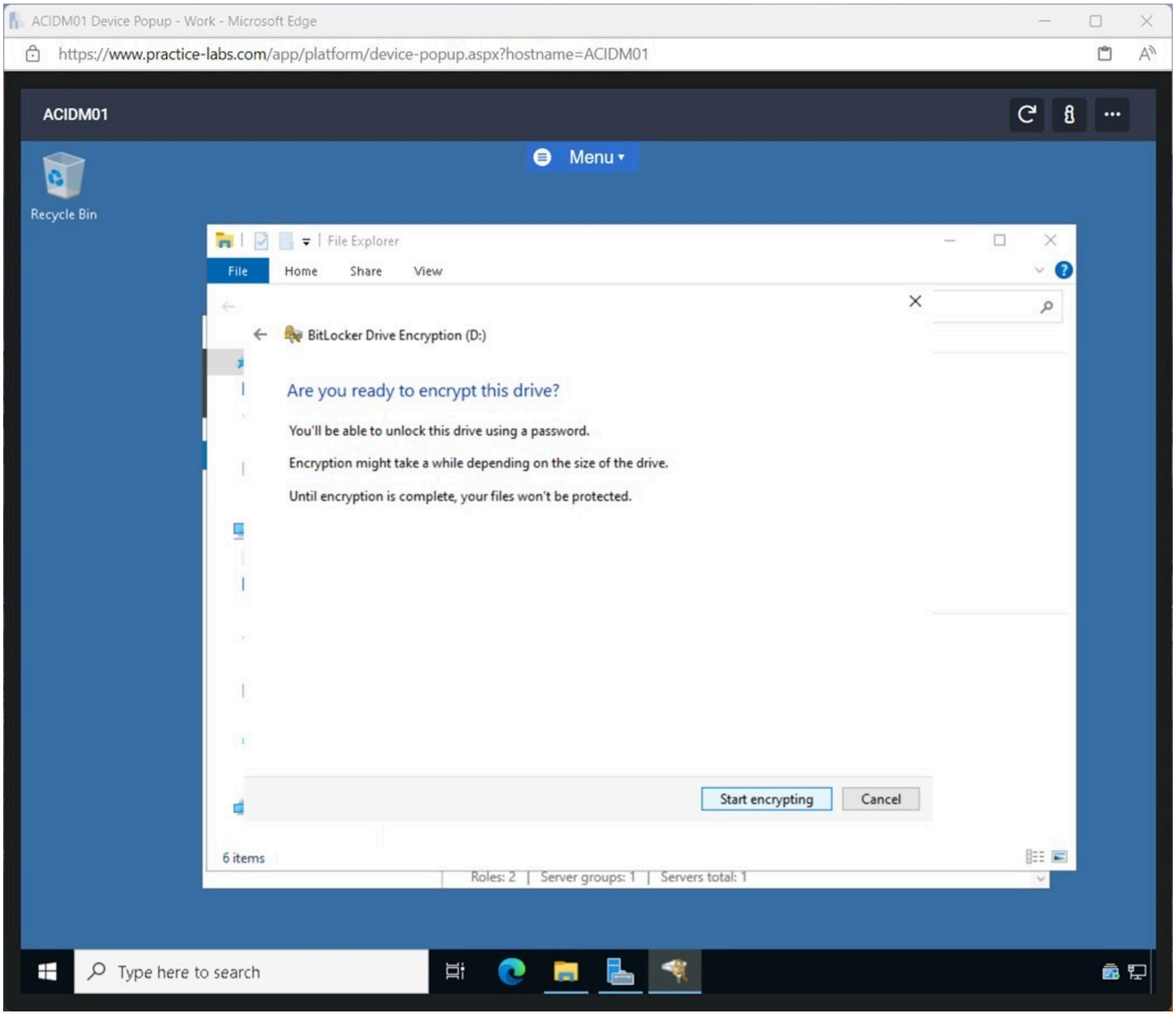


Figure 2.18 Screenshot of ACIDM01: Displaying clicking Start encrypting on the Are you ready to encrypt this drive? pane.

Step 10

Close **File Explorer**.

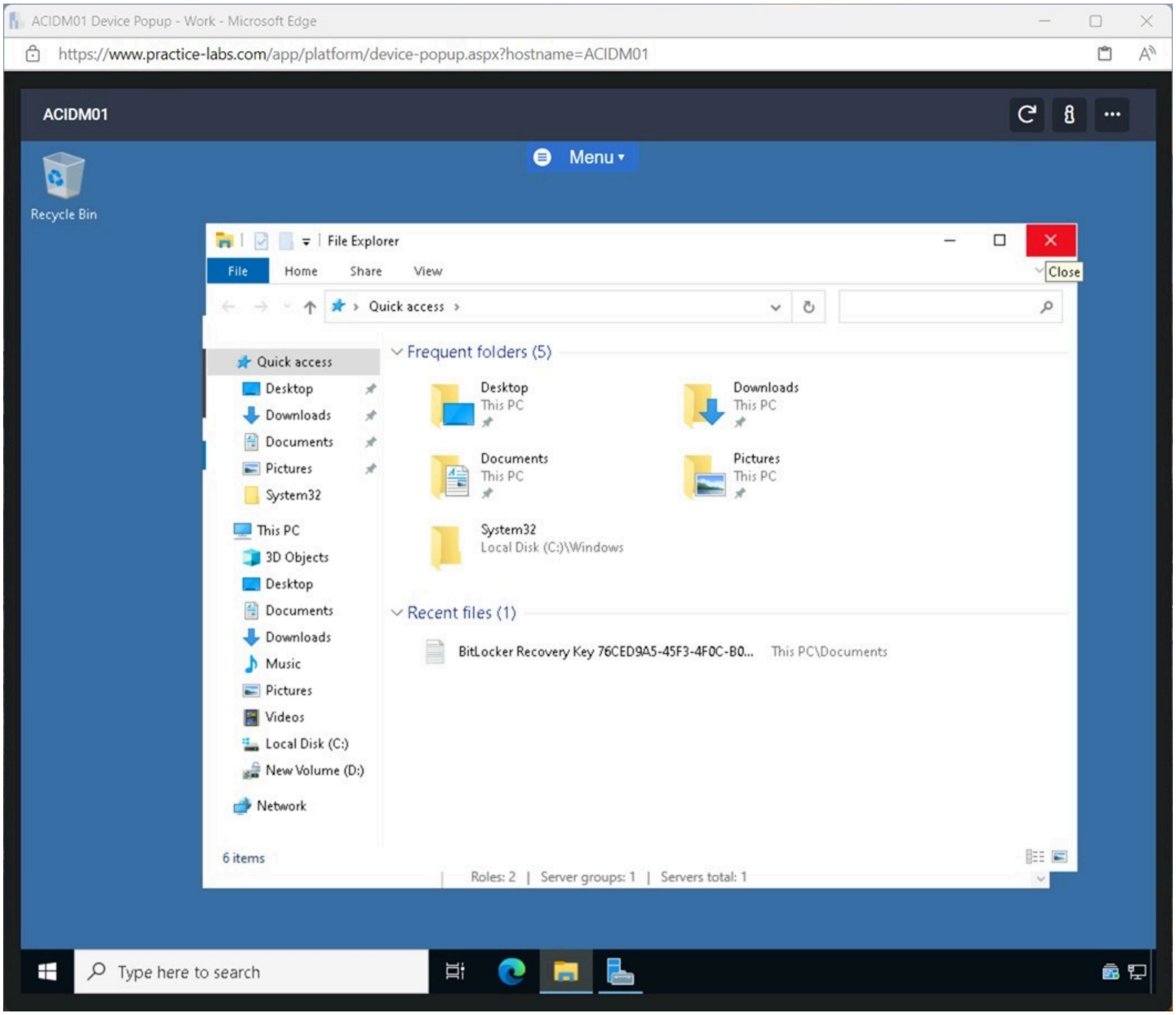


Figure 2.19 Screenshot of ACIDM01: Displaying closing File Explorer.

> ***Note:*** *In File Explorer, the New Volume (D:) icon has changed to a lock, indicating that the drive has successfully been encrypted.*

Exercise 3 - Enable Multifactor Authentication

Multifactor Authentication can be enabled to add an additional layer of security to protect user accounts. Several different methods can be used to implement multifactor authentication. These include smart cards, using an authentication application, or a secure USB key.

Multifactor Authentication is used in conjunction with the user's password to ensure a secure login.

In this exercise, a user will be created, and multifactor authentication will be enabled.

Learning Outcomes

After completing this exercise, you should be able to:

- Enable Multifactor Authentication
- Verify Multifactor Authentication

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01 -** Windows Server 2022 - Domain Controller
- **ACIWIN11 -** Windows 11 - Domain Member Workstation

Task 1 - Enable Multifactor Authentication

User accounts in Active Directory can be secured by enabling multifactor authentication for the accounts.

In this task, a user will be created, and Multifactor Authentication will be enabled.

Step 1

Connect to **ACIDC01** in **Server Manager**, click **Tools** and select **Active Directory Users and Computers**.

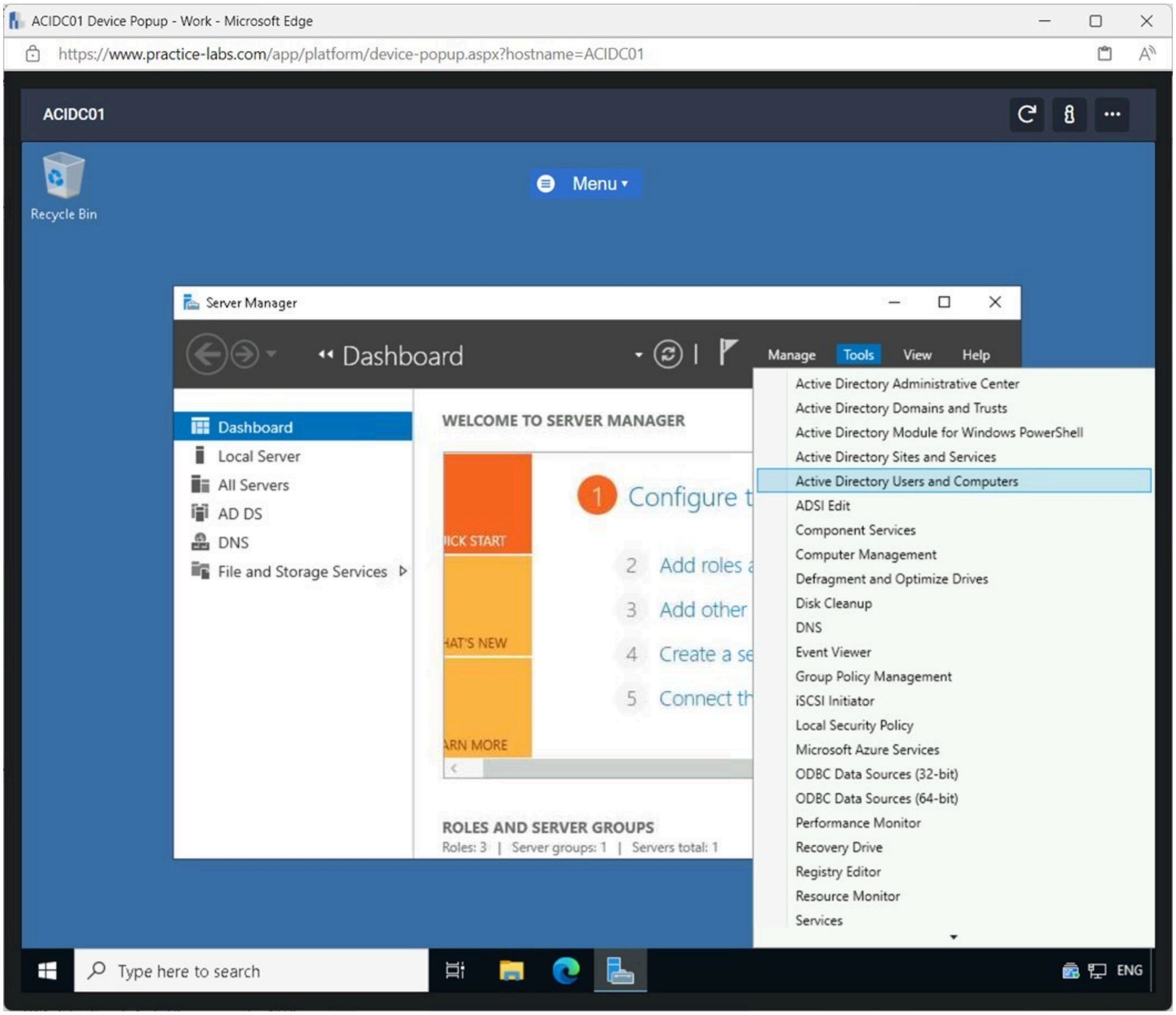


Figure 3.1 Screenshot of ACIDC01: Displaying clicking Tools in Server Manager and selecting Active Directory Users and Computers.

Step 2

In the **Active Directory Users and Computers** pane on the left expand **ACIPLAB.COM** if it's not expanded. Right-click **Users**, select **New** and click **User**.

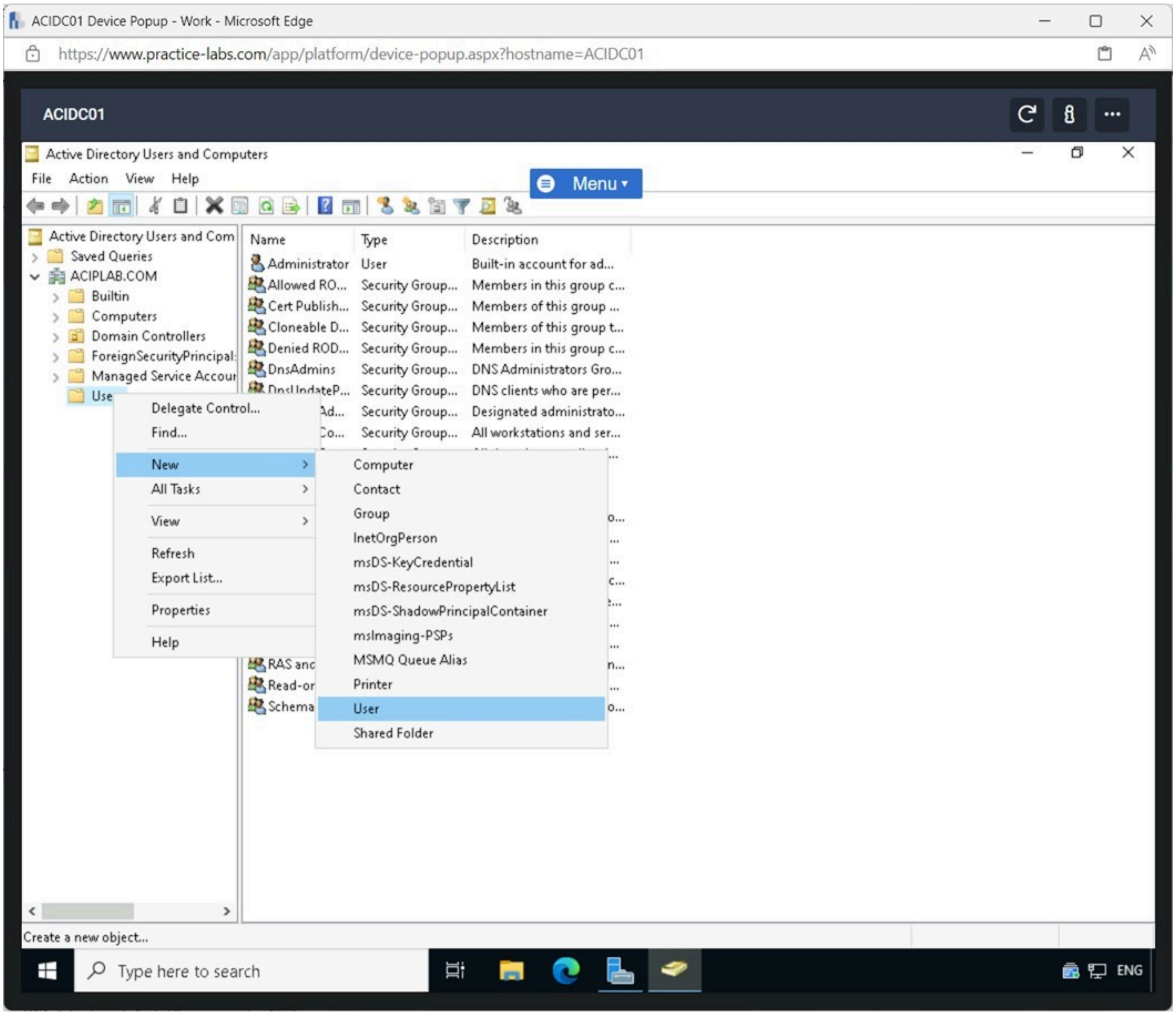


Figure 3.2 Screenshot of ACIDC01: Displaying right-clicking Users selecting New and clicking User.

Step 3

On the **New Object - User** complete the following:

```
First name: Subi
User logon name: Subi
```

Click **Next**.

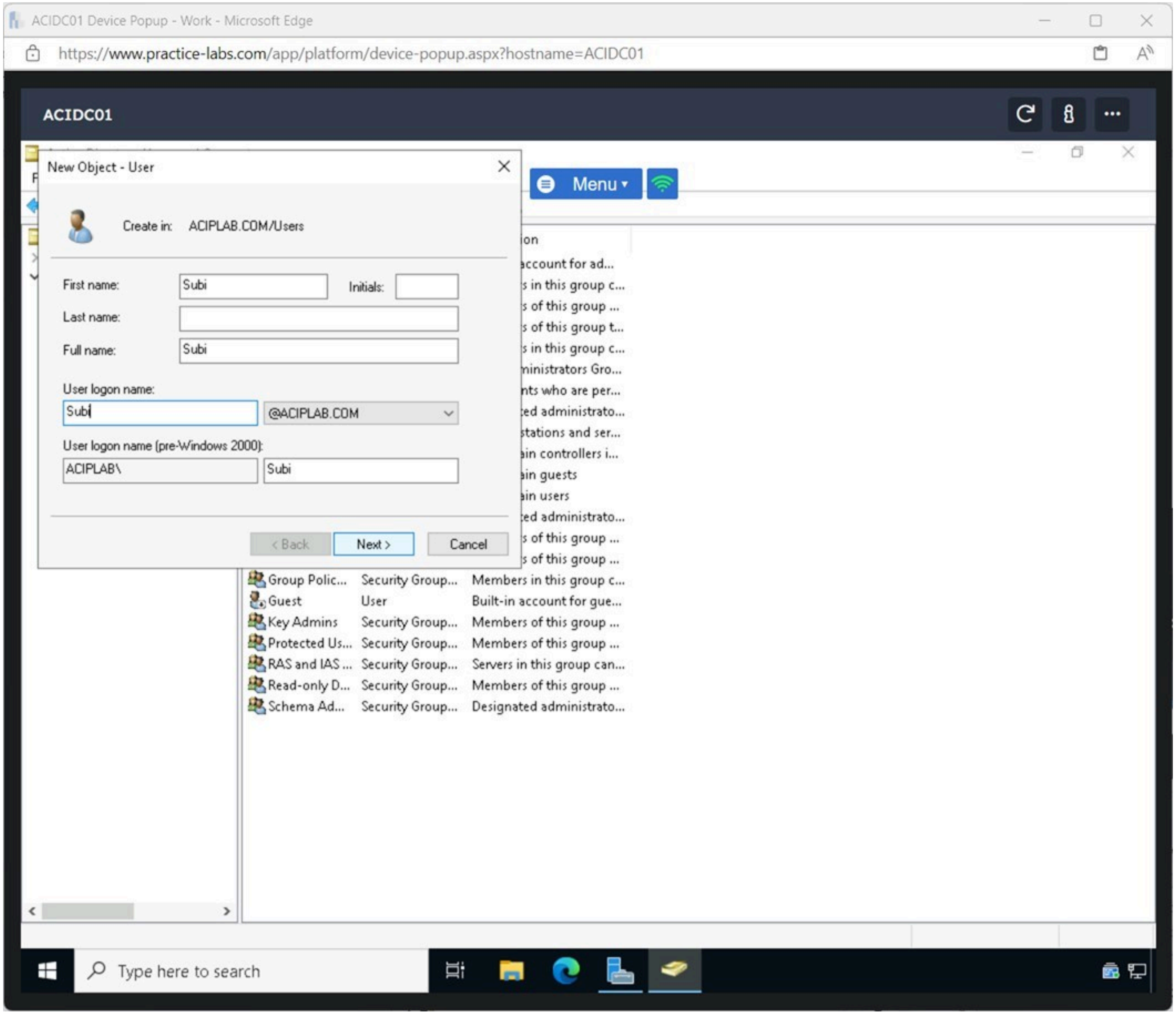


Figure 3.3 Screenshot of ACIDC01: Displaying completing the specified fields in the New Object - User window and clicking Next.

Step 4

On the next screen, complete the following fields:

```
Password:        Passw0rd
Confirm password:   Passw0rd
```

Untick **User must change password at next logon** checkbox

Click **Next**.

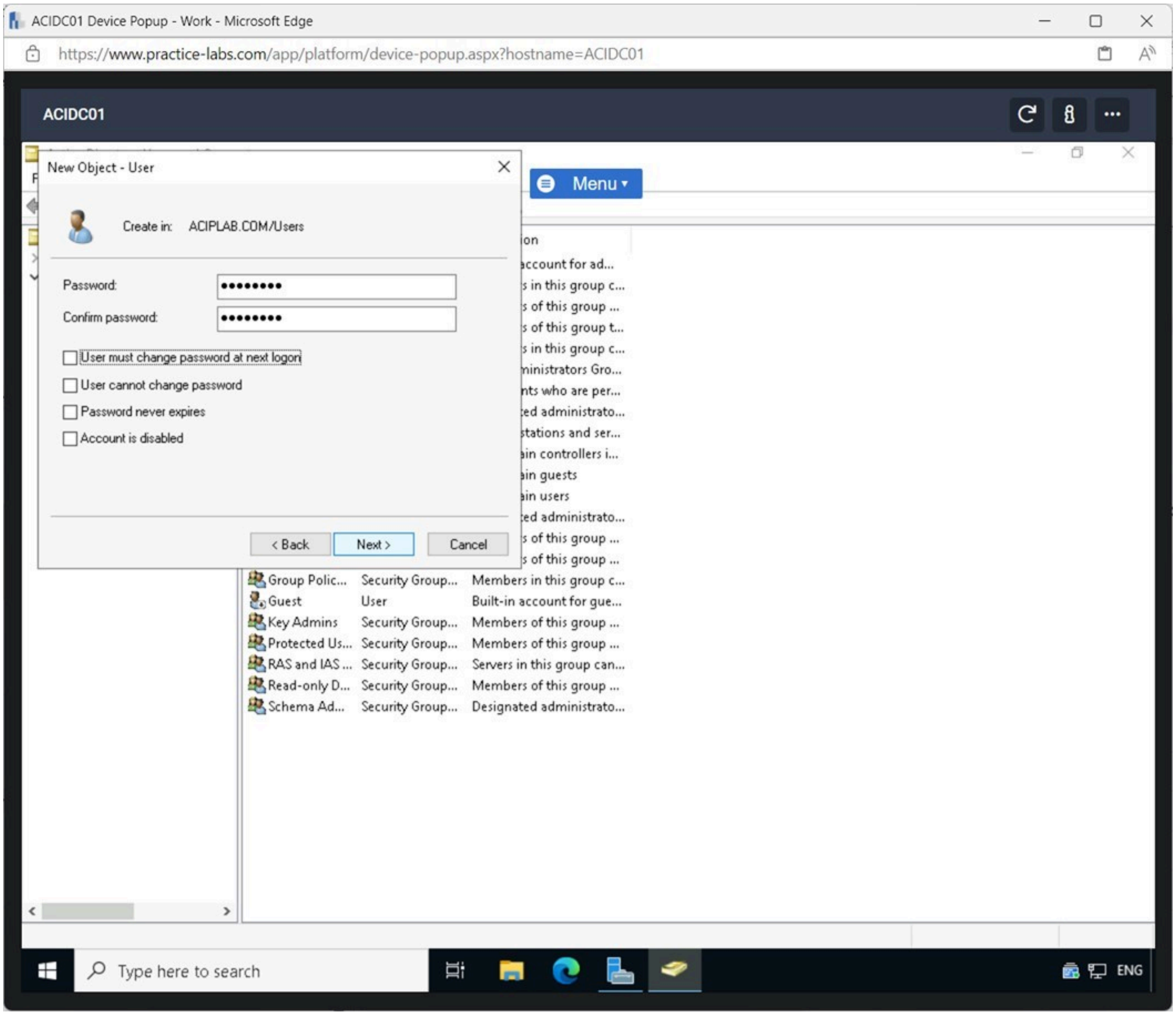


Figure 3.4 Screenshot of ACIDC01: Displaying entering the password unticking the tickbox, and clicking Next.

Step 5

Click **Finish** on the **New Object - User** window.

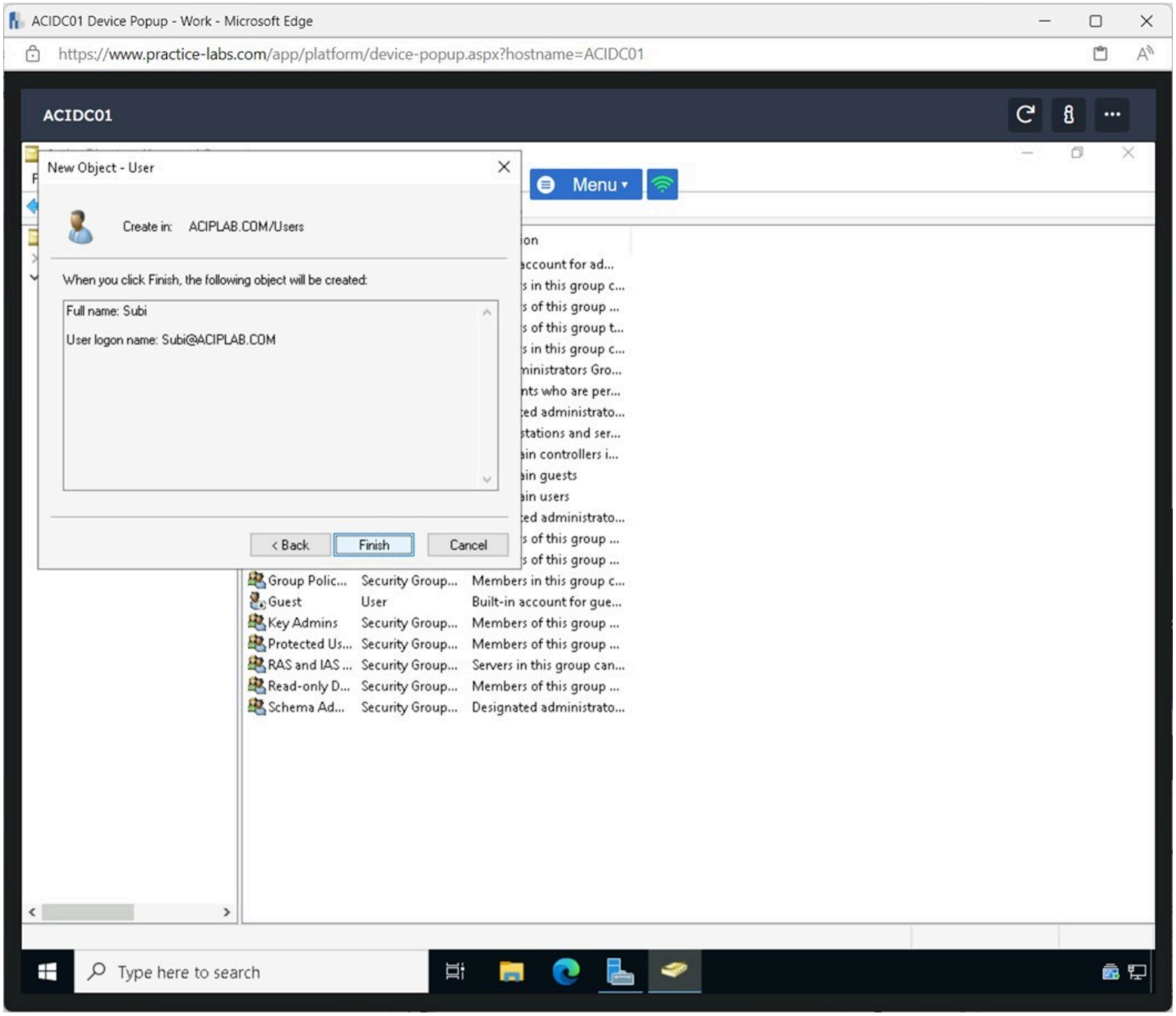


Figure 3.5 Screenshot of ACIDC01: Displaying clicking Finish on the New Object - User window.

Step 6

Double-click the **Users** folder on the left pane. Right-click **Subi** and select **Properties** in the **Active Directory Users and Computers** window.

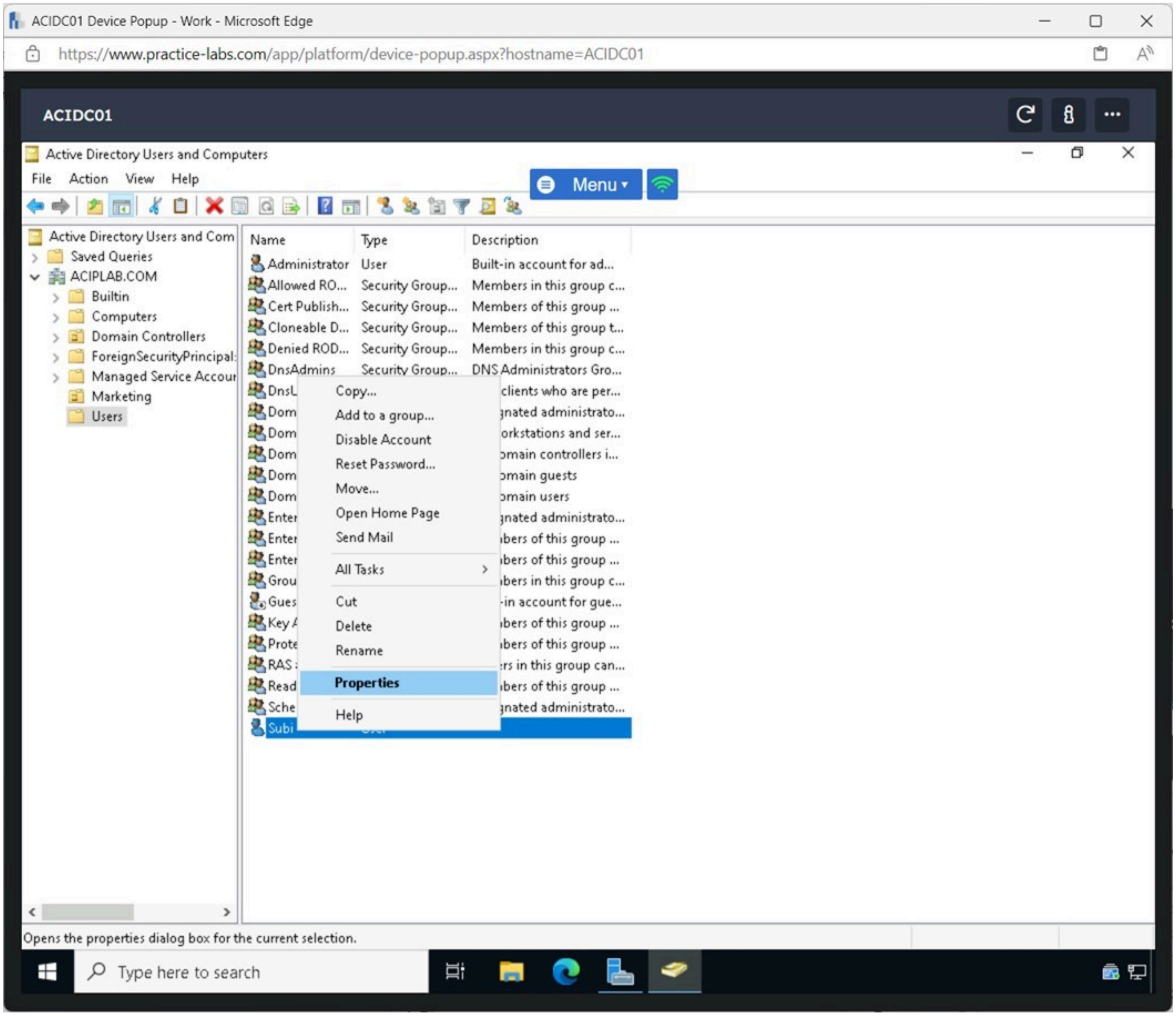


Figure 3.6 Screenshot of ACIDC01: Displaying right-clicking Subi and selecting Properties.

Step 7

On the **Subi Properties** window, select the **Account** tab.

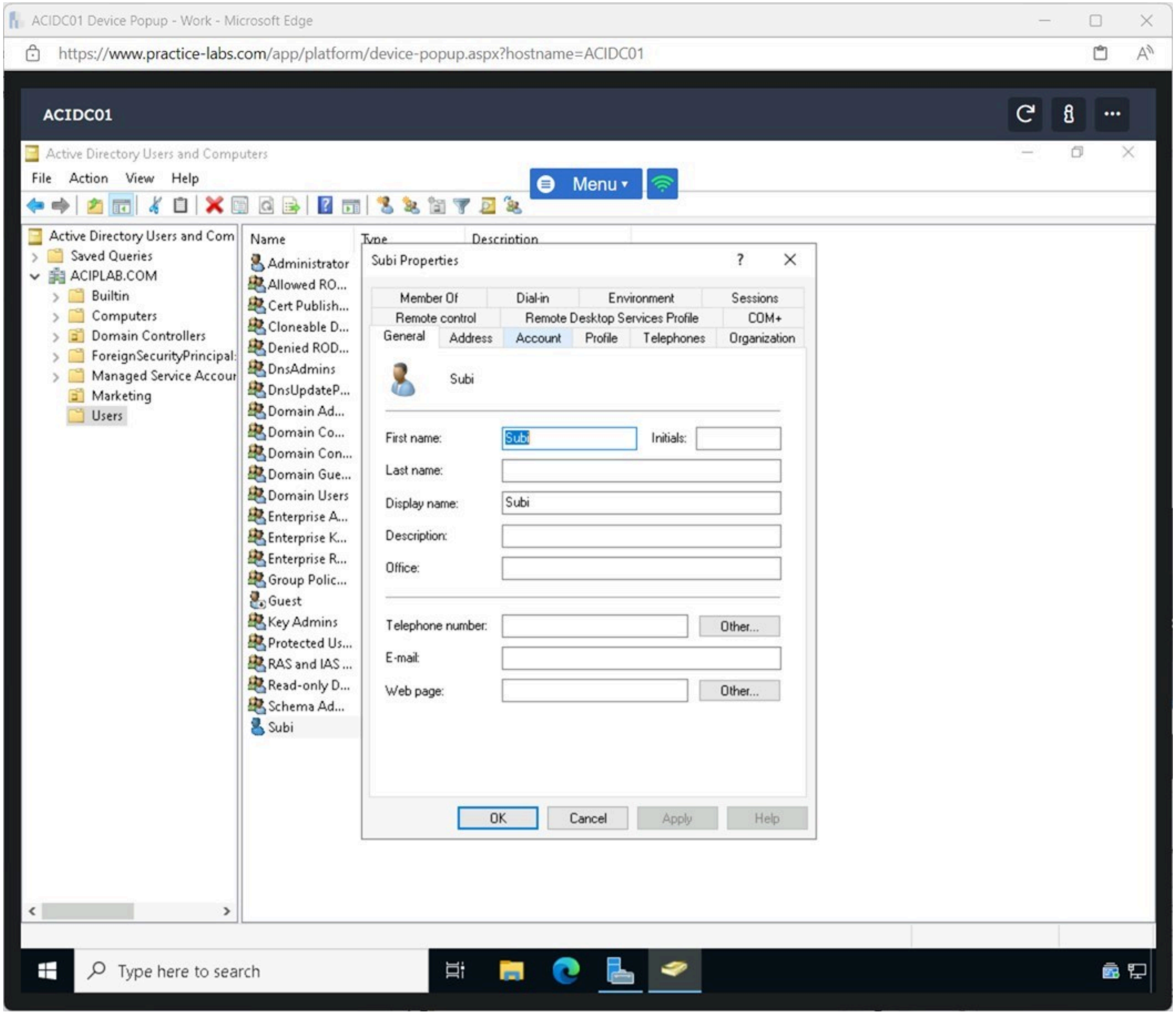


Figure 3.7 Screenshot of ACIDC01: Displaying selecting the Account tab in the Subi Properties window.

Step 8

On the **Properties Properties** window in the **Account options** pane, scroll down and tick the **Smart card is required for interactive logon** tickbox.

Click **OK**.

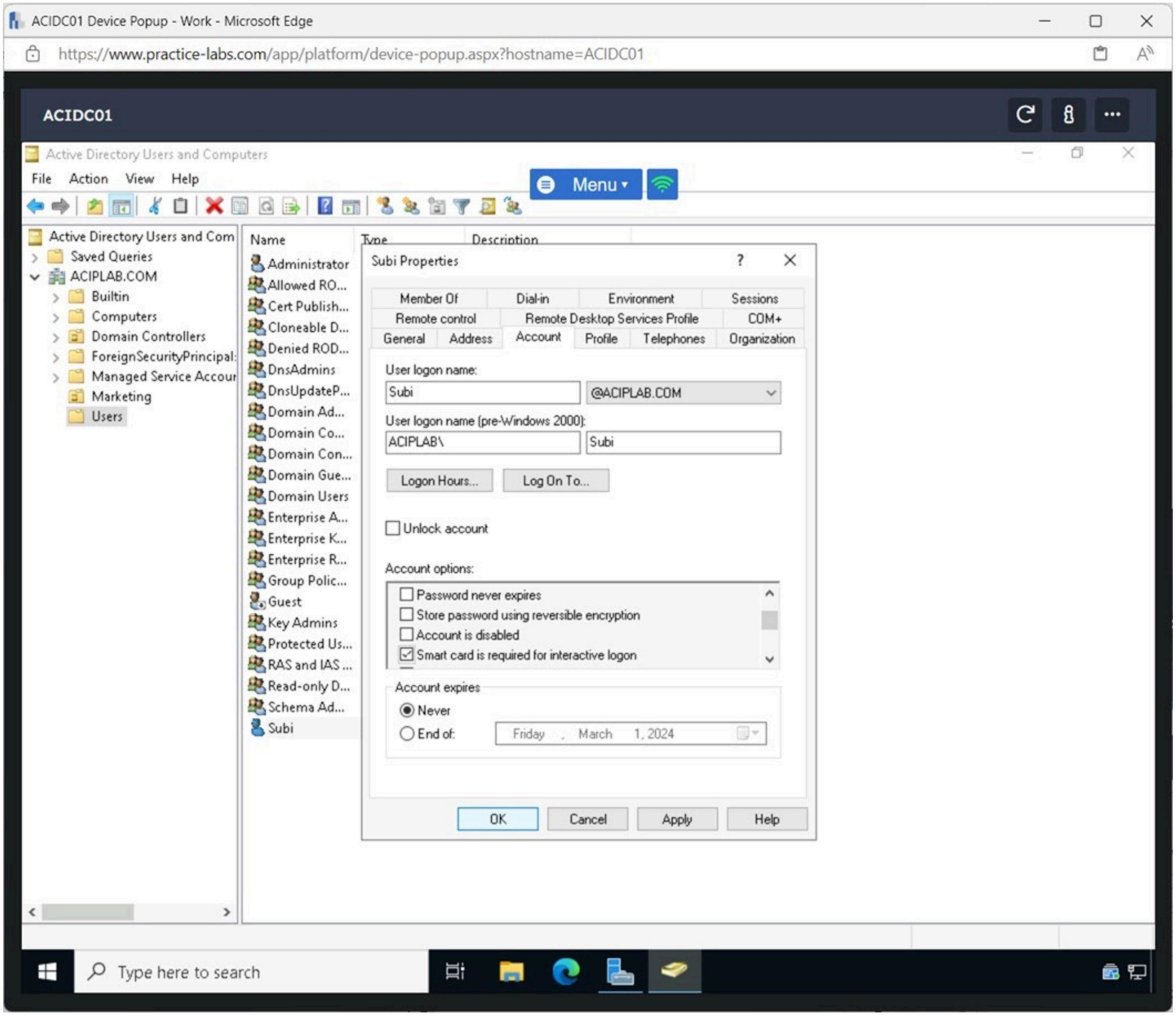
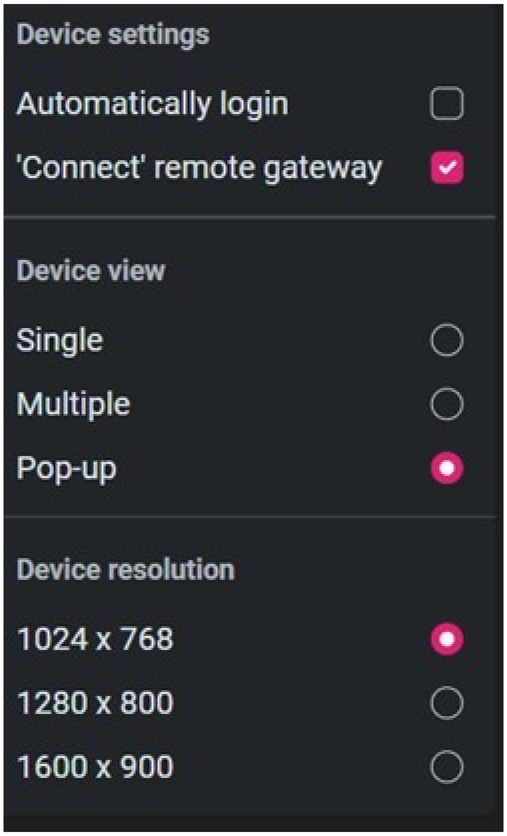


Figure 3.8 Screenshot of ACIDC01: Displaying enabling the Smart card is required for interactive logon option and clicking OK.

Task 2 - Verifying Multifactor Authentication

In this task, Multifactor Authentication for the newly created user will be verified.

> **Alert:** Before continuing with this task, the lab platform's automatic login feature must be disabled. This can be achieved by clicking the Settings cog and unticking the tickbox.



Step 1

Connect to **ACIWIN11**, and click **Other user** on the login screen.

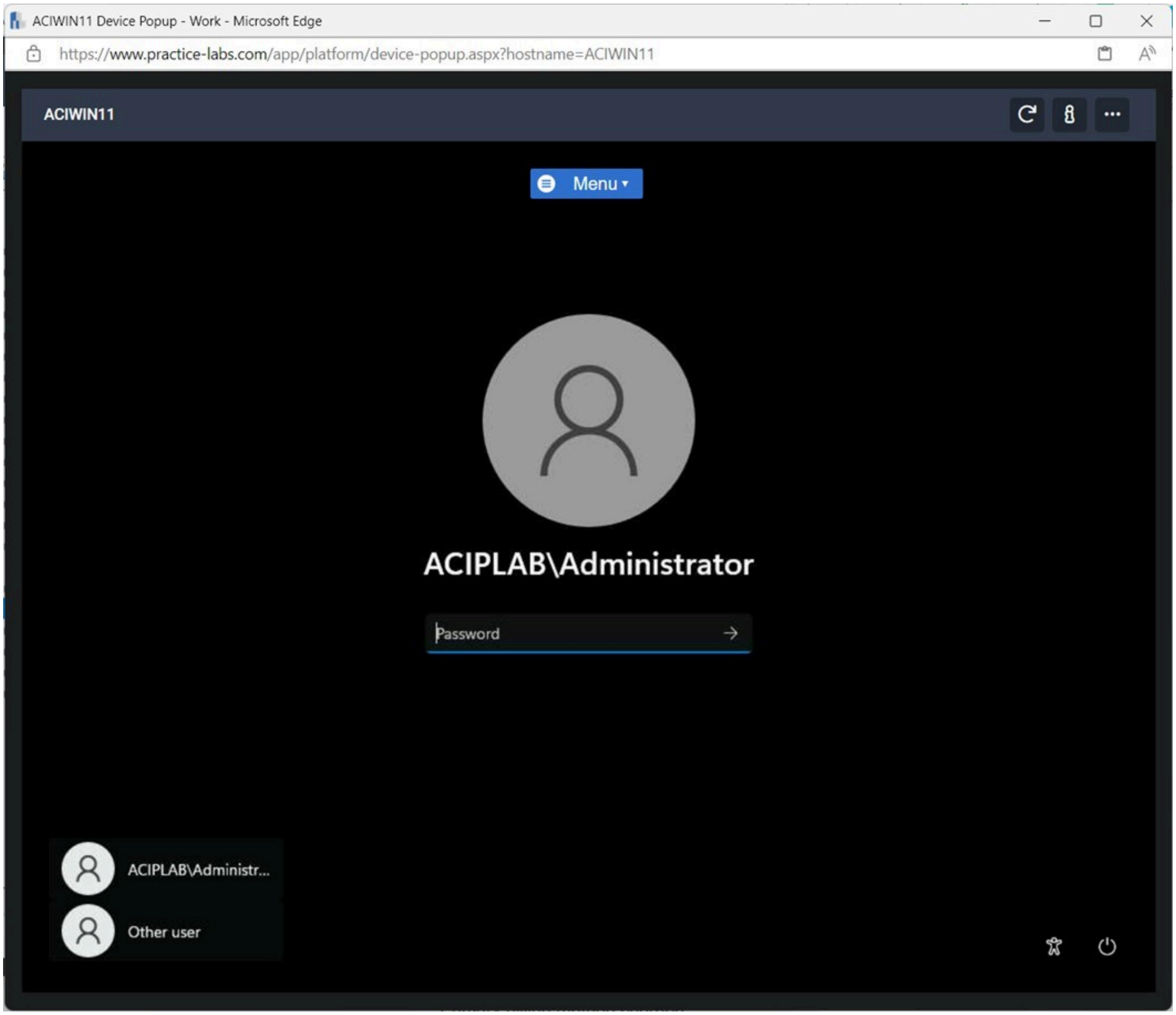


Figure 3.9 Screenshot of ACIWIN11: Displaying the login screen and clicking Other user.

Step 2

Enter the following on the login screen:

```
Username: Subi
Password: Passw0rd
```

Press **Enter**.

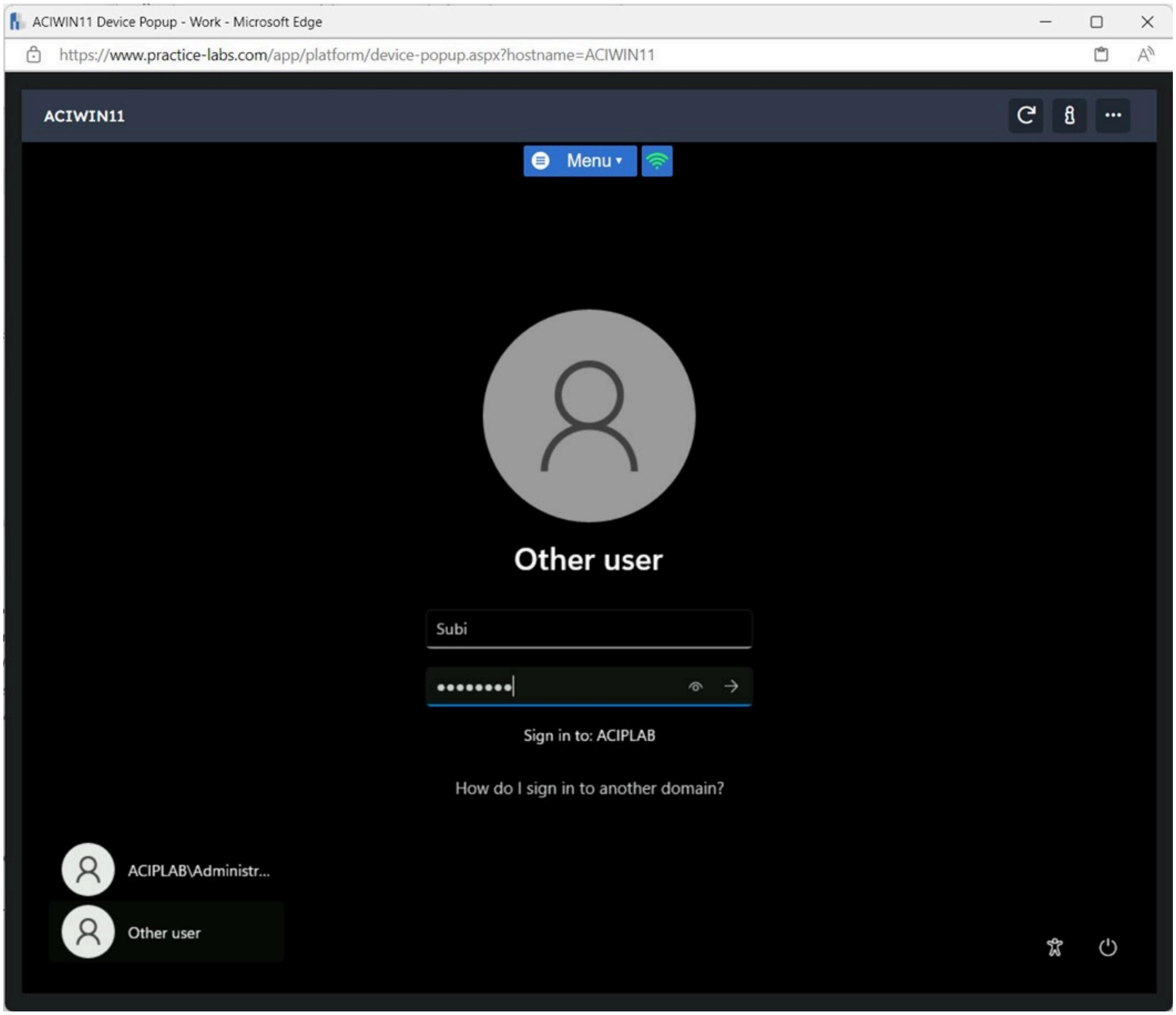


Figure 3.10 Screenshot of ACIWIN11: Displaying entering the username and password and pressing Enter.

Step 3

Notice the error message when trying to log in with the user account.

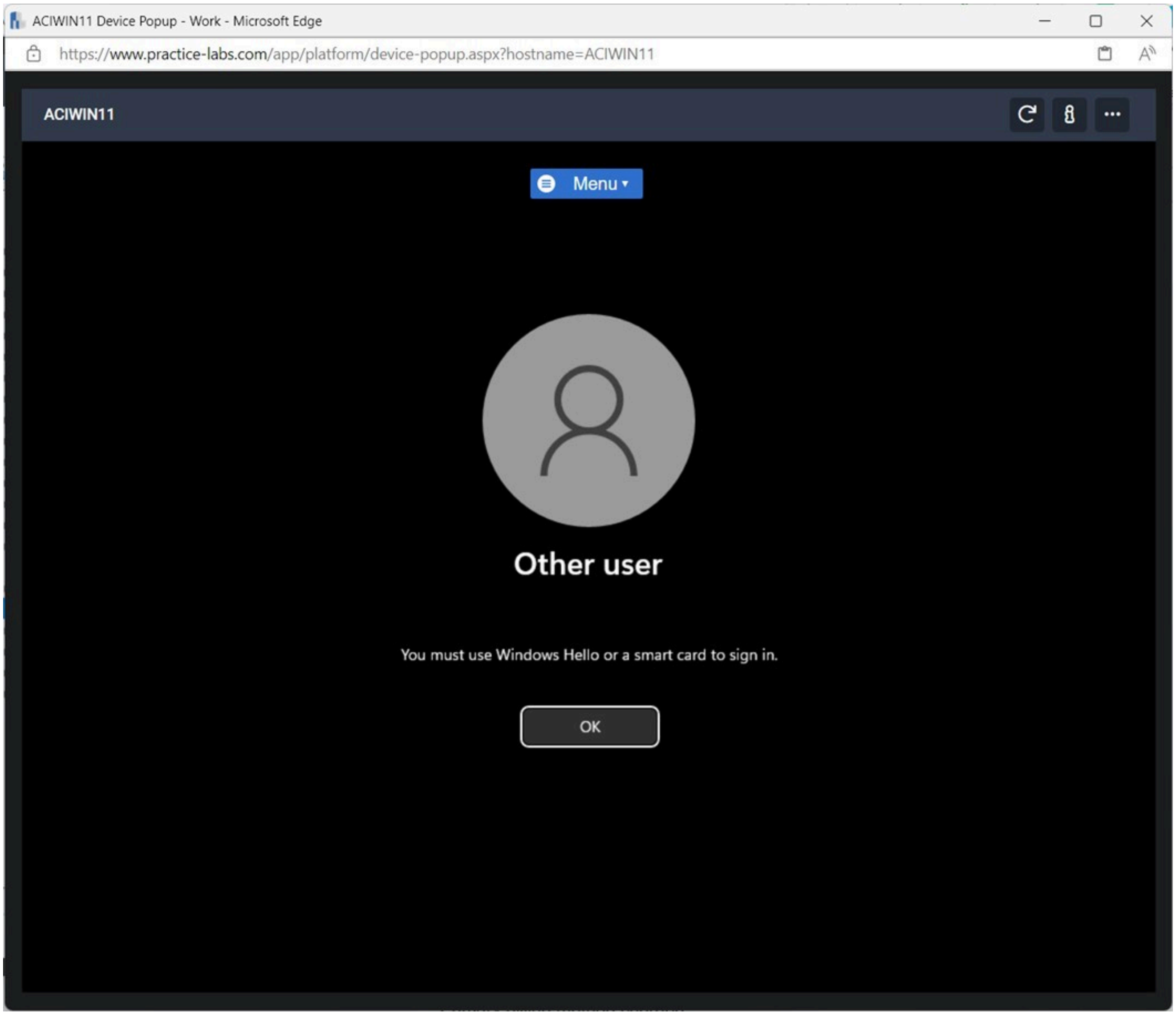


Figure 3.11 Screenshot of ACIWIN11: Displaying the user can not log in to the device.

Note: *Due to the lab environment's physical restriction, multifactor authentication can not be further applied.*

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **System & Network Security Implementation Concepts** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Log Collection with Splunk
- Exercise 2 - Encrypting Sensitive Data
- Exercise 3 - Enable Multifactor Authentication

You should now be able to:

- Install and Configure the Splunk Enterprise Application
- Configure a Splunk Client
- Collect Logs using Splunk Enterprise
- Install the BitLocker Drive Encryption Feature
- Encrypt a Local Drive on a Server
- Enable Multifactor Authentication
- Verify Multifactor Authentication

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.